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Lecture Notes in Physics

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Chapter 1

Geometry and Kinematics in a Plane

We start our investigations into physics and geometry by considering the simplest geometry that is not as trivial as a line. Most of the physics you have encountered in secondary school was, one way or another, one-dimensional. The extension into two dimensions is a good way of getting some affinity with the new methods and concepts that you will encounter when studying the three-dimensional world. **All numbered equations in this chapter are also valid in three dimensions.** When we do physics or geometry we will need to be able to describe things which have *size* and *direction*. Mathematics has developed a tool for this, the language of *vector-algebra*, in the late nineteenth, early twentieth century. In this course we will use a modern and very efficient formulation of *linear algebra*.

1.1. Drawing and Geometry

Drawing typically is a two-dimensional enterprise. Making drawings can help you to understand what is actually going on in a physical system. Drawings are always sketchy as you should never rely on a drawing to give you the solution to a problem. For simple problems that may be possible, but in general a drawing is always flawed in one way or another. Nevertheless they do have their uses, and so we start with a little drawing lesson.

1.1.1. Origin, coordinates and Bases

You may consider a plane to be a set of points that are all equivalent. To be able to identify different points we usually choose *coordinates*. Often this is done by picking one arbitrary point, called the *origin*, denoted by $\mathcal{O}$. In a physical problem *nothing* should depend on the choice of origin.

In a second step you may *relate* the other points in the plane to your origin so as to make them identifiable. Let me give two specific examples.

Cartesian coordinates

Suppose you choose two *perpendicular* directions $\mathbf{e}_1$ and $\mathbf{e}_2$ and represent them by little arrows emanating from the origin. Now draw two infinite lines through the origin in the directions of $\mathbf{e}_1$ and $\mathbf{e}_2$. Usually we call these the x_1 - and the x_2 -axes. Every point $\mathbf{x}$ can now be related to $\mathcal{O}$ by two numbers:

- x_1 = the distance you need to travel in the $\mathbf{e}_1$ to get from $\mathcal{O}$ to $\mathbf{x}$.
- x_2 = the distance you need to travel in the $\mathbf{e}_2$ to get from $\mathcal{O}$ to $\mathbf{x}$.

That is why many books introduce the notation $\mathbf{x} = \{x_1, x_2\}$. These numbers are called the *cartesian coordinates* of the point $\mathbf{x}$ in the plane. However, the values of the numbers x_j will depend on your choice of basis directions $\mathbf{e}_j$. That is why in this text I use

$$\mathbf{x} = x_1\mathbf{e}_1 + x_2\mathbf{e}_2 . \quad (1.1.1)$$

Problem 1: Do it

Make the drawing as described in the text above, choose a few arbitrary points next to $\mathcal{O}$ and determine their cartesian components.

Ofcourse cartesian components are not the only possible coordinates.

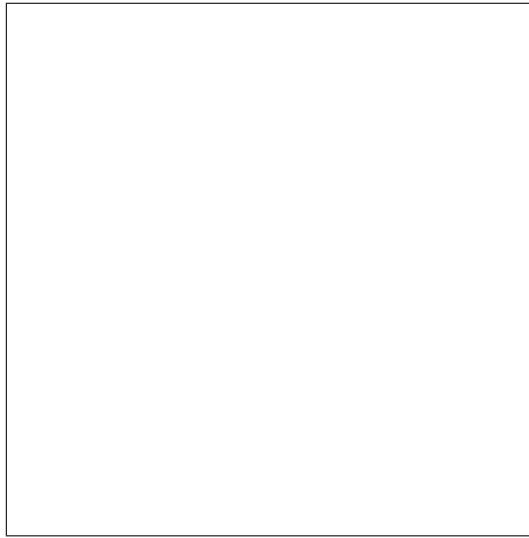


fig. 1: Cartesian and Polar coordinates in a plane

Sine, Cosine and Polar coordinates

Consider again the cartesian coordinates. Convince yourself that the *distance* r from the point $\mathbf{x}$ to the origin $\mathcal{O}$ is given by Pythagoras' theorem, so

$$r = \sqrt{x_1^2 + x_2^2} . \quad (1.1.2)$$

So if we would specify r as a coordinate then you would at least know on which circle around $\mathcal{O}$ the point $\mathbf{x}$ lies. So we introduce a second coordinate θ : we choose the intersection between the x_1 -axis and the circle as $\theta = 0$, i.e. θ is the angle between the x_1 -axis and the line connecting $\mathcal{O}$.

Problem 2: Cosine and Sine

- a: Look up the geometric definitions of the Sine and the Cosine functions.
 b: Check for yourself that with the definitions given in the text above

$$\begin{aligned}x_1 &= r \cos(\theta) , \\x_2 &= r \sin(\theta) .\end{aligned}$$

- c: Why are $\{r, \theta\}$ good coordinates for any point but $\mathcal{O}$?

The coordinates we have now constructed are called *polar coordinates*.

1.1.2. Reflections and Rotations

If you draw a point on a piece of paper and a line then it is not difficult to reflect the point in the line. It works as follows: measure the perpendicular distance between the point and the line, and then draw the reflected point at the same perpendicular distance on the opposite side of the line.

Problem 3: Reflections

Draw two perpendicular x - and y -axes on a piece of paper and draw the line $x = y$ in a different colour. Choose three different, but arbitrary points and reflect them in the line $x = y$.

When you do such reflections you'll see that different points are mapped by a varying degree depending on whether they are close to, or far away from, the line in which you reflect. This changes when you do a double reflection.

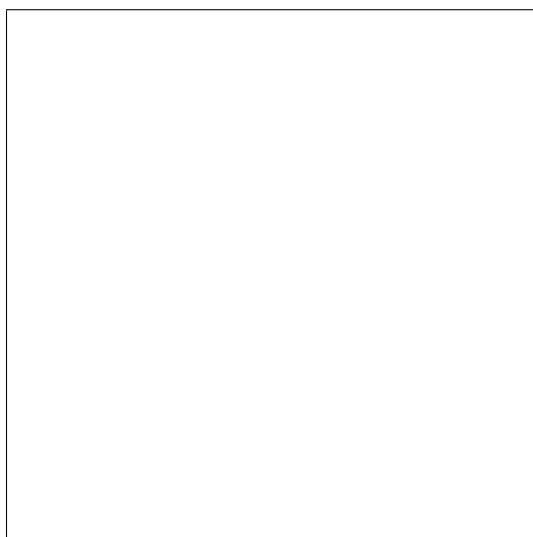


fig. 2: Constructing a reflection

Problem 4: Double Reflections

Again draw two perpendicular x - and y -axes on a piece of paper and now draw the lines $x = \frac{1}{2}y$ and $x = 2y$ in different colours. Choose two different, but arbitrary points and reflect first them in the line $x = \frac{1}{2}y$ and then in the line $x = 2y$.

If you measure the angle by which each point has been displaced after a double reflection in lines through the origin, you will see all reflected points have been *rotated* around the origin by the same angle! In fact, if you compare that angle to the angle between the lines you will see that the angle of rotation is twice the angle between the lines.

Problem 5: $\alpha \rightarrow 2\alpha$

Convince yourself of the point made in the text concerning the angle of rotation. Make several drawings, do reflections in different pairs of lines with varying angles between them and compare the results to the angles of rotation.

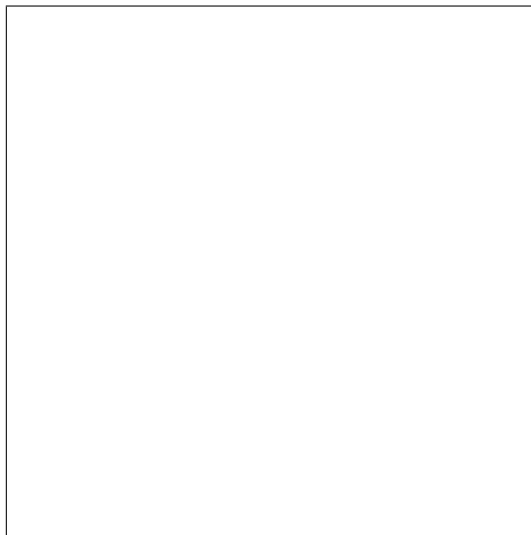


fig. 3: Double reflections and rotation

1.2. Vectors and Algebra

Objects which have size and direction are called *vectors*. Usually they are introduced in secondary school as little arrows that can be drawn. The disadvantage of relying on drawings is that we sometimes face situations that go far beyond our artistic skills, either because so many different vectors are involved, or because the number of *dimensions* is larger than we can picture, i.e. larger than 3. So we would like to be able to express relations among vector by the use of *algebra*. Algebra is just the *grammar* of mathematics, nothing more and nothing less. I will assume that you have a glancing familiarity with vectors. As they have a size it makes sense to talk about the *length* of a vector. And for all I care, in this chapter you may start out by thinking that this size is actually measured by putting your rule next to the vector. From now on we will always denote vectors with bold

small case letters, like $\mathbf{a}$ and $\mathbf{b}$. There is one special class of vectors, those with *unit-length*. To distinguish these in notation from the others, we give them a hat: for example $\hat{\mathbf{n}}$. Now ofcourse every vector $\mathbf{a}$ can be split up in its length a and its direction $\hat{\mathbf{a}}$ as

$$\mathbf{a} = a\hat{\mathbf{a}} . \quad (1.2.1)$$

Unit vectors will play an inportant role in our algebraic handling of vectors and geometry.

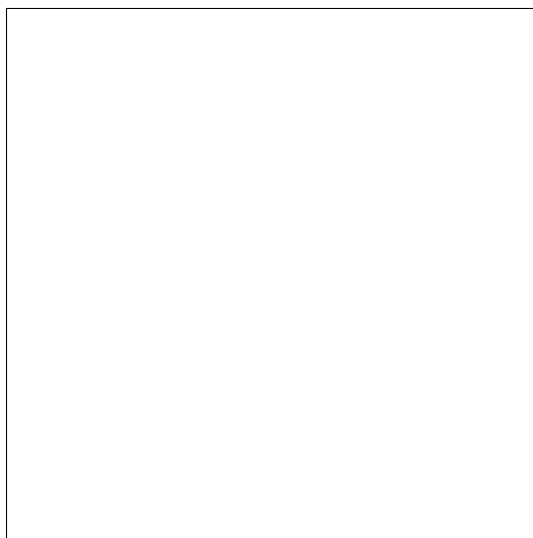


fig. 4: $\mathbf{a}$ and $\hat{\mathbf{a}}$.

Dot- or Innerproduct

You will have learned how to add vectors, graphically, and probably you can guess what it means if we multiply a vector by a *number*. The vector $5\mathbf{a}$ is just five times longer than $\mathbf{a}$ but has the same direction as $\mathbf{a}$. We will now introduce the dotproduct between unit-vectors, first in words, and then as an equation.

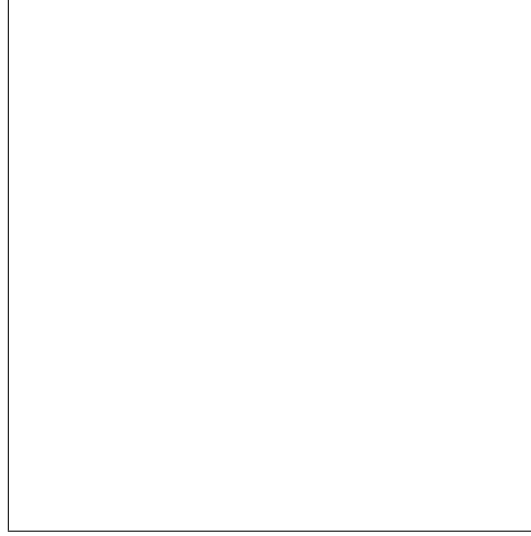
- The dotproduct $\hat{\mathbf{a}} \cdot \hat{\mathbf{b}}$ is the length of the part of $\hat{\mathbf{a}}$ that is along $\hat{\mathbf{b}}$.
- When you put both unit-vectors in a drawing on a circle of radius 1 around the origin you will easilly see that $\hat{\mathbf{a}} \cdot \hat{\mathbf{b}} = \cos(\theta)$, where θ is the angle between the two vectors.

Note that this means that for any two vectors $\mathbf{a}$ and $\mathbf{b}$ with lengths a and b this means that

$$\mathbf{a} \cdot \mathbf{b} = ab \cos(\theta) . \quad (1.2.2)$$

This is probably the definition that you have learned in secondary school about vectors and their inner-product. It is still valid in this course! Now that is good news isn't it.

Perpendicularity and components

fig. 5: Drawing $\hat{\mathbf{a}} \cdot \mathbf{b}$.

Before we get started, what I need to address is what we get when two vectors are perpendicular. Suppose $\mathbf{a}$ and $\mathbf{b}$ are perpendicular, then we will have

$$\mathbf{a} \cdot \mathbf{b} = 0 . \quad (1.2.3)$$

In a plane we can always find two unit-vectors that are mutually perpendicular. We will give these two directions two standard symbols: $\mathbf{e}_1$ and $\mathbf{e}_2$ for which

$$\mathbf{e}_1 \cdot \mathbf{e}_1 = 1 , \quad (1.2.4)$$

$$\mathbf{e}_2 \cdot \mathbf{e}_2 = 1 , \quad (1.2.5)$$

$$\mathbf{e}_2 \cdot \mathbf{e}_1 = \mathbf{e}_1 \cdot \mathbf{e}_2 = 0 . \quad (1.2.6)$$

These vectors are called *basis*-vectors. If you can memorize these rules you have mastered the most difficult part of this course! We can make even more use of our dot-product as follows. Suppose I give you some arbitrary vector $\mathbf{a}$. You know that you can, again graphically, write this vector as a sum of two perpendicular vectors. Graphically you know how to get the components of this vector if I give you two perpendicular vectors $\mathbf{e}_1$ and $\mathbf{e}_2$. But now notice that if we define the *numbers* a_j

$$\mathbf{a} \cdot \mathbf{e}_j = a_j , \quad (1.2.7)$$

then

$$\mathbf{a} = a_1 \mathbf{e}_1 + a_2 \mathbf{e}_2 . \quad (1.2.8)$$

The use of components makes the addition of vectors, and multiplication by a number, very easy, for example

$$\mathbf{a} + \mathbf{b} = (a_1 + b_1) \mathbf{e}_1 + (a_2 + b_2) \mathbf{e}_2 , \quad (1.2.9)$$

$$c \mathbf{a} = c a_1 \mathbf{e}_1 + c a_2 \mathbf{e}_2 . \quad (1.2.10)$$

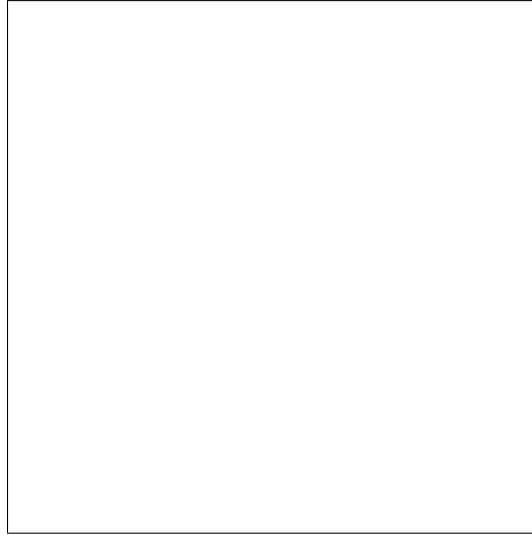


fig. 6: Vector addition and scalar-multiplication

Finally I want to show one more thing, take a look at what we can do with the dot-product if we use the basis-vectors there to,

$$\begin{aligned}
 \mathbf{a} \cdot \mathbf{b} &= (a_1 \mathbf{e}_1 + a_2 \mathbf{e}_2) \cdot (b_1 \mathbf{e}_1 + b_2 \mathbf{e}_2) , \\
 &= a_1 \mathbf{e}_1 \cdot b_1 \mathbf{e}_1 + a_1 \mathbf{e}_1 \cdot b_2 \mathbf{e}_2 + a_2 \mathbf{e}_2 \cdot b_1 \mathbf{e}_1 + a_2 \mathbf{e}_2 \cdot b_2 \mathbf{e}_2 , \\
 &= a_1 b_1 + a_2 b_2 .
 \end{aligned} \tag{1.2.11}$$

In this last line we simply used the properties of the basis-vectors.

Problem 6: Lists of Components

Many books on vector algebra represent vectors by *lists of components*; so they write $\mathbf{a} = \{a_1, a_2\}$ and never deal with the basis-vectors themselves.

a: Calculate the dot-product between $\mathbf{a} = \{1, 2\}$ and $\mathbf{b} = \{2, -1\}$.

b: How would we write down the vectors $\mathbf{a}$ and $\mathbf{b}$.

The dot-product can thus be expressed directly in terms of the components. As a special case we get

$$\mathbf{a} \cdot \mathbf{a} = a_1^2 + a_2^2 = a^2 , \tag{1.2.12}$$

which shows that the length-squared of a vector is just the dot-product of a vector with itself; Pythagoras is all you need to understand this result.

Problem 7: Lengths

Use the vectors $\mathbf{a}$ and $\mathbf{b}$ from the previous Problem and calculate their lengths.

Now this possibly was a lot of information, but it wasn't too difficult I guess. This is also usually where vector algebra stops: addition and multiplication by a number written in algebraic terms.

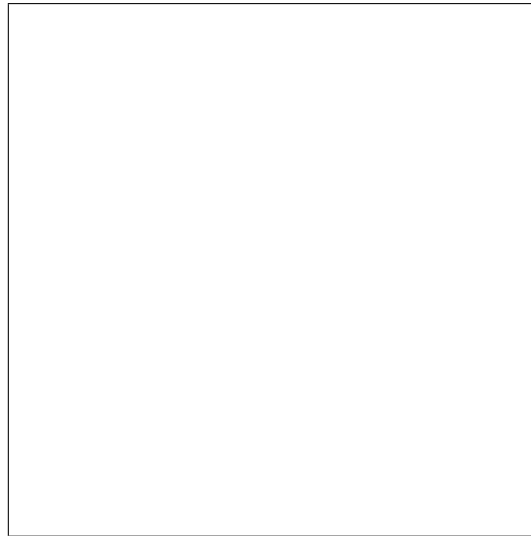


fig. 7: Components in a basis

Problem 8: Adding and multiplying vectors

Given the following vectors in a basis of $\mathbf{e}_1$ and $\mathbf{e}_2$,

$$\mathbf{a} = \mathbf{e}_1 - 3\mathbf{e}_2 ,$$

$$\mathbf{b} = 3\mathbf{e}_1 + \mathbf{e}_2 ,$$

$$\mathbf{c} = \frac{1}{2}\mathbf{e}_1 - 2\mathbf{e}_2 ,$$

$$\mathbf{d} = \mathbf{e}_1 + \mathbf{e}_2 .$$

a: Calculate the vectors $\mathbf{a} + \mathbf{b}$ and $\mathbf{a} - \mathbf{c}$.

b: Calculate whether $\mathbf{a}$ and $\mathbf{b}$ are perpendicular. Draw then in a sketch and confirm your graphically calculation.

c: Calculate the following products; $\mathbf{a} \cdot \mathbf{c}$, $\mathbf{b} \cdot \mathbf{d}$, and $\mathbf{d} \cdot \mathbf{d}$.

But there is more to geometry than just this. We have seen reflections in a line, and rotations as double reflections. Remarkably the *algebra of vectors* also covers these operations.

Problem 9: Parallel and Perpendicular

Consider two vectors $\mathbf{a}$ and $\mathbf{b}$.

a: If $a^2 = \mathbf{a} \cdot \mathbf{a}$, then calculate the components of $\hat{\mathbf{a}}$ given by

$$\hat{\mathbf{a}} = \frac{\mathbf{a}}{a} \quad (1.2.13)$$

in some basis $\mathbf{e}_1, \mathbf{e}_2$.

b: Show by computation that the vector $\hat{\mathbf{a}}$ is a unit-vector parallel to $\mathbf{a}$.

c: Assume that $\mathbf{a}$ and $\mathbf{b}$ are *not* parallel. Show that the vector

$$\mathbf{b}_{//} = \hat{\mathbf{a}}(\hat{\mathbf{a}} \cdot \mathbf{b}) , \quad (1.2.14)$$

is the part of $\mathbf{b}$ that is parallel to $\mathbf{a}$.

d: Show now that the vector

$$\mathbf{b}_{\perp} = \mathbf{b} - \mathbf{b}_{//} ,$$

is perpendicular to $\mathbf{a}$ by computing $\mathbf{a} \cdot \mathbf{b}_{\perp}$.

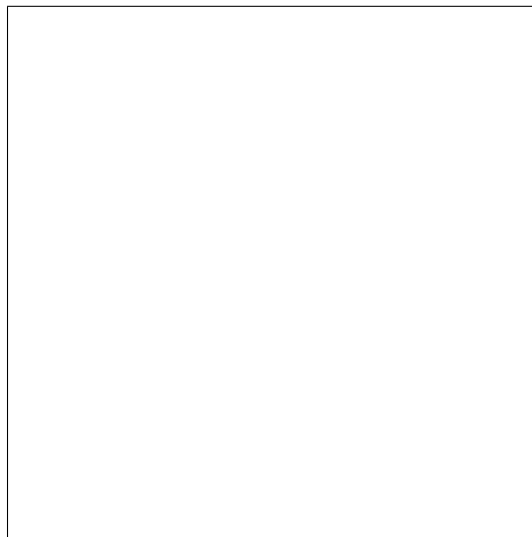


fig. 8: Parallel and perpendicular parts

1.2.1. Linear Maps

The standard way in linear algebra of discussing things like rotations and reflections is by means of so-called *linear maps*. A *linear map* L on vectors, like $\mathbf{a}$, is a function that maps the vector $\mathbf{a}$ onto a new vector $L(\mathbf{a})$ such that it satisfies

$$L(\mathbf{a} + \mathbf{b}) = L(\mathbf{a}) + L(\mathbf{b}) , \quad (1.2.15)$$

$$L(c\mathbf{a}) = cL(\mathbf{a}) . \quad (1.2.16)$$

Linear maps are very common in applications in physics, economics or other areas of applied mathematics. Sometimes a linear map takes us from one space to a different space. But in this text we will always deal with maps that is a map on the space itself.

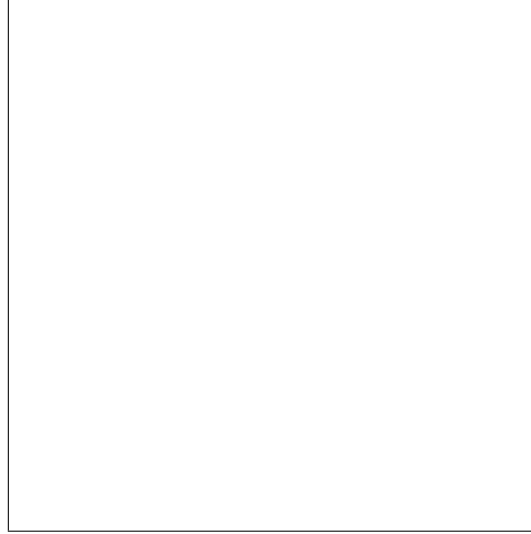


fig. 9: A linear map between spaces

An important characteristic of linear maps is that to study describe effect on vectors all we need to know is how they affect the basis vectors $\mathbf{e}_j$. This gives rise to the notions of *components* of a linear map.

Problem 10: Linear maps and Components

a: Given two vectors $\mathbf{a}'$ and $\mathbf{a}$, give expressions for the components a'_i and a_j of both vectors in the basis $\mathbf{e}_1, \mathbf{e}_2$.

b: We now define *the components* L_{ij} of the linear map L in this basis as

$$L_{ij} = \mathbf{e}_i \cdot L(\mathbf{e}_j) . \quad (1.2.17)$$

c: How many components are there of a linear map in two dimensions? And in three dimensions? And in n dimensions?

d: Let the vector $\mathbf{a}'$ be the image of $\mathbf{a}$ under the linear map L , i.e. $\mathbf{a}' = L(\mathbf{a})$. Show that

$$a'_i = \sum_{n=1}^2 L_{in} a_n . \quad (1.2.18)$$

The components of a linear map can be written as a square array of number with 2 rows and two collums (in two dimensions, there are n rows and n collums in n dimensions). This square array is usually called the *matrix* representing the linear map, and it is denoted simply by L_{ij} . Most textbooks in physics rely on matrix methods to treat linear maps. Most (pure) mathematics textbooks prefer to use the abstract symbols, like L in this problem, because they do not refer to any choice of basis.

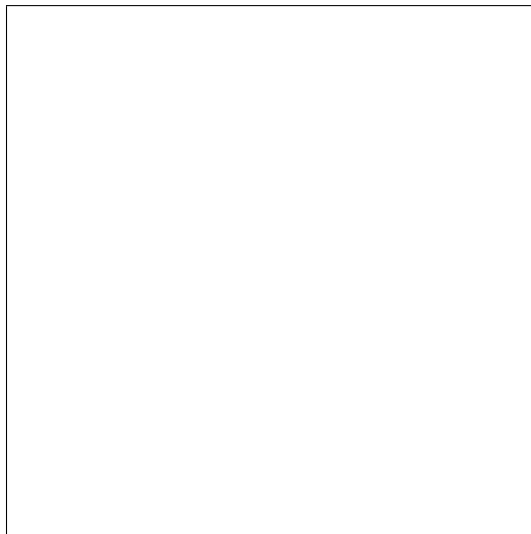


fig. 10: Components of a linear map

Problem 11: Examples of Maps

In this problem we are going to look at a few easy examples.

- a: Consider the map $L(\mathbf{a}) = \mathbf{a}$; calculate the components L_{ij} .
- b: Consider a map L of which the components are given by $L_{11} = L_{22} = 0$ and $L_{12} = -L_{21} = 1$. Calculate the result of $L(\mathbf{a})$ for an arbitrary vector $\mathbf{a}$.
- c: Is this map a reflection?

Using the techniques of this problem you can also attempt the following. It is a bit more work, but not really harder than the previous one.

Problem 12: Products of linear Maps

Let both M and K be linear maps, and define a new linear map H by $H(\mathbf{a}) = M(K(\mathbf{a}))$. Show that the components of H and those of K and M are related through

$$H_{ij} = \sum_n M_{in} K_{nj} . \quad (1.2.19)$$

Those of you familiar with matrices may check that this is just the standard product between matrices.

Problem 13: The *kernel* of a map

Consider a linear map A and its action on vectors $\mathbf{a}$ from some vectorspace. The *kernel* K_A of a A is the set of vectors $\mathbf{b}$ for which $A[\mathbf{b}] = 0$. Now suppose that the vector $\mathbf{b}$ is a solution to the equation $A[\mathbf{b}] = \mathbf{c}$. Show that if the kernel of A is not an empty set, then $\mathbf{b}$ is not the only solution.

1.2.2. Algebra of geometry

We are going to start by *assuming* the existence of a more general, and in fact

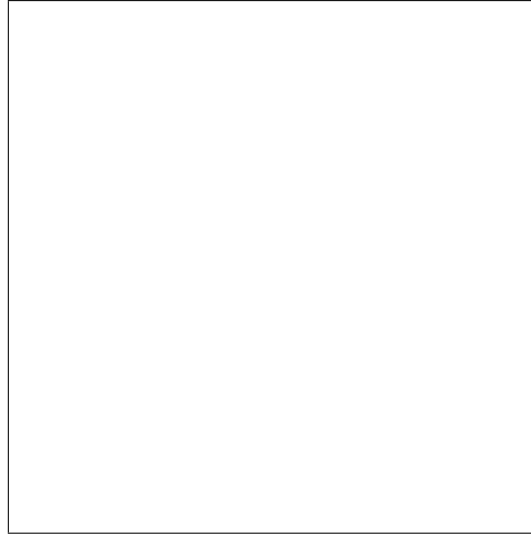


fig. 11: A product of maps

more powerfull product among vector, the *geometric product*. I will denote the geometric product between two vectors $\mathbf{a}$ and $\mathbf{b}$ as $\mathbf{ab}$. All we will require of the geometric product is that the square of a vector gives its length squared:

$$\mathbf{aa} = \mathbf{a} \cdot \mathbf{a} . \quad (1.2.20)$$

This restriction has the very important consequence that the dot-product between two vectors can be rewritten in terms of the geometric product as

$$\mathbf{a} \cdot \mathbf{b} = \frac{1}{2}(\mathbf{ab} + \mathbf{ba}) . \quad (1.2.21)$$

Now there is nothing which tells us that $\mathbf{ab} = \mathbf{ba}$ so we will not assume this. Rather, we will define the other half of the geometric product as the so-called *wedge-product*

$$\mathbf{a} \wedge \mathbf{b} = \frac{1}{2}(\mathbf{ab} - \mathbf{ba}) . \quad (1.2.22)$$

On the basis of these two definitions we have

$$\mathbf{ab} = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \wedge \mathbf{b} , \quad (1.2.23)$$

so the geometric product is the ordinary dot and somethign extra. What is this extra bit? Does it make sense, or is it just something from our imagination?

The wedge

The wedge is a member of the rogue squadron of mathematics. It's tricky and very powerfull if you know how to use it; in the right place, at the right time. Let is first note that by definition we have

$$\mathbf{a} \wedge \mathbf{b} = -\mathbf{b} \wedge \mathbf{a} . \quad (1.2.24)$$

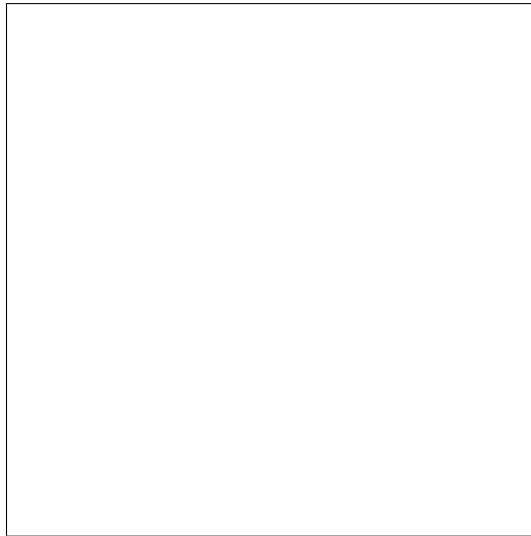


fig. 12: The wedge and orientations

Because for basis vectors we have $\mathbf{e}_1 \cdot \mathbf{e}_2 = 0$ we get

$$\mathbf{e}_1 \mathbf{e}_2 = -\mathbf{e}_2 \mathbf{e}_1 . \quad (1.2.25)$$

This result is really all you need to know in order to be able to work with geometric products. Because when we now compute the wedge between two vectors $\mathbf{a}$ and $\mathbf{b}$ in this basis we find

$$\mathbf{a} \wedge \mathbf{b} = (a_1 b_2 - a_2 b_1) \mathbf{e}_1 \mathbf{e}_2 . \quad (1.2.26)$$

So what is the strange thing $\mathbf{e}_1 \mathbf{e}_2$ and what is the meaning of the *number*

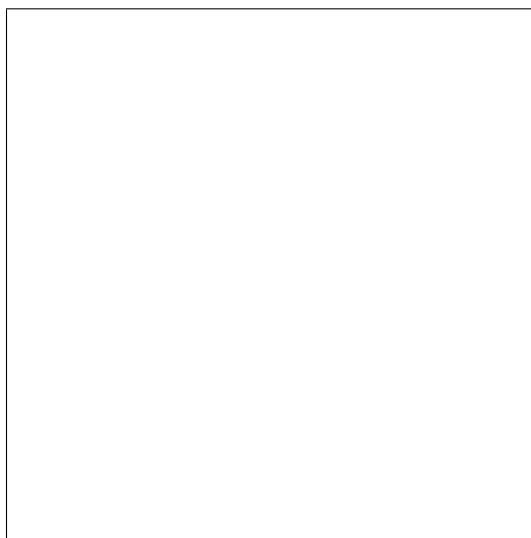


fig. 13: The wedge and surface areas

$a_1 b_2 - a_2 b_1$? If we make a clever choice of basis we can choose one of the vectors,

say $\mathbf{e}_1$ to be along the vector $\mathbf{a}$, then we find

$$\begin{aligned} a_1 b_2 - a_2 b_1 &= a_1 b_2 , \\ &= ab \sin(\theta) , \end{aligned} \quad (1.2.27)$$

and it represents the *surface area of the parallelogram* spanned by the two vectors. Now that is a geometrically meaningful quantity.

Problem 14: Parallel and Perpendicular

Consider two vectors $\mathbf{a}$ and $\mathbf{b}$.

a: If $\mathbf{a}$ and $\mathbf{b}$ are parallel, show that that would imply $\mathbf{a} \wedge \mathbf{b} = 0$.

b: Look at the definition of perpendicular part from the previous section and then show that

$$\mathbf{b}_\perp = \hat{\mathbf{a}}(\hat{\mathbf{a}} \wedge \mathbf{b}) , \quad (1.2.28)$$

is true. (Hint: use that $\hat{\mathbf{a}}^2 = 1$ to write $\mathbf{b} = \hat{\mathbf{a}}\mathbf{a}\mathbf{b}$ and then use the expression for $\mathbf{b}_\perp$ from problem 2).

What about this *thing* $\mathbf{e}_1\mathbf{e}_2$; well as we don't know what it is let's start by giving it a name: it's a bi-vector! We give it a special symbol too, and call it

$$J_3 = \mathbf{e}_1\mathbf{e}_2 . \quad (1.2.29)$$

Now unearthing what J_3 starts us on a wonderful trip into the realm of Geometry. Much of the power of the algebra comes from the fact that things such as J_3 exist and have fantastic properties. Let us use our new knowledge.

We now know that multiplying two vectors $\mathbf{a}\mathbf{b}$ generally gives a result that contains a number $\mathbf{a} \cdot \mathbf{b}$ and a bivector $\mathbf{a} \wedge \mathbf{b}$. Can we use this knowledge?

Problem 15: Inverse vectors

Vectors can also be used to divide by. From ordinary numbers c you know that they have an *inverse* c^{-1} defined by $cc^{-1} \equiv 1$. Only if we use the *geometric product* we can also invert vectors. Consider a vector $\mathbf{a}$ for which $a^2 = \mathbf{a}^2$.

a: The vector $\mathbf{a}^{-1}$ is defined to satisfy $\mathbf{a}\mathbf{a}^{-1} = 1$, then why must $\mathbf{a}^{-1}$ be parallel to $\mathbf{a}$?

b: Show that

$$\mathbf{a}^{-1} = \frac{\mathbf{a}}{a^2} . \quad (1.2.30)$$

The only reason why the inversion is unique is because for $\mathbf{a}\mathbf{a}^{-1} = 1$ to be true it is required that $\mathbf{a} \wedge \mathbf{a}^{-1} = 0$. This restriction allows the unique identification of one, and only one, vector $\mathbf{a}^{-1}$ as the inverse of $\mathbf{a}$.

Problem 16: Computing inverses

In this problem you will apply the results of the previous problem.

a: Take the vectors $\mathbf{a}$ through $\mathbf{d}$ from problem 8, and compute their inverses $\mathbf{a}^{-1}$ through $\mathbf{d}^{-1}$.

b: Calculate the following wedges; $\mathbf{a} \wedge \mathbf{b}$, $\mathbf{b} \wedge \mathbf{d}$, and $\mathbf{d} \wedge \mathbf{d}$.

c: Compute $\mathbf{a}^{-1} \wedge \mathbf{b}^{-1}$; what is the relation between this bivector and $\mathbf{a} \wedge \mathbf{b}$?

The bivector

We saw that the component of J_3 in a wedge-product was related to a surface area defined by the two vectors. Furthermore we saw that the presence of bivectors makes the geometric product *invertible*. But the *thing* J_3 may still seem rather mysterious to you. So what *is* this bivector really?

If you don't know what something is you can always try and square it. If we do that we get

$$\begin{aligned} J_3^2 &= \mathbf{e}_1 \mathbf{e}_2 \mathbf{e}_1 \mathbf{e}_2 , \\ &= -\mathbf{e}_1 \mathbf{e}_1 \mathbf{e}_2 \mathbf{e}_2 , \\ &= -1 . \end{aligned} \tag{1.2.31}$$

Wow! J_3 is something that squares to -1 , so its something like $\sqrt{-1}$ that surely is weird. Some people (about 90 percent of all physicists) call this an *imaginary unit* and usually denote it with i without realising that it is a bivector. We will see lots of applications for this in the sections to come.

1.2.3. Reflections and Rotations

The great utility of the geometric product lies in the fact that operations such as rotating a vector, and reflecting it in a line, are expressed with so much ease and without the need to return to matrices.

Reflections

Consider some arbitrary vector $\mathbf{a}$ and the line through the origin in the direction of $\mathbf{e}_1$. Now note the following little computation,

$$\begin{aligned} \mathbf{e}_1 \mathbf{a} \mathbf{e}_1 &= \mathbf{e}_1 (a_1 \mathbf{e}_1 + a_2 \mathbf{e}_2) \mathbf{e}_1 , \\ &= a_1 \mathbf{e}_1 \mathbf{e}_1 \mathbf{e}_1 + a_2 \mathbf{e}_1 \mathbf{e}_2 \mathbf{e}_1 , \\ &= a_1 \mathbf{e}_1 - a_2 \mathbf{e}_2 . \end{aligned}$$

Hence the vector that we obtain by *sandwiching* $\mathbf{a}$ between two equal unit-vectors, is the vector $\mathbf{a}$ reflected in the line through the origin along the unit-vector we have used. Therefor we will from now on use: let $\mathbf{a}'$ be the reflection of $\mathbf{a}$ in a line through the origin with the direction $\hat{\mathbf{b}}$, then

$$\mathbf{a}' = \hat{\mathbf{b}} \mathbf{a} \hat{\mathbf{b}} . \tag{1.2.32}$$

This is a useful equation whenever we want to calculate with reflections.

Rotations from Reflections

We saw in the very first section that rotations are nothing but double reflections. In this section we have learned how to do reflections algebraically. Now we can also do two reflections.

Let $\mathbf{c}$ be a vector that we rotate this way into a new vector $\mathbf{c}'$, by a reflection in two lines through the origin in the directions $\hat{\mathbf{a}}$ and $\hat{\mathbf{b}}$. We get

$$\mathbf{c}' = \hat{\mathbf{b}} \hat{\mathbf{a}} \mathbf{c} \hat{\mathbf{a}} \hat{\mathbf{b}} .$$

Note that the product of unit-vector multiplying $\mathbf{c}$ from the right has the *reversed order* when compared to the factors on the left. This will always happen, and because of that it is convenient to introduce the operation of *reversal* denoted by a tilde, so

$$\begin{aligned}\widetilde{(\hat{\mathbf{b}}\hat{\mathbf{a}})} &= \hat{\mathbf{a}}\hat{\mathbf{b}} , \\ \tilde{J}_3 &= -J_3 , \\ \tilde{\mathbf{c}} &= \mathbf{c} , \\ \tilde{d} &= d .\end{aligned}$$

Reversing means that in a given expression the order of all vectors in a geometric product is reversed.

Problem 17: Reversing the Bivector

Show that the reverse of J_3 is indeed equal to $-J_3$.

Now suppose that the vector $\hat{\mathbf{b}}$ is just $\mathbf{e}_1$ and $\hat{\mathbf{a}} = \cos(\theta)\mathbf{e}_1 + \sin(\theta)\mathbf{e}_2$. Then it is easy to compute that

$$\hat{\mathbf{b}}\hat{\mathbf{a}} = \cos(\theta) - \sin(\theta)\mathbf{e}_1\mathbf{e}_2 .$$

So we can write the rotation as

$$\mathbf{c}' = (\cos(\theta) - \sin(\theta)\mathbf{e}_1\mathbf{e}_2)\mathbf{c}(\cos(\theta) + \sin(\theta)\mathbf{e}_1\mathbf{e}_2) ,$$

where θ is the angle between $\hat{\mathbf{a}}$ and $\hat{\mathbf{b}}$. As we know from our drawings this generates a rotation of $\mathbf{c}$ about the angle 2θ . The algebra also produces that result.

Problem 18: Rotating $\mathbf{e}_1$.

Use $\mathbf{c} = \mathbf{e}_1$ in the above equation and check by doing the algebra that $\mathbf{e}_1$ is indeed rotated by an angle 2α .

Hence the *thing* that rotates a vector about an angle α is therefore

$$R(\alpha) = \cos\left(\frac{\alpha}{2}\right) - \sin\left(\frac{\alpha}{2}\right)J_3 , \quad (1.2.33)$$

and it is called a *Rotor*. The minus-sign ensures that the rotor rotates vectors in the with the orientation as J_3 has. The rotated vector is found through

$$\mathbf{c}' = R(\alpha)\mathbf{c}\tilde{R}(\alpha) . \quad (1.2.34)$$

This is the way in which all the rotations are handled. In two dimensions, but also in three and more. It is fairly easy to show that

$$J_3 = R(\alpha)J_3\tilde{R}(\alpha) . \quad (1.2.35)$$

This indicates that the bivector represents something that does not change under rotations. As we already know that it relates to surfaces, we can now finally give

an interpretation of this bivector: it represents the entire plane!

Problem 19: Rotating J_3 .

- a: First give an argument why you would expect J_3 not to change under a rotation.
 b: Rotate the basis vectors $\mathbf{e}_j$ and compute the wedge of the rotated basis vectors. Does that result confirm the invariance of J_3 ?

Now look at how nice this is:

- Numbers or *scalars*: they represent objects which only have a size or magnitude.
- Vectors represent objects which have a *size* and a *direction*.
- Bivectors represent flat surfaces that that have an *area* and lie in some *plane*.

The *Geometry of two-dimensional space* consists of objects which can be written as a sum of numbers, vectors and the bivector

$$M = a + \mathbf{b} + cJ_3 , \quad (1.2.36)$$

called *multivectors*, and that can be added and multiplied.

1.3. Kinematics in a plane

In classical, non-relativistic, physics the *trajectory* of an object is the function that assigns a position $\mathbf{x}(t)$ to every time t . The goal of classical physics is to *predict* the trajectory of every particle involved in a physical process from a given *initial* state of the system. Classical physics, in its late nineteenth century form is *reductionist*, i.e. it assumes that *all properties of a material system* can be understood from the trajectories of all the constituents of the system.

Kinematics is *the description of motion*, and as such kinematics makes no attempt at explaining motion. It merely puts together a number of concepts that are relevant in describing how particles move. There are two types of motion that we will deal with separately, translational motion and rotational motion.

Point particles

The simplest objects are so-called *point particles*. They by definition do not have a size and only have mass, a position and a velocity, and possibly things like *electric charge*. Their motion is described in terms of position, velocity and acceleration. The laws of *dynamics*, i.e. Newton's laws, relate these aspects of motion to the properties of the particle and the forces acting on a particle. Extended systems are believed to be made up out of elementary point-particles. Originally these were believed to be *molecules* although twentieth century atomic and nuclear physics have shown us that truly *elementary* particles are only to be found deep inside the atoms: the quarks and leptons.

Yet molecules as point particles are good enough for practically all applications of classical mechanics. Next to that many macroscopic objects can be considered to be point particles in the relevant context. Let me give a few examples of this. Although molecules are definitely not point particles in the strict sense, they can

usually be approximated as such when we calculate the properties of gasses, liquids and solids at ordinary temperatures. When we calculate the orbits of planets in the Solar System, the planets themselves can be considered to be point particles in the Sun's gravitational field. When we study the evolution and dynamics of our galaxy, our Solar System can be taken as a point particle. Finally, if we study the later stages of the Big Bang the universe can be well approximated treating each galaxy as a point particle.

1.3.1. Linear Motion

In kinematics we start out by assuming we know the trajectory $\mathbf{x}(t)$ of a particle. As the *position* of a particle with respect to some reference point $\mathcal{O}$, called the *origin*, has both a direction and a magnitude it is represented by a vector. The *velocity* $\mathbf{v}(t)$ of the particle is the *rate* at which its position changes. Expressed mathematically this is achieved by differentiating the position with respect to time, i.e.

$$\mathbf{v}(t) = \frac{d}{dt}\mathbf{x}(t) . \quad (1.3.1)$$

As you have seen in the first chapter these vectors can be expanded in terms of a *fixed* basis $\mathbf{e}_j$. In two dimensions we would therefor be able to write

$$\begin{aligned} \mathbf{v}(t) &= \frac{d}{dt}\{x_1(t)\mathbf{e}_1 + x_2(t)\mathbf{e}_2\} , \\ &= \frac{d}{dt}x_1(t)\mathbf{e}_1 + \frac{d}{dt}x_2(t)\mathbf{e}_2 , \\ &= v_1(t)\mathbf{e}_1 + v_2(t)\mathbf{e}_2 . \end{aligned}$$

We call the functions $v_j(t)$ the components of the velocity in the corresponding direction. Velocity should be distinguished from *speed*. We can define the speed $v(t)$ of a particle as

$$v(t) = \sqrt{\mathbf{v}(t) \cdot \mathbf{v}(t)} , \quad (1.3.2)$$

and it is just the magnitude of the velocity.

Problem 20: Calculating velocity

Consider a particle whose position is given as

$$\mathbf{x}(t) = R \cos(\omega t)\mathbf{e}_1 + R \sin(\omega t)\mathbf{e}_2 .$$

- a: Compute the velocity of this particle.
- b: Compute the speed of the particle.
- c: Is its kinetic energy constant?

The next and final kinematic concept is that of *acceleration*. It is the *rate* at which the velocity changes. As a result it too is found by differentiation,

$$\mathbf{a}(t) = \frac{d}{dt}\mathbf{v}(t) , \quad (1.3.3)$$

and is also a vector quantity. Because of the relation between velocity and position, we see that acceleration is found by differentiating the position *twice* with respect

to time. We can denote this as

$$\mathbf{a}(t) = \frac{d^2}{dt^2} \mathbf{x}(t) .$$

Here our discussion of kinematics ends, but this is not trivial. In principle we could go on differentiating forever. But we do not need to. The reason for that is as deep as it is useful.

Problem 21: Calculating acceleration

Consider the same particle again whose position was given in the previous Problem.

a: Compute the acceleration $\mathbf{a}(t)$ of this particle.

b: Compute $\mathbf{v}(t) \cdot \mathbf{a}(t)$ for this particle.

c: Sketch the particle orbit, its velocity and acceleration at several points?

Newton's first law

Newton's first law tells us that *if no net force acts on a particle with mass, then its velocity is constant*. It is called the principle of inertia; it defines constant velocity as the trivial state of affairs.

Problem 22: Uniform linear motion

Consider a particle moving along a straight line with a constant velocity $\mathbf{v}_0$.

a: Show that $\mathbf{a}(t) = 0$.

b: Show that $\mathbf{x}(t) = \mathbf{x}(0) + \mathbf{v}_0 t$.

c: Show that $\mathbf{x}(t) \wedge \mathbf{v}(t)$ is constant.

1.3.2. Rotational motion

Suppose that some point particle is restricted to move only at a fixed distance from the origin. In other words, $\mathbf{x}(t)^2 = \text{constant}$. This happens when, for example, the particle we are studying is part of a larger and rigid body that rotates around the origin. In such a case notice that we have,

$$\begin{aligned} 0 &= \frac{d}{dt} \text{constant} , \\ &= \frac{d}{dt} \mathbf{x}(t) \cdot \mathbf{x}(t) , \\ &= 2\mathbf{x}(t) \cdot \mathbf{v}(t) . \end{aligned}$$

In other words, position and velocity vector are perpendicular.

Let us consider rotational motion in two dimensions. In that case we know that there is a *rotor* $R(t)$ such that it rotates the initial position $\mathbf{x}(0)$ into its position at time t ,

$$\mathbf{x}(t) = R(t)\mathbf{x}(0)\tilde{R}(t) . \quad (1.3.4)$$

These rotors they satisfy $R\tilde{R} = 1$ and so differentiating this gives

$$\begin{aligned} 0 &= \frac{d}{dt} 1 , \\ &= \frac{d}{dt} R(t)\tilde{R}(t) , \end{aligned}$$

$$= \left\{ \frac{d}{dt} R(t) \right\} \tilde{R}(t) + R(t) \left\{ \frac{d}{dt} \tilde{R}(t) \right\} ,$$

and so we find that

$$\begin{aligned} \Omega(t) \equiv -2 \left\{ \frac{d}{dt} R(t) \right\} \tilde{R}(t) &= 2R(t) \left\{ \frac{d}{dt} \tilde{R}(t) \right\} , \\ &= -\tilde{\Omega}(t) . \end{aligned} \quad (1.3.5)$$

The quantity $\Omega(t)$ is a bivector and it both represents the *plane* in which the rotation occurs as well as the *rate* at which the rotation occurs and its orientation. We call it the *angular velocity*.

Problem 23: Calculating angular velocity

Consider the particle again whose position was given as

$$\mathbf{x}(t) = R \cos(\omega t) \mathbf{e}_1 + R \sin(\omega t) \mathbf{e}_2 .$$

- a: Find a rotor $R(t)$ what produces this rotation.
b: Compute the angular velocity of the particle.

To understand this better let us first calculate $\mathbf{v}(t)$ in terms of $\Omega(t)$,

$$\begin{aligned} \frac{d}{dt} \mathbf{x}(t) &= \frac{d}{dt} R(t) \mathbf{x}(0) \tilde{R}(t) , \\ &= \left\{ \frac{d}{dt} R(t) \right\} \mathbf{x}(0) \tilde{R}(t) + R(t) \mathbf{x}(0) \frac{d}{dt} \tilde{R}(t) , \\ &= \left\{ \frac{d}{dt} R(t) \right\} R(t) \tilde{R}(t) \mathbf{x}(0) \tilde{R}(t) + R(t) \mathbf{x}(0) \frac{d}{dt} \tilde{R}(t) R(t) \tilde{R}(t) , \\ &= -\frac{1}{2} \Omega(t) R(t) \mathbf{x}(0) \tilde{R}(t) + \frac{1}{2} R(t) \mathbf{x}(0) \tilde{R}(t) \Omega(t) , \end{aligned}$$

so we find that

$$\mathbf{v}(t) = \frac{1}{2} \{ \mathbf{x}(t) \Omega(t) - \Omega(t) \mathbf{x}(t) \} \equiv [\mathbf{x}(t), \Omega(t)] . \quad (1.3.6)$$

This result is important and you will see that such expressions will keep popping up everywhere. When you've seen them often enough, you will start to understand what they mean. The key to understanding it is seeing that only if the vector $\mathbf{x}(t)$ is *not in the plane* $\Omega(t)$, then the bivector and the vector commute, i.e. $\Omega(t) \mathbf{x}(t) = \mathbf{x}(t) \Omega(t)$!

Problem 24: Checking velocity and angular velocity

Consider the particle from the previous Problem.

- a: Compute the velocity of this particle by differentiation of $\mathbf{x}(t)$.
b: Compute the velocity again by using the above equation involving the angular velocity.
Do the results match?

In two dimensions there is only one bivector J_3 , and all others are proportional to it. So we can introduce this proportionality

$$\Omega(t) = \omega(t) J_3 ,$$

and using that any vector in the plane must obviously anti-commute with J_3 one can calculate with a bit of algebra

$$\begin{aligned}\mathbf{x}(t) \wedge \mathbf{v}(t) &= x(t)^2 \omega(t) J_3 , \\ x(t) &= \sqrt{\mathbf{x}(t) \cdot \mathbf{x}(t)} .\end{aligned}$$

Note that for pure rotational motion the scalar $x(t)$ is indeed constant. The scalar $\omega(t)$ specifies the *rate* at which angle between $\mathbf{x}(0)$ and $\mathbf{x}(t)$ changes and is normally called the *angular speed*. Using that $J_3^2 = -1$ we can write

$$\omega(t) = \sqrt{-\Omega(t)^2} , \quad (1.3.7)$$

mimicking the relation between ordinary velocity and ordinary speed. I would like to ask you to consider the bivector $\Omega(t)$ to be the angular velocity. Just like ordinary velocity it not only represents the magnitude of the velocity, but also its orientation. In two dimensions there is only one possible orientation for $\Omega(t)$ and so distinguishing it from $\omega(t)$ seems awkward. It is analogous to the distinction between ordinary speed and velocity in one-dimensional problems. In three dimensions it will also become evident why $\Omega(t)$ is the angular velocity and $\omega(t)$ the *angular speed*.

Problem 25: Circular Orbits

The orbit of a particle is given by

$$\begin{aligned}R(t) &= \cos(-3t) - \sin(-3t)J_3 , \\ \mathbf{x}(t) &= R(t)\mathbf{e}_1\tilde{R}(t) .\end{aligned}$$

- a: What is the angular velocity of the particle?
- b: Calculate $\mathbf{v}(t)$ by direct differentiation.
- c: Calculate $\mathbf{x}(t) \wedge \mathbf{v}(t)$.
- d: Does the angular velocity depend on time?

The rotor-equation

The rotor $R(t)$ describing the rotation is related to $\Omega(t)$ through the definition given earlier, which can be rephrased as

$$\frac{d}{dt}R(t) = -\frac{1}{2}\Omega(t)R(t) . \quad (1.3.8)$$

This equation is known as a rotor-equation and in many cases of interest it can be solved. It obtains special significance in the theories of relativity and quantum physics.

Angular acceleration

Formally we can define a quantity known as *angular acceleration* $A(t)$, also a bivector and related to $\Omega(t)$ through differentiation,

$$A(t) = \frac{d}{dt}\Omega(t) . \quad (1.3.9)$$

Problem 26: Angular and ordinary acceleration

Can a particle have $\mathbf{a}(t) \neq 0$, while still having no angular acceleration?

The purpose of this whole enterprise is that the laws of physics will allow us to write down an equation for $\Omega(t)$. This rotational equivalent of Newton's second law is called *Euler's equation*, and we will come around to discussing it in a later chapter.

Problem 27: Wobbles

Consider a trajectory given by

$$\begin{aligned} R(t) &= \cos\left(-\frac{1}{2}\theta_0 \cos(t)\right) - \sin\left(-\frac{1}{2}\theta_0 \cos(t)\right)J_3, \\ \mathbf{x}(t) &= R(t)\mathbf{e}_1\tilde{R}(t). \end{aligned}$$

- a: Sketch the trajectory.
- b: Calculate the angular velocity.
- c: Calculate the angular acceleration.
- d: Can you imagine a physical system that shows this type of behaviour?

This now concludes our discussion of kinematics.

1.4. Assignments

In this section you will find a few more problems that illustrate applications of the material covered in the preceding sections. You are expected to hand in the solutions to these problems at the end of Part I.

Problem 28: Lines in the Sand

Consider a vector $\mathbf{x}_0$ and a unit-vector $\hat{\mathbf{u}}$.

- a: Show that the function

$$\mathbf{x}(\lambda) = \mathbf{x}_0 + \lambda\hat{\mathbf{u}},$$

of some parameter λ , running from $-\infty$ to ∞ , describes all the points on a straight line going through $\mathbf{x}_0$ in the direction $\hat{\mathbf{u}}$.

- b: Show that for any point on this line the equation

$$\hat{\mathbf{u}} \wedge \mathbf{x}(\lambda) = \hat{\mathbf{u}} \wedge \mathbf{x}_0$$

is true.

- c: Show that if $\hat{\mathbf{u}} \wedge \mathbf{x}(\lambda) = 0$ the line goes through the origin.

- d: Show, if d is the shortest distance between the line and the origin, that

$$\hat{\mathbf{u}} \wedge \mathbf{x}_0 = dJ_3. \quad (1.4.1)$$

Problem 29: Working with lines

Use the vectors defined in problem 8 and consider the three lines going through the point $\mathbf{x}_0 = \mathbf{a}$ that have $\mathbf{u}$ equal to $\mathbf{b}$, $\mathbf{c}$ or $\mathbf{d}$. Solve the following problems *algebraically*.

a: Does any of the three lines go through the origin?

b: Which of the lines passes closest to the origin?

c: Show that the vector $\mathbf{u}_0$ given by

$$\mathbf{u}_0 = \frac{1}{2}\{\mathbf{x}_0 J_3 - J_3 \mathbf{x}_0\} ,$$

is perpendicular to $\mathbf{x}_0$.

d: Show that for the line given by

$$\mathbf{x}(\lambda) = \mathbf{x}_0 + \frac{\lambda}{2}\{\mathbf{x}_0 J_3 - J_3 \mathbf{x}_0\} ,$$

the point $\mathbf{x}_0$ is closer to the origin than all other points on the line.

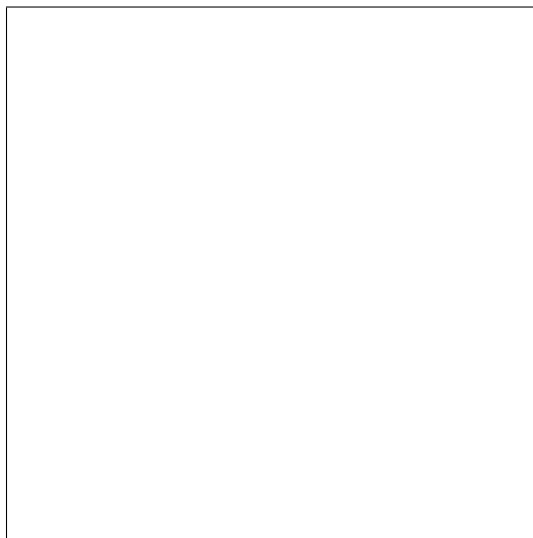


fig. 14: Lines in the sand

Problem 30: Components of a reflection

We have discussed linear maps previously. Recall that if you do not remember.

a: Show that the map $L(\mathbf{a}) = \mathbf{e}_1 \mathbf{a} \mathbf{e}_1$ is a linear map.

b: Compute the components of this map.

Problem 31: Reflections in many mirrors

In this exercise we are going to study an example from optics. Consider a set of N mirrors each with a orientation specified by a unit-vector $\hat{\mathbf{m}}_i$ that is perpendicular to the mirror i .

a: A lightray with an initial direction $\mathbf{r}_0$ hits the first mirror, draw the situation with the incoming and outgoing lightray, the vector $\hat{\mathbf{m}}_1$ and the origin at the point where the ray strikes the mirror.

b: Show that the reflected lightray has a new direction $\mathbf{r}_1$ given by

$$\mathbf{r}_1 = -\hat{\mathbf{m}}_1 \mathbf{r}_0 \hat{\mathbf{m}}_1 . \quad (1.4.2)$$

c: If we define the multivector $T(N) = \hat{\mathbf{m}}_N \hat{\mathbf{m}}_{N-1} \dots \hat{\mathbf{m}}_2 \hat{\mathbf{m}}_1$, then show that

$$T(N) \tilde{T}(N) = 1 . \quad (1.4.3)$$

d: Prove that after N reflections the direction of the lightray has become

$$\mathbf{r}_N = (-1)^N T(N) \mathbf{r}_0 \tilde{T}(N) . \quad (1.4.4)$$

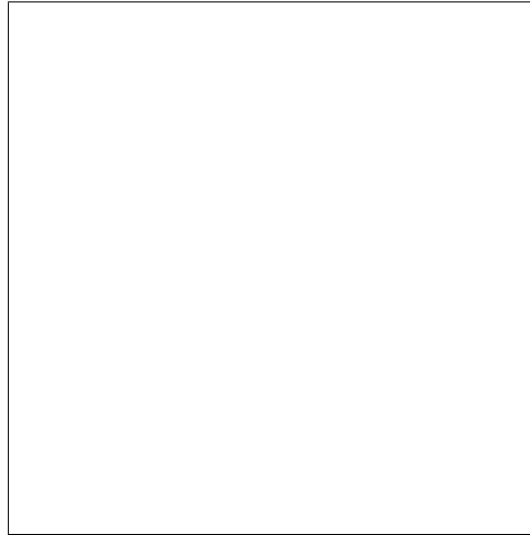


fig. 15: Lines perpendicular to $\mathbf{x}_0$.

Problem 32: Determinants of Maps

Consider the map $\mathbf{a}' = A[\mathbf{a}]$ described on the components by a matrix with the components $A_{11} = a, A_{12} = b, A_{21} = c, A_{22} = d$.

a: Compute the results of $\mathbf{e}'_1 = A[\mathbf{e}_1]$ and $\mathbf{e}'_2 = A[\mathbf{e}_2]$.

b: Give a decisive but non-computational argument why $J'_3 = \mathbf{e}'_1 \wedge \mathbf{e}'_2$ must be proportional to J_3 ?

c: Set $J'_3 = D(A)J_3$ and calculate the number $D(A)$. It is called the *determinant* of the matrix A_{ij} , or the map A .

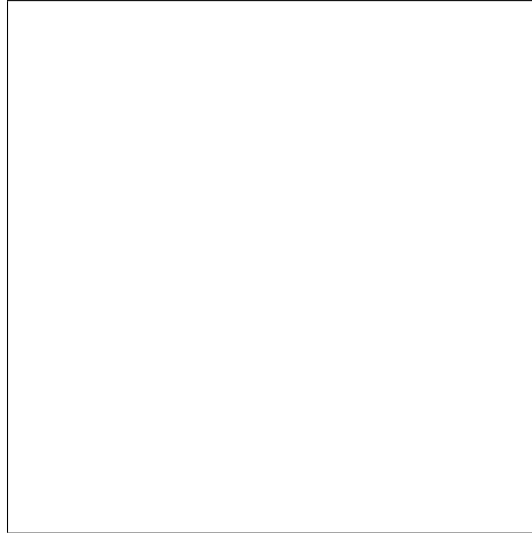


fig. 16: Area deformation and the determinant

Problem 33: Intersecting lines

Consider two lines given by the parametrizations

$$\begin{aligned}\mathbf{x}_1(\lambda_1) &= \mathbf{a}_1 + \lambda_1 \mathbf{u}_1, \\ \mathbf{x}_2(\lambda_2) &= \mathbf{a}_2 + \lambda_2 \mathbf{u}_2.\end{aligned}$$

- a: If there is an intersection point why do we expect that there we have $\mathbf{a}_1 + \lambda_1 \mathbf{u}_1 = \mathbf{a}_2 + \lambda_2 \mathbf{u}_2$.
 b: Show that *at the intersection point* the following two equations hold

$$\begin{aligned}\mathbf{u}_1 \wedge \mathbf{a}_1 &= \mathbf{u}_1 \wedge \mathbf{a}_2 + \lambda_2 \mathbf{u}_1 \wedge \mathbf{u}_2, \\ \mathbf{u}_2 \wedge \mathbf{a}_1 &= \mathbf{u}_2 \wedge \mathbf{a}_2 + \lambda_1 \mathbf{u}_1 \wedge \mathbf{u}_2.\end{aligned}$$

- c: Show that this results in the following two equations for the scalar parameters λ_1 and λ_2

$$\begin{aligned}\frac{\mathbf{u}_1 \wedge \{\mathbf{a}_1 - \mathbf{a}_2\}}{\mathbf{u}_1 \wedge \mathbf{u}_2} &= \lambda_2, \\ \frac{\mathbf{u}_2 \wedge \{\mathbf{a}_1 - \mathbf{a}_2\}}{\mathbf{u}_1 \wedge \mathbf{u}_2} &= \lambda_1.\end{aligned}$$

- d: Explain in words why this means that we have solved the problem of finding the intersection-point of two arbitrary lines and identify the situations in which there is no solution (Hint: Can the λ_j become infinite?).

No-

tice that in this last exercise we did not use any where that we were working in two dimensions. Hence the solution is valid in any number of dimensions.

Chapter 2

Dynamics and Conservation laws

The *explanation* of motion, or time-evolution in general, is the field of *dynamics*. The approach we take in this text is to study time-evolution by means of conservation laws.

2.1. Conserved quantities

We can use the ordinary and rotational velocity to construct other useful quantities. Most notable are *momentum* and energy. We will see in this section that these are quantities that satisfy so-called *conservation laws*.

2.1.1. Momentum and Newton's second law

A quantity that describes the *amount* of motion of a body is the *linear momentum* usually called *momentum* only. The momentum vector $\mathbf{p}$ of a particle with a mass m moving with a velocity $\mathbf{v}(t)$ is defined as

$$\mathbf{p}(t) = m\mathbf{v}(t) . \quad (2.1.1)$$

When Newton studied the laws of mechanical motion he associated the concept of a *force* with the rate-of-change- of momentum. We will see in the next section how this works, but for the time being we will simply write this idea down: when a force $\mathbf{f}$ acts on a body with mass m it's momentum satisfies

$$\frac{d}{dt}\mathbf{p}(t) = \mathbf{f} . \quad (2.1.2)$$

The reason why momentum is such a crucial property in physics is that it is a *conserved quantity*.

Problem 34: More familiar form of Newton's 2nd

Show that if the mass is time-independent Newton's second law becomes

$$m\mathbf{a}(t) = \mathbf{f} .$$

Newton's third law

Consider two particles A and B with momenta $\mathbf{p}_A$ and $\mathbf{p}_B$ exerting forces on one another. Let the force that A exerts on B be denoted by $\mathbf{f}_{AB}$, and the force that B exerts on A by $\mathbf{f}_{BA}$. Newton's third law, often called the *action = reaction*-law now requires that at any time and at any position we must have

$$\mathbf{f}_{AB} = -\mathbf{f}_{BA} . \quad (2.1.3)$$

Problem 35: Apple and Earth

Let $\mathbf{e}_3$ be the upward direction perpendicular to the surface of the Earth. When an object with a mass m is close to the surface of the Earth it experiences a *gravitational* force $\mathbf{f}_G = -mg\mathbf{e}_3$ due to the Earth, where the constant $g = 9.81\text{m/s}^2$.

- a: If you have an apple with a mass of 200 gr; find the force it exerts on the Earth.
- b: Compare the relative accelerations the Earth and the Apple get.

Now $\mathbf{f}_{AB}$ is a force on A and thus is the rate-of-change of the momentum $\mathbf{p}_A$. The same relation holds between $\mathbf{f}_{BA}$ and the momentum $\mathbf{p}_B$. So what NEWton's third law states is

$$\begin{aligned} \mathbf{f}_{AB} + \mathbf{f}_{BA} &= 0 , \\ \frac{d}{dt}\mathbf{p}_A + \frac{d}{dt}\mathbf{p}_B &= 0 , \\ \frac{d}{dt}\{\mathbf{p}_A + \mathbf{p}_B\} &= 0 . \end{aligned} \quad (2.1.4)$$

So the total momentum of the two particles is time-independent, i.e. remains conserved! If we have a system with many particles, each interacting with the others, then the total momentum remains conserved for the entire system.

Problem 36: Apollo 13 has a problem

Due to an explosion aboard the Apollo 13 service module, during its free-fall trajectory at about 8 km/s towards the Moon, the mission had to be cancelled. Suppose that the entire service module had a mass of 10^4 kg. After the explosion a 100 kg panel was ejected from the service module with a speed of 10 m/s, in a direction perpendicular to the original direction of motion.

- a: What was the change in velocity of the service module due to the loss of the panel?
- b: Estimate by how many km Apollo 13 would it miss its target after a flight time of 3 days.
- c: After the explosion oxygen leaked from tanks inside the surface module. If you assume that 250 gr of oxygen escaped each second with a speed of 200 m/s in the direction of motion, compute the force acting on Apollo 13 due to this leak (you can ignore the change in Apollo's mass).

The rate of change of momentum thus is what we call a *force*. In most cases forces give rise to a change in velocity, but occasionally a change in mass also occurs.

2.1.2. Energy

When a moving body strikes another object it is capable of doing damage, i.e. doing *work* in deforming the object. That is why we associate a certain amount of

energy with the motion of an object; the *kinetic energy* E_k . For a particle of mass m it is given by

$$E_k(t) = \frac{m}{2} \mathbf{v}(t)^2 = \frac{\mathbf{p}(t)^2}{2m} , \quad (2.1.5)$$

Note that this kinetic energy in general will be a function of time! Particles moving under the influence of forces can experience a change in kinetic energy that we associate with the *work* done by the force. We also distinguished between conservative and non-conservative forces. For conservative forces the total work done on a particle only depends on the initial and final position, and *not* on the path taken. For conservative forces we can introduce a new concept that will make understanding their effect on particles a lot easier.

Kinetic Energy and Work

Recall the kinetic energy

$$E_k = \frac{m}{2} \mathbf{v}(t)^2 .$$

If we differentiate it with respect to time, and use Newton's second law we find,

$$\frac{d}{dt} E_k = \mathbf{v}(t) \cdot \mathbf{f} . \quad (2.1.6)$$

So generally the action of the force will be to change the kinetic energy of the particle. We say that the force does *work*.

Problem 37: Exponential loss of Energy

Consider a particle moving with a velocity $\mathbf{v}(t)$ through a fluid under the influence of a linear drag force $\mathbf{f}(t) = -\gamma \mathbf{v}(t)$.

a: Show that the rate of change of kinetic energy satisfies

$$\frac{d}{dt} E_k(t) = -\frac{2\gamma}{m} E_k(t) .$$

b: Show that the function $E_k(t) = E_k(0) \exp(-\frac{2\gamma}{m}t)$ satisfies this equation.

The *rate* at which this kinetic energy changes is the *rate* at which the force does work and it is called the *power* $P(t)$. From the above expressions we have

$$P(t) = \mathbf{v}(t) \cdot \mathbf{f} . \quad (2.1.7)$$

Evidently the power does not only depend on the magnitude and direction of the force, but also on the rate at which the particle traverses the forcefield.

Problem 38: Losing Energy due to Friction

Consider a particle whose position is given as

$$\mathbf{x}(t) = R \cos(\omega t) \mathbf{e}_1 + R \sin(\omega t) \mathbf{e}_2 ,$$

and suppose it is immersed in a fluid.

a: Compute the velocity of this particle.

b: The friction force by the fluid is given by $\mathbf{f}(t) = -\gamma \mathbf{v}(t)$. Why must this particle lose energy?

c: Calculate the rate at which the particle loses energy.

The total work done between a time t_1 and a time t_2 is just the integral of this power, i.e.

$$W(t_2, t_1) = \int_{t_1}^{t_2} dt \mathbf{v}(t) \cdot \mathbf{f} .$$

A more appealing form for this can be found by realising that $dt \mathbf{v}(t)$ is just the *displacement* $d\mathbf{x}$ of the particle along its trajectory during the infinitesimally small time-interval dt . So as the particle moves along its trajectory $\mathbf{x}(t)$ from $\mathbf{x}(t_1)$ to $\mathbf{x}(t_2)$ we can write

$$W(\mathbf{x}(t)) = \int_{\mathbf{x}(t_1)}^{\mathbf{x}(t_2)} d\mathbf{x} \cdot \mathbf{f} . \quad (2.1.8)$$

Complicated as this may seem, many of you will have already encountered this result; for *constant* forces and trajectories which are *straight* lines. In that case, if $\Delta \mathbf{x}$ is the total displacement, you have

$$\begin{aligned} W &= \Delta \mathbf{x} \cdot \mathbf{f} , \\ &= \Delta x f \cos(\theta) . \end{aligned}$$

Problem 39: Work along a straightline

Consider a particle moving along a straight line give by

$$\mathbf{x}(t) = \mathbf{e}_1 + vt \mathbf{e}_2 ,$$

and consider the force $\mathbf{f} = -mg\mathbf{e}_1 - m\gamma \mathbf{v}(t)$.

Compute the work done on the particle between $t = 0$ and $t = 1$.

We can use our more general concept of work to distinguish between two types of forces; those that are *conservative* and those that are not. For a conservative force the work done *only depends on the initial and final position* and not on the path in between. These forces are called *conservative* because if we take a particle from some position $\mathbf{x}_1$ go through a closed trajectory to return at $\mathbf{x}_1$ then the total work done will be zero! Non-conservative forces will yield a non-vanishing amount of work in this case. Let me give you some examples of conservative and non-conservative forces.

- The gravitational force of the Sun on a planet in the Solar System is a conservative force.

- The friction force between a floor and a moving block is not a conservative force.
- The buoyancy force is a conservative force.
- The elastic force of a spring is conservative for as long as you do not deform the spring. When you stretch it too much though, the elastic force becomes non-conservative. This so-called *harmonic oscillator* is an important physical system for several reasons. First of all it is very common in nature. Many systems behave approximately like the harmonic oscillator. Secondly, we can solve it exactly.

Problem 40: The Harmonic Oscillator

The standard example of a harmonic oscillator is a mass m connected to a spring with spring-constant k . Suppose the particle can only move along the x -axis and is at $x = 0$ when the spring is relaxed. For other x the force on the particle is given by $\mathbf{f}(x) = -kx$.

a: Write down Newton's second law for this system.

b: Show that for $\omega = \sqrt{\frac{k}{m}}$ the function $x(t) = x(0) \cos(\omega t)$ satisfies Newton's second law.

c: Sketch this trajectory $x(t)$ and explain why ω is the *frequency* of the oscillation.

d: Compute the velocity of the particle and sketch it too.

- The force experienced by a charged particle in a time-varying electromagnetic field is non-conservative.
- The linear drag-force experienced by a small particle moving in a liquid is non-conservative.

Problem 41: Lost along a Circle

Consider a particle moving along a circle line given by

$$\mathbf{x}(t) = \cos(\omega t)\mathbf{e}_1 + \sin(\omega t)\mathbf{e}_2 ,$$

and consider the force $\mathbf{f} = -mg\mathbf{e}_1 - m\gamma\mathbf{v}(t)$.

a: Compute the work done on the particle during one complete revolution.

b: What causes the dependence of the work on ω ?

c: If $\gamma \rightarrow 0$, can you explain the result?

Potential Energy

In nature you typically encounter non-conservative forces when the particle is likely to lose energy to its surroundings. Friction is therefore a very typical example. Consider a particle moving under the influence of a conservative force $\mathbf{f}(\mathbf{x})$ which we can safely assume to be independent of the velocity (why?). Now irrespective of the exact trajectory of the particle, we know that the change in the kinetic energy between its starting point $\mathbf{x}(0)$ and some point $\mathbf{x}$ along its trajectory is given by

$$W(\mathbf{x}) = \int_{\mathbf{x}(0)}^{\mathbf{x}} d\mathbf{x}' \cdot \mathbf{f}(\mathbf{x}') . \quad (2.1.9)$$

If its initial kinetic energy was E_0 then its kinetic energy E_k at $\mathbf{x}$ is thus $E_0 + W(\mathbf{x})$. In other words the quantity $E_k - W(\mathbf{x})$ is *constant*.

Problem 42: Work in one dimension.

Consider a particle that can only move in one dimension, i.e. along the x -axis. Now suppose that the particle moves in a forcefield $F(x)$ from $x = 0$ to $x = 5$.

- a: What is the total work done when $F(x) = F_0$?
 b: What is the total work done when $F(x) = kx$?

The fact that $W(\mathbf{x})$ only depends on $\mathbf{x}$ allows us to define a new quantity called the *potential energy* $U(\mathbf{x})$ at $\mathbf{x}$,

$$U(\mathbf{x}) = U_0 - W(\mathbf{x}) . \quad (2.1.10)$$

U_0 is some physically irrelevant constant energy. Using this we can now define the total energy of the particle E as

$$E(t) = \frac{m}{2} \mathbf{v}(t)^2 + U(\mathbf{x}(t)) , \quad (2.1.11)$$

and by construction this total energy is conserved! This is how the potential energy arises from the presence of a conservative force.

Problem 43: Work and Energy in one dimension.

Consider again the particle that can only move in one dimension, i.e. along the x -axis. The particle moves in a forcefield $F(x)$ from $x = 0$ to $x = 5$. When the particle was at $x = 0$ its velocity was v .

- a: What is the velocity of the particle at $x = 5$ when $F(x) = F_0$?
 b: What is the largest distance the particle can move away from the origin when $F(x) = kx$?

Let me give you a few examples of forces, some of which result from a potential energy:

- The elastic force on an object hanging on a spring.

Problem 44: The harmonic oscillator at work

Consider the harmonic oscillator again. If the force on a particle of mass m at a distance x from the origin of the x -axis equals $\mathbf{f}(x) = -kx$. The what is the amount of work needed to displace the particle from $x = 0$ to x ?

- On the surface of the Earth, with $\mathbf{e}_1$ perpendicular to this surface and upward, the gravitational pull on a particle with mass m is represented by a force $\mathbf{f} = -mg\mathbf{e}_1$ arising from a linear potential. This force is constant gives rise to so-called *free-fall* trajectories.

Problem 45: Free fall

Consider the force of gravity, $\mathbf{f} = -mg\mathbf{e}_1$, close to the Earth's surface.

- a: Calculate $\mathbf{v}(t)$ if $\mathbf{v}(0) = 0$.
 b: Compute $\mathbf{x}(t)$ using the result under a: and assuming that $\mathbf{x}(1) = h_0\mathbf{e}_1$.

- A macroscopic object of mass m and volume V falling under the influence of gravity, but immersed in a fluid of density ρ , is subject to three forces: gravity $-mg\mathbf{e}_1$, the buoyancy force $g\rho_w V\mathbf{e}_1$, and a friction force $-\gamma\mathbf{v}(t)$ proportional but opposite to the objects' velocity. The constant γ depends on the characteristics of the fluid (viscosity) and the object (cross-section). The number ρ_w is the density of water. Friction is non-conservative, the force of gravity stems from a linear potential and the buoyancy force is a result of a pressure gradient, i.e. its potential energy is proportional to the hydrostatic pressure.

Problem 46: Buoyancy and Eureka

- a: Give the force acting on a body of volume V in a fluid of density ρ_f .
- b: What is the gravitational force on this body if it is made of material with a density ρ_B ?
- c: Express the net force on the body, when at rest, in terms of the density difference between the fluid and the body.
- d: This body is then released in the fluid. Why will its velocity become constant after some (possibly very long) time?
- e: Express this *final* velocity of the body in terms of the density difference.

- A massive block that moves across a not too smooth floor usually experiences only a single, frictional force that only depends on the direction of its velocity, $-\kappa\hat{\mathbf{v}}(t)$. The constant κ depends on the details of the floor and the block. As a result the rate at which the fact that it is proportional to $\hat{\mathbf{v}}(t)$ the work done by this friction force is proportional to the distance the block has travelled.

Problem 47: Sliding the block

Consider the example given in the text for a block sliding across a rough surface.

- a: Give the general expression for the work done by this friction force between $t = 0$ and t .
- b: Why won't this force change the direction the block is moving in?
- c: Why is the work done proportional to the displacement?
- d: Is it right to say, as many textbooks do, this friction force is a *constant* force?

- If you take a bicycle ride through the rain your mass will change as a result of your coat soaking up the rainwater. If you want to persist at a constant velocity you will need to exert a force. So although there is no acceleration you do need a force. This is due to the changing mass.
- A planet with mass m at a distance r from the Sun experiences a gravitational force directed *towards* the Sun with a magnitude given by $\frac{GmM}{r^2}$. Here M is the mass of the Sun. This force does not depend on velocity but only on position. If you calculate the orbits of bodies in the solar system for extremely long intervals of time you need to take into account that the masses of the bodies slowly change due to either a periodic loss of material (comets) or an almost constant gain by gathering interplanetary dust (planets).
- Two charged particles that moves with respect to each other exert electric and magnetic forces on one another. The magnetic forces do not respect Newton's third law, but momentum remains conserved as the electromagnetic field

itself also carries an amount of momentum which accounts for the discrepancy occurring in Newton's third law.

Problem 48: Linear Drag

Consider the force experienced by a small object traversing a fluid at moderate velocity $\mathbf{v}(t)$; $\mathbf{f} = -m\gamma\mathbf{v}(t)$.

a: Calculate $\mathbf{v}(t)$ if $\mathbf{v}(0) = -v\mathbf{e}_1$.

b: Although the velocity does not become 0 in a finite time, there is nevertheless a maximum value for the distance the object can travel. Given $\mathbf{x}(0) = 0$, compute $\mathbf{x}(\infty)$.

Once you know the forces acting on a particle, the principle goal of classical mechanics is to find a trajectory $\mathbf{x}(t)$ that will satisfy Newton's second law. In other words, this law should *predict* the motion of the particle! Because of this, the law is sometimes also referred to as an *equation of motion*. Solving this law is sometimes rather difficult, and sometimes completely undoable. The energy-considerations are other ways to gain some understanding of the motion of a particle without solving this equation.

2.1.3. Rotational Energy

We can also express the kinetic energy due to the rotational motion in terms of $\omega(t)$. This is useful in cases where the motion of the particle is either strictly rotational, or where a clean separation between rotational and linear motion simplifies the analysis.

Problem 49: Rotational and translation kinetic energy

Consider an arbitrary trajectory $\mathbf{x}(t)$.

a: Why can we write $\mathbf{x}(t) = x(t)\hat{\mathbf{x}}(t)$?

b: Why does $\hat{\mathbf{x}}(t)$ only change in time by a rotation?

c: Why is $\frac{d}{dt}\hat{\mathbf{x}}(t) = -\Omega(t)\hat{\mathbf{x}}(t) = \hat{\mathbf{x}}(t)\Omega(t)$, in two dimensions?

d: Show that

$$\begin{aligned}\frac{d}{dt}\mathbf{x}(t) &= \hat{\mathbf{x}}(t)\left[\frac{d}{dt}x(t) + x(t)\Omega(t)\right] . \\ &= \left[\frac{d}{dt}x(t) - x(t)\Omega(t)\right]\hat{\mathbf{x}}(t) .\end{aligned}$$

e: Show that the kinetic energy of the motion is

$$\begin{aligned}E_k(t) &= \frac{m}{2}\left[\frac{d}{dt}x(t) + x(t)\Omega(t)\right]\left[\frac{d}{dt}x(t) - x(t)\Omega(t)\right] , \\ &= \frac{m}{2}\left(\frac{d}{dt}x(t)\right)^2 - \frac{m}{2}x(t)^2\Omega(t)^2 .\end{aligned}$$

The last equation shows that for an arbitrary particle rotational kinetic energy can be separated from the *radial* kinetic energy. For a mass m that is strictly rotating the above expression for the velocity leads to

$$E_k = -\frac{1}{2}mx^2\Omega(t)^2 = -\frac{1}{2}I\Omega(t)^2 .$$

The minus sign appears because $\Omega(t)^2 < 0$. The quantity $I = mx^2$ is usually called

the moment of inertia. We will have great use for it in the later chapters on rigid rotations. If we have a body consisting of N objects with different masses m_j at different distances x_j from the common axis of rotation we get

$$I = \sum_{j=1}^N \frac{1}{2} m_j x_j^2 . \quad (2.1.12)$$

When we compare rotational kinetic energy to ordinary kinetic energy we see that I plays the same role as mass m . This analogy suggests that we could also define an *angular momentum*, L , via

$$L = I\Omega(t) .$$

This is indeed the case and its total angular momentum also turns out to be a conserved quantity. We will study this in detail in the next chapter and consider its application in analyzing the orbits of planets in the solar system.

2.2. Dynamics

In the previous sections we have seen how to describe motion and how the conserved quantities give us relations between kinematical variables. The study of dynamics, however, does not only encompass the use of conserved quantities. Newton's 2nd law all by itself allows us to find trajectories of particles once we know the forces acting on them.

Put simply, a force tells a particle in which direction it must to accelerate. The inertial mass m is a measure of how susceptible the particle is to the action of the force. According to Newton's second we have

$$m\mathbf{a}(t) = \mathbf{f}(\mathbf{x}(t)) ,$$

for a force that is allowed to depend on the particle's position. Conservative forces are merely telling the particle to move to the minimum of potential energy. Below you will see how to find the conservative force corresponding to a given potential energy.

2.2.1. The Gradient of the potential Energy

Now we will be tracing the steps back to the force, but taking the potential energy as a given function. We know that the total energy is a conserved quantity, so

$$\frac{d}{dt}E(t) = 0 .$$

As a result we get

$$\frac{d}{dt} \frac{m}{2} \mathbf{v}(t)^2 = - \frac{d}{dt} U(\mathbf{x}(t)) .$$

Now we can evaluate the l.h.s. of this equation, but what about the r.h.s? Well, it's just a matter of using the chain rule. We will do so in two dimensions,

$$\frac{d}{dt} U(\mathbf{x}) = \frac{\partial}{\partial x_1(t)} U(\mathbf{x}(t)) \frac{d}{dt} x_1(t) + \frac{\partial}{\partial x_2(t)} U(\mathbf{x}(t)) \frac{d}{dt} x_2(t) .$$

Here the symbols

$$\frac{\partial}{\partial x_j} U(\mathbf{x}) ,$$

represent taking a derivative of $U(\mathbf{x})$ with respect to x_j as if x_j is an ordinary argument while we keep any other x_k , so $k \neq j$, constant. These derivatives are called *partial derivatives*. You work with them in basically the same way work with ordinary derivatives.

Problem 50: Partial Derivatives.

Consider the following potentials that are functions of x_1 and x_2 and compute the partial derivatives $\frac{\partial}{\partial x_1}$ and $\frac{\partial}{\partial x_2}$;

a: $U(x_1, x_2) = \cos(k_1 x_1 + k_2 x_2)$ for constants k_j ?

b: $U(x_1, x_2) = \exp -\alpha(x_1^2 + x_2^2)$ for a constant α ?

We are now going to see where this chainrule business lead us. If we look at the expression we recognize the components $\frac{d}{dt}x_j(t)$ of the velocity $\mathbf{v}(t)$

$$\mathbf{v}(t) = \frac{d}{dt}x_1(t)\mathbf{e}_1 + \frac{d}{dt}x_2(t)\mathbf{e}_2 .$$

The other factor in each term, $\frac{\partial}{\partial x_j} U(\mathbf{x})$, can be used to form another vector

$$\nabla U(\mathbf{x}) = \mathbf{e}_1 \frac{\partial}{\partial x_1} U(\mathbf{x}) + \mathbf{e}_2 \frac{\partial}{\partial x_2} U(\mathbf{x}) .$$

This vector is called the *gradient* of $U(\mathbf{x})$, the ∇ is called the *vector-derivative* usually pronounced as "nabla".

Problem 51: Gradients.

Consider the following potentials that are functions of x_1 and x_2 and compute $\nabla U(\mathbf{x})$;

a: $U(\mathbf{x}) = \cos(\mathbf{k} \cdot \mathbf{x})$ for a constant vector $\mathbf{k}$?

b: $U(\mathbf{x}) = \exp -\alpha(\mathbf{x}^2)$ for a constant α ?

With this gradient we can rewrite the rate of change of the potential energy in a very appealing way as

$$\frac{d}{dt}U(\mathbf{x}) = \mathbf{v}(t) \cdot \nabla U(\mathbf{x}) . \quad (2.2.1)$$

If we now see what conservation of the total energy means, we get

$$\mathbf{v} \cdot \mathbf{f} = -\mathbf{v} \cdot \nabla U .$$

This statement is completely independent of the trajectory the particle is taking, so we can only conclude that

$$\mathbf{f}(\mathbf{x}) = -\nabla U(\mathbf{x}) . \quad (2.2.2)$$

What is the interpretation of this equation?

Suppose that the function $U(\mathbf{x})$ at $\mathbf{x}$ is constant in the direction $\mathbf{e}_1$. In that case the derivative with respect to x_1 will vanish and the only derivative that may yield something is with respect to x_2 . As a result $\nabla U(\mathbf{x})$ will point in the positive or negative $\mathbf{e}_2$ -direction. Now the derivative of $U(\mathbf{x})$ with respect to x_2 will be positive if $U(\mathbf{x})$ increases in the $\mathbf{e}_2$ -direction. In other words, $\nabla U(\mathbf{x})$ is a vector at $\mathbf{x}$ that points in the direction of increasing $U(\mathbf{x})$ and is perpendicular to lines (or planes) of constant $U(\mathbf{x})$. So for the relation between a conservative force and the corresponding potential energy this gives, due to the minus sign: *The force points in the direction of maximally decreasing potential energy.*

Problem 52: Potential and Force

- a: A particle with mass m close to the Earth's surface experiences a constant gravitational force $\mathbf{f} = -mg\mathbf{e}_1$, where the vector $\mathbf{e}_1$ is a unit-vector pointing upwards from the ground. Give a potential that generates this force.
- b: A particle that can move in plane but is attached to the origin with a spring has a potential energy given by $U(\mathbf{x}) = \frac{k}{2}\mathbf{x}^2$. Calculate the force acting on this particle if its position is $\mathbf{x}(t)$.
- c: If a particle is at the position $\mathbf{x}$ we denote its distance from the origin by $r(\mathbf{x}) = \sqrt{\mathbf{x}^2}$. Calculate the gradient of $r(\mathbf{x})$.
- d: If a particle experiences a potential energy given by $U(r) = \frac{k}{r}$, where k is a constant; calculate the force on the particle.

Are there other physical examples of gradients? Yes, let me give you a few:

- Fluids or gasses tend to flow from regions of high pressure to regions of low pressure. Minus the gradient of the pressure gives a force acting on the local flow.
- Heat tends to flow from hot places to cold places. Heat transfer through heat conduction usually follows the minus temperature gradient.
- When a drop of ink is immersed in water it diffuses from places with a high-density of ink to places with a low density of ink. The ink-current follows minus the density-gradient.

So gradients are important in many applications of physics such as meteorology, physical oceanography, engineering, bulk chemistry, and even the analysis of traffic congestion.

Problem 53: Diffusion.

- Consider a drop of ink in water. The density of ink ρ is given as $\rho(\mathbf{x}) = \exp(-\alpha\mathbf{x}^2)$.
- a: What is the meaning of α in this context?
- b: Compute $\nabla\rho(\mathbf{x})$ and explain what it represents?

In many instances the forces in physics are conservative. In general we cannot say much about non-conservative forces, so do not expect a general theory about them.

Problem 54: Potential Energy and Equilibrium

Consider a particle of mass m moving in a plane under the influence of a potential energy $U(\mathbf{x})$ given by

$$U(\mathbf{x}) = \frac{k}{2}\mathbf{x}^2 + \frac{\lambda}{4}(\mathbf{x}^2)^2 .$$

- a: Write out the potential energy in terms of the two components x_1, x_2 of $\mathbf{x}$ with respect to some basis $\mathbf{e}_j$, $j = 1, 2$.
- b: Sketch this potential in a 3d-plot if you assume that both $k > 0$ and $\lambda > 0$.
- c: At what $\mathbf{x}_0$ does the potential energy have its lowest value?
- d: Show that the force on a particle vanishes at that point $\mathbf{x}_0$.

There is one type of force that we have not considered so far: forces that do not do any work.

2.2.2. Lorentz-type Forces

When a force is *perpendicular* to the velocity it does not do work. This implies that the velocity, if it changes at all, undergoes some rotation. In other words, given an initial velocity $\mathbf{v}(0)$, the velocity at a later time is specified by some kind of rotor

$$\mathbf{v}(t) = R(t)\mathbf{v}(0)\tilde{R}(t) . \quad (2.2.3)$$

Hey ... we have seen this before haven't we? If we now compute the acceleration $\mathbf{a}(t)$ we get

$$\begin{aligned} \mathbf{a}(t) &= \frac{1}{2}\{\mathbf{v}(t)\Omega(t) - \Omega(t)\mathbf{v}(t)\} , \\ \frac{d}{dt}R(t) &= -\frac{1}{2}\Omega(t)R(t) . \end{aligned}$$

Hence, a force that never does any work can be described by a *bivector* G such that in Newton's second law we would have

$$m\mathbf{a}(t) = \frac{1}{2}\{\mathbf{v}(t)G - G\mathbf{v}(t)\} .$$

So in general we can split up forces into those that *do* do work, $\mathbf{f}$ and those that *never* do work G . These two types of forces enter into Newton's second law in different ways, but together they form

$$m\mathbf{a} = \mathbf{f} + \frac{1}{2}\{\mathbf{v}G - G\mathbf{v}\} . \quad (2.2.4)$$

One could call this a *Lorentz-type* of force as the most well known example of such a force is the so-called Lorentz force experienced by a particle with mass m and electric charge q moving in an electric field $\mathbf{E}$ and a magnetic field B , for in that case

$$m\mathbf{a} = q\mathbf{E} + \frac{q}{2}\{\mathbf{v}B - B\mathbf{v}\} . \quad (2.2.5)$$

For now I will leave you with the surprising effect that the magnetic field, of which you always thought it was a vector, has suddenly changed into a bivector. We will

be able to understand this strange phenomenon properly once we have arrived in three dimensions.

Problem 55: A constant magnetic Field

Consider a Lorentz-type force $B = B_3 \mathbf{e}_1 \mathbf{e}_2$ for some constant B_3 .

a: What is the force experienced by a charged particle with charge q and a velocity $\mathbf{v}(t)$?

b: Show that the following is a solution to the equation of motion

$$R(t) = \cos\left(-\frac{qB}{2m}t\right) - \sin\left(-\frac{qB}{2m}t\right)J_3, \mathbf{v}(t) = R(t)\mathbf{v}(0)\tilde{R}(t).$$

c: Calculate $\mathbf{v}(t)$.

2.3. Assignments

From the following problems 3 must be handed in at the end of part I of the course.

Problem 56: Ordinary Exponential functions

In secondary school you have encountered the exponential function e^x , or $\exp(x)$, as the function satisfying

$$\exp(0) = 1 \quad \frac{d}{dx} \exp(x) = \exp(x).$$

Here you will uncover new properties of this function.

a: Compute the derivative of $\exp(\alpha x)$ with respect to x .

b: I now define the following functions

$$\begin{aligned} \cosh(\alpha x) &= \frac{\exp(\alpha x) + \exp(-\alpha x)}{2}, \\ \sinh(\alpha x) &= \frac{\exp(\alpha x) - \exp(-\alpha x)}{2}. \end{aligned}$$

Compute their derivatives with respect to x and compare the results to these definitions.

c: Show that $\cosh(\alpha x)^2 - \sinh(\alpha x)^2 = 1$.

d: Show that $\exp(\alpha x) = \cosh(\alpha x) + \sinh(\alpha x)$.

Problem 57: Extraordinary Exponential functions

Consider the two-dimensional geometric algebra, its bivector and the rotor function

$$f(\alpha) = \cos(\alpha) + \sin(\alpha)J_3 .$$

a: Show that the following is true

$$\frac{d}{d\alpha}f(\alpha) = J_3f(\alpha) .$$

b: Give the rules for differentiating the exponential function $\exp(\alpha x)$ and comment on the proposal to write

$$\exp(\alpha J_3) = \cos(\alpha) + \sin(\alpha)J_3 .$$

c: Show that if we use this, we can write a rotation about an angle α in the plane as

$$\mathbf{a}' = \exp\left(-\frac{\alpha}{2}J_3\right)\mathbf{a}\exp\left(\frac{\alpha}{2}J_3\right) .$$

Problem 58: Trigonometric identities

We will now make use of the proposal in the previous exercise to uncover some relations you will have seen before. Consider

$$\exp(\alpha J_3) = \cos(\alpha) + \sin(\alpha)J_3 ,$$

and evaluate the following expressions;

a: Calculate $\exp(2\alpha J_3)$ by using the relation above.

b: Calculate $\exp(2\alpha J_3) = [\exp(\alpha J_3)]^2$ by using the relation above on *both factors* in the product on the r.h.s.

c: Compare the two calculations of $\exp(2\alpha J_3)$ and give a relation between the scalar- and the bivector-terms on both sides.

Problem 59: Commutator brackets

In the text we defined the shorthand

$$[A, B] \equiv AB - BA .$$

These *commutator brackets* are very common in physics, in particular in quantum mechanics.

So it is useful to point out a few remarkable properties.

a: Show that $[A, B] = -[B, A]$.

b: Show that $[AB, C] = [A, C]B + A[B, C]$.

c: Show that $[[A, B], C] + [[B, C], A] = [[A, C], B]$.

Problem 60: Symmetry breaking

Consider again the potential from problem 30. Now assume that $k < 0$.

- a: Sketch the potential in this case.
- b: Find the possible values of $\mathbf{x}$ for which the potential energy takes a minimum value.
- c: The point $\mathbf{x}_0$ that you found in problem 30 is no longer a minimum. Yet show that the force on a particle at $\mathbf{x}_0$ still vanishes.
- d: The point $\mathbf{x}_0$ is now called an *unstable* equilibrium. Explain that name.
- e: If the particle is at rest in a stable equilibrium, a symmetry must be broken. Which symmetry is lost when the particle is at rest in a stable equilibrium?

Problem 61: Fluctuations

Consider again the potential from problem 28. Now assume that $k > 0$ but $\lambda < 0$.

- a: Sketch the potential in this case.
- b: There is only a small area where the particle can stay without running off to infinity. Determine that area and draw it.
- c: There is only a restricted set of velocities that the particle can have if it has to stay near the origin. What is the maximal speed the particle can have without leaving for infinity.
- d: If we have a collection of such particles in the small potential well. Now suppose these particles collide regularly among each other. What do you think will eventually happen?

Problem 62: Two particles on a rigid rod

Consider two particles, A and B, of mass m connected by a rigid rod of fixed length d . Let the vector $\mathbf{d}(t)$ denote the rod connecting particle A with B.

- a: Argue that if particle A has the position $\mathbf{x}(t)$ then particle B is at $\mathbf{x}(t) + \mathbf{d}(t)$.
- b: Sketch the system.
- c: Now both particles experience a potential energy $U(\mathbf{x}) = k\mathbf{x}^2$. Compute the forces on both particles separately if it is given that particle A is at $\mathbf{x}(t)$.
- d: For which cases are the particles in an equilibrium situation?
- e: Express the total kinetic energy of the particles in terms of the position of particle A and $\mathbf{d}(t)$.
- f: Why do we expect that $\mathbf{d}(t)$ only changes with time due to rotations?

Problem 63: Potentials in Polar Coordinates

If we discuss two-dimensional systems the so-called cartesian $\{x_1, x_2\}$ -coordinates are very useful. Using them a point in the 2d plane is represented by the vector $\mathbf{x} = x_1 \mathbf{e}_1 + x_2 \mathbf{e}_2$. However sometimes *other* coordinates are also useful. We give an example here.

a: We introduce the so-called *polar* coordinates $\{r, \theta\}$ by writing

$$\mathbf{x} = r \cos(\theta) \mathbf{e}_1 + r \sin(\theta) \mathbf{e}_2 . \quad (2.3.1)$$

Can every point in the plane be expressed in terms of these two coordinates?

b: Show that the symmetry-breaking potential from problem 30 is given by $U(r) = \frac{k}{2}r^2 + \frac{\lambda}{4}r^4$ in polar coordinates.

c: Show that we can write for the position of an arbitrarily moving object in the plane

$$\mathbf{x}(t) = r(t) \exp\left(-\frac{\theta(t)}{2} J_3\right) \mathbf{e}_1 \exp\left(\frac{\theta(t)}{2} J_3\right) .$$

d: Show that $\mathbf{x}(t) \wedge \mathbf{v}(t) = r(t)^2 \frac{d}{dt} \theta(t) J_3$.

Chapter 3

Extended Systems

What orbits do planets and moons follow, and why are these orbits as they are. To properly describe these things I will need the new conserved quantity called *angular momentum*. But before we get started, we first quickly expand our mathematical apparatus to three dimensions.

3.1. Geometry and Algebra in 3d

In a previous chapter we have studied the $d = 2$ algebra. All those results will remain usefull when we expand to three dimensions. All that changes is that one more direction becomes available and that we will have more than one bi-vector. Furthermore you will encounter a new object, called a *tri-vector*. It neatly falls into the scheme of things.

Let us assume that we have been given a basis $\mathbf{e}_i$ where the index i now runs from 1 through 3. For this basis we acain have

$$\begin{aligned}\mathbf{e}_i \cdot \mathbf{e}_j &= 1 \text{ if } i = j , \\ &= 0 , \text{ otherwise .}\end{aligned}$$

Let us start by discussing the new element of the algebra of $d = 3$, the *tri-vector*

$$i = \mathbf{e}_1 \mathbf{e}_2 \mathbf{e}_3 . \quad (3.1.1)$$

Like bivectors represent a *surface area* the tri-vector represents the geometrical properties of a *volume element*. The tri-vector displays a number of interesting properties. When we square it also yields $i^2 = -1$.

Problem 64: Prove

Check for yourself this property that $i^2 = -1$.

I have denoted the tri-vector by i , using notation as if it were a scalar. Isn't that confusing? Note the following, consider for example the $\mathbf{e}_1$ vector and its product with i , we find

$$i\mathbf{e}_1 = \mathbf{e}_1 i . \quad (3.1.2)$$

So the tri-vector i *commutes* with vectors. This is a property that it shares with scalar quantities! So using scalar notation seems appropriate. If you take a second

look at the equation above you will note an actually even more surprising thing. Note that we have

$$i\mathbf{e}_3 = \mathbf{e}_1\mathbf{e}_2 = J_3 , \quad (3.1.3)$$

and so we can write in general

$$i\mathbf{e}_j = J_j . \quad (3.1.4)$$

This equation provides a close correspondence between vectors and bi-vectors. It associates to every vector $\mathbf{a}$ a *plane* represented by the bivector $i\mathbf{a}$ that is perpendicular to the vector. This phenomenon is typical for $d = 3$, it does not exist in higher dimensions. Ofcourse we can also go the other way. If we are given a plane represented by a bivector A , multiplication by $-i$ allows us to swap between a surface-element A and the vector $-iA$ which is normal to it and whose length coincides with the area of A . The minus-sign is required because i squares to -1 . So, in particular we have

$$\mathbf{e}_j = -iJ_j . \quad (3.1.5)$$

These relations allow us to identify the *cross product* between vectors with the wedge-product.

The Cross Product

In the course book you will see that the authors introduced another product, $\mathbf{a} \times \mathbf{b}$: the *cross-product*. It is closely related to our *wedge-product*. Our wedge-product generates a bivector, and not a vector. But we have just seen how we can turn a bivector into a vector by multiplying by i . This *only works in three dimensions*. Therefor, for this particular situation we can write the traditional cross product in our new geometric language as:

$$\mathbf{a} \times \mathbf{b} \equiv -i\mathbf{a} \wedge \mathbf{b} . \quad (3.1.6)$$

The result of a cross-product between two vectors is a third vector *that is perpendicular* to the surface element represented by the outer-product of the two vectors.

Problem 65: The righthanded Basis

In the coursebook, in chapter 1, you will encounter the following relationship between the basis-vectors $\mathbf{e}_j$ for a *righthanded basis*

$$\mathbf{e}_1 \times \mathbf{e}_2 = \mathbf{e}_3 .$$

Verify this relationship for the definition of the cross product that was given here.

Note that the cross-product only exists in $d = 3$. In higher dimensions one must return to the wedge-product which exists in any number of dimensions.

Electric and Magnetic fields

Using these algebraic properties of 3d geometry we can clarify the nature of the *Lorentz-type* forces that I introduced in chapter 2. Recall, that a force that never does any work can be represented by a bivector G that enters Newton's 2nd law like

$$m\mathbf{a} = \frac{1}{2}\{\mathbf{v}G - G\mathbf{v}\} .$$

Now define the vector $\mathbf{g} = -iG$ and use the cross-product to find

$$m\mathbf{a} = \mathbf{v} \times \mathbf{g} .$$

As a result the force on a charged particle in an electromagnetic field can also be written

$$\mathbf{f}_{em} = q\mathbf{E} + q\mathbf{v} \times \mathbf{B} , \quad (3.1.7)$$

where the magnetic field vector $\mathbf{B}$ and the magnetic field bivector B are related as $\mathbf{B} = -iB$. In three dimensions this is the natural way to do things because the electric field is obviously a vector. However in four-dimensional spacetime the electric field becomes a bivector too!

Problem 66: Translate Angular Momentum

The coursebook associates angular momentum L with a vector $\mathbf{L}$.

a: Translate the coursebook definition $\mathbf{L} = \mathbf{x} \times \mathbf{p}$ to the definition I gave in chapter 3 of these notes.

b: We discussed angular momentum in two dimensions as well. If you strictly follow the cross-product definition, then is there angular momentum in two dimensions?

Pseudo-scalars, Axial vectors and Determinants

One other important property of i needs to be discussed to fully appreciate how it differs from ordinary scalars. Consider a reflection in the origin of the vectorspace. This takes every unit vector $\mathbf{e}_j$ to $-\mathbf{e}_j$. This is called a *parity* transformation. The tri-vector changes as $i \rightarrow -i$ under parity. Scalars do not do this. They have no sense of orientation at all. Because of the fact that i is so similar to a scalar except for the property under parity i is often called the unit-*pseudo-scalar*. Books which refrain from using algebra must introduce pseudo-scalars as "scalars that transform with an extra minus-sign under parity".

As I said earlier, most textbooks desperately try to identify surface areas, i.e. bivectors, with vectors to! If you check what happens to a bivector under parity, it doesn't change at all. Ordinary vectors however go like $\mathbf{a} \rightarrow -\mathbf{a}$. To us it is obvious that $i\mathbf{a} \rightarrow i\mathbf{a}$ because both i and $\mathbf{a}$ get a minus sign. However, if you use a method that calls a pseudo-scalar a number that transforms with an extra minus sign under parity, then you must also call a bivector "a vector that transforms with an extra minus sign under parity". These vectors are then usually called *axial vectors*. In practically all textbooks you will see statements like "angular momentum is an axial vector" or "the magnetic field is an axial vector". It is not always easy to recognize a priori whether a vector is axial or not. It is never a problem to identify bivector.

Last but not least, let $\mathbf{a}' = H(\mathbf{a})$ be a linear map taking 3d vectors into 3d vectors, then its determinant $\text{Det}[H]$ is found through,

$$i' = \mathbf{e}'_1 \wedge \mathbf{e}'_2 \wedge \mathbf{e}'_3 = \text{Det}[H]i .$$

This equation is the 3d equivalent of the one discussed in problem 14 of chapter 1. It allows a very neat compact equation for the determinant

$$\frac{i'}{i} = \text{Det}[H] , \quad (3.1.8)$$

this is valid in any number of dimensions.

Problem 67: Determinant of reflection and rotation in a plane

If we want to reflect a vector $\mathbf{a}$ in a plane represented by the unit-bivector J_3 , then all we need to calculate is $\mathbf{a}' = J_3 \mathbf{a} J_3$.

a: Show that the determinant of this transformation equals -1 .

b: Consider the transformation $\mathbf{a}' = \mathbf{e}_3 \mathbf{a} \mathbf{e}_3$. Draw the effect of this transformation for two arbitrary vectors in the J_3 -plane and note that it is a *rotation* by π radians around the $\mathbf{e}_3$ axis.

c: Show that the determinant of this rotation is $+1$.

d: Use the relationship between $\mathbf{e}_3$ and J_3 to show that what we are really doing is $\mathbf{a}' = -J_3 \mathbf{a} J_3$, which indeed represents a rotation by an angle π in the J_3 -plane.

Bivector Algebra

Let us write down explicitly define the three bi-vectors that we can make from the three basis vectors

$$\begin{aligned} J_1 &= \mathbf{e}_2 \wedge \mathbf{e}_3 , \\ J_2 &= \mathbf{e}_3 \wedge \mathbf{e}_1 , \\ J_3 &= \mathbf{e}_1 \wedge \mathbf{e}_2 . \end{aligned} \tag{3.1.9}$$

Each of these three squares to -1 and in the geometry of the plane they correspond to they play the same role as J_3 from the previous chapters. All results obtained there can be simply taken over. It is a simple exercise to establish the following relations among the bi-vectors

$$\begin{aligned} J_1 J_2 &= -J_3 , \\ J_2 J_3 &= -J_1 , \\ J_3 J_1 &= -J_2 . \end{aligned} \tag{3.1.10}$$

As bi-vectors correspond to the generators of rotations, these relations may be referred to as the *algebra of rotations*.

Problem 68: The Algebra is real

In this Problem you are expected to convince yourself that the algebra of bivectors really describes something real, and is not just another stew of fuzzy mathematical ingredients. a: Show that a rotation over π radians in the J_k plane is represented by the rotor $R = J_k$.

b: Argue that the equation $J_1 J_2 = -J_3$ is an equation about the result of performing subsequent rotations about perpendicular axes.

c: Pick up a book and see whether you can reproduce the equation by making rotations over π radians with the book around perpendicular axes.

You will encounter them again, albeit in a different context, in your quantum mechanics courses.

Rotors in 3d

Rotations can be performed in 3d just as in 2d. Again we will have a bivector Ω specifying a plane in which the rotation occurs, and the angle of the rotation. The

rotor then becomes

$$R(\alpha) = \exp\left(-\frac{\alpha}{2}\hat{\Omega}\right) = \cos\left(\frac{\alpha}{2}\right) - \sin\left(\frac{\alpha}{2}\right)\hat{\Omega} , \quad (3.1.11)$$

and it acts on vectors and other elements of the algebra by sandwiching. However, unlike the situation in 2-dimensions, where all bivectors were proportional to J_3 we now can have any combination of bivectors, so

$$\Omega = \omega_1 J_1 + \omega_2 J_2 + \omega_3 J_3 . \quad (3.1.12)$$

Fortunately in three dimensions *every bivector represents a plane*. Recall that a plane was represented by the wedgeproduct of two non-parallel unit-vectors. In three dimensions every bivector can be written as the wedgeproduct of two vectors.
*

Problem 69: Commutation relations

Consider a time-dependent bivector given by $\Omega(t) = \omega_1 J_1 + \omega_2 J_2$.

Show that in general $\Omega(0)\Omega(t) \neq \Omega(t)\Omega(0)$. What does that mean?

Parallel and Perpendicular

In order to get some feeling for the algebra of rotations it is good to consider what happens to vectors that are perpendicular to the plane of rotation. Let $\hat{\Omega}$ represent the plane of rotation, and $\hat{\Omega}^2 = -1$. A vector $\mathbf{a}$ perpendicular to $\hat{\Omega}$ satisfies

$$\mathbf{a}\hat{\Omega} - \hat{\Omega}\mathbf{a} = 0 . \quad (3.1.13)$$

We say that it *commutes* with the bivector. When the vector $\mathbf{a}$ commutes with $\hat{\Omega}$ it is easy to verify that

$$R(\alpha)\mathbf{a}\tilde{R}(\alpha) = R(\alpha)\tilde{R}(\alpha)\mathbf{a} = \mathbf{a} .$$

When we are given a plane bivector $\hat{\Omega}$ and an arbitrary vector $\mathbf{b}$, we can split it up in a part which is perpendicular to the plane, and a part which is parallel. This works as follows

$$\mathbf{b}_{//} = \frac{1}{2}\{\mathbf{b} - \hat{\Omega}\mathbf{b}\hat{\Omega}\} , \quad (3.1.14)$$

$$\mathbf{b}_{\perp} = \frac{1}{2}\{\mathbf{b} + \hat{\Omega}\mathbf{b}\hat{\Omega}\} . \quad (3.1.15)$$

Problem 70: Prove

Check these relations as exercises.

Differentiation

*This is no longer true in the four dimensions of spacetime.

We already saw the gradient in 2d. Now we simply add a direction and thus, if the x_i are the coordinates pertaining to the basis $\mathbf{e}_i$ then we can write,

$$\nabla = \sum_{j=1}^3 \mathbf{e}_j \frac{\partial}{\partial x_j} . \quad (3.1.16)$$

You can take this ∇ to represent differentiation with respect to the position vector $\mathbf{x}$.

Problem 71: ∇ Gymnastics

Consider the three dimensional ∇ and an arbitrary but constant vector $\mathbf{a}$.

a: Show that $\nabla \mathbf{x} = \mathbf{3}$.

b: Show that $\nabla \mathbf{a} \cdot \mathbf{x} = \mathbf{a}$.

c: Calculate $\nabla \mathbf{a} \wedge \mathbf{x}$.

The ∇ has vector character and so if it acts on vectors $\mathbf{a}(\mathbf{x})$ this should somehow show up in the "product" $\nabla \mathbf{a}(\mathbf{x})$. Indeed this is the case. We naturally find the following two

$$\nabla \cdot \mathbf{a}(\mathbf{x}) = \sum_{j=1}^3 \mathbf{e}_j \cdot \left\{ \frac{\partial}{\partial x_j} \mathbf{a}(\mathbf{x}) \right\} , \quad (3.1.17)$$

$$\nabla \wedge \mathbf{a}(\mathbf{x}) = \sum_{j=1}^3 \mathbf{e}_j \wedge \left\{ \frac{\partial}{\partial x_j} \mathbf{a}(\mathbf{x}) \right\} . \quad (3.1.18)$$

The first is called *the divergence of $\mathbf{a}(\mathbf{x})$* , the second is called *the curl of $\mathbf{a}(\mathbf{x})$* . They will play an important conceptual role in later chapters.

An arbitrary vector like $\mathbf{b}$ can be expanded in a basis as

$$\mathbf{b} = \sum_{j=1}^3 b_j \mathbf{e}_j ,$$

where the b_j are the components again. If we have some arbitrary function $f(\mathbf{b})$ we can differentiate it with respect to $\mathbf{b}$ as follows

$$\partial_{\mathbf{b}} f(\mathbf{b}) = \sum_{j=1}^3 \mathbf{e}_j \frac{\partial}{\partial b_j} f(\mathbf{b}) . \quad (3.1.19)$$

We denote this derivative by a different symbol $\partial_{\mathbf{b}}$ because we reserve ∇ for derivatives with respect to *position* vectors like $\mathbf{x}$. In the Problem below you will investigate a very powerful application of this differentiation with respect to arbitrary vectors.

Problem 72: Frame free Trace

Consider some linear map H . There is a very important property of H called it's trace, $Tr[H]$. We first give a standard definition of the trace and subsequently a very neat one that algebra allows us to do.

a: The trace of the linear map H is defined as

$$Tr[H] = \sum_{j=1}^3 \mathbf{e}_j \cdot H(\mathbf{e}_j) = \sum_{j=1}^3 H_{jj} . \quad (3.1.20)$$

Argue that the *number* $Tr[H]$ does not change under rotations of the basis-vectors.

b: Show that the above definition is equivalent to the following

$$Tr[H] = \partial_{\mathbf{b}} \cdot H(\mathbf{b}) . \quad (3.1.21)$$

3.2. Angular momentum and Central Forces

If a particle moves along a straight line, then the result of problem **21** implies that for any point $\mathbf{x}$ along the path the product $\mathbf{x} \wedge \mathbf{v}$ takes the same value. This bivector represents the plane in which motion takes place, but also the area swept out in this plane, per second, by the position vector $\mathbf{x}$. Let us study this bivector and investigate what its properties are.

Consider a particle path $\mathbf{x}(t)$ and a force, $\mathbf{F}$, acting on the particle . We now define the bivector L as

$$L(t) = \mathbf{x}(t) \wedge \mathbf{p}(t) = m\mathbf{x}(t) \wedge \mathbf{v}(t) . \quad (3.2.1)$$

This bivector is called *angular momentum* and it is related to the angular momentum vector $\mathbf{L}$ introduced in [1]. We will see the exact relation in the next chapter. Now note what happens when we study the rate at which angular momentum changes under the influence of the force $\mathbf{F}$. Differentiating we find,

$$\frac{d}{dt}L(t) = \left[\frac{d}{dt}\mathbf{x}(t)\right] \wedge \mathbf{p}(t) + \mathbf{x}(t) \wedge \left[\frac{d}{dt}\mathbf{p}(t)\right] = \mathbf{x}(t) \wedge \mathbf{F} . \quad (3.2.2)$$

The quantity $T = \mathbf{x} \wedge \mathbf{F}$ is called the *Torque*[†] exerted by the force.

What is more interesting however is that if the force acts in the direction of $\mathbf{x}(t)$, i.e. if $\mathbf{F} \parallel \mathbf{x}(t)$, then

$$\frac{dL}{dt} = 0 , \quad (3.2.3)$$

then angular momentum is a conserved quantity! This means that under certain restrictions on the forces acting between particles, we have an additional conserved quantity. We could use this quantity in phenomenological considerations if we can obtain a proper understanding how to use it. To get a thorough feeling for it we will study one application in the next section where it is important: orbits in the

[†]In Dutch: (Kracht)-moment.

solar system.

Problem 73: See-Saw

Choose $\mathbf{e}_1$ to be the horizontal basisvector pointing towards the right, and $\mathbf{e}_2$ the vertical basis vector pointing upward. Now consider a perfectly rigid *see-saw*, supported in the origin, and with two arms with length l_1 to the right, and length l_2 to the left. At the end of each arms there is mass m_1 to the right and m_2 to the left. The force of gravity on a mass m is given by $\mathbf{f} = -mg\mathbf{e}_2$.

- a: If the see-saw is in its horizontal position, what is the total torque at the origin?
b: If the see-saw makes an angle α with the $\mathbf{e}_1$ direction, then show that the positions of the two masses are give by

$$\begin{aligned}\mathbf{x}_1 &= l_1 \cos(\alpha)\mathbf{e}_1 + l_1 \sin(\alpha)\mathbf{e}_2, \\ \mathbf{x}_2 &= l_2 \cos(\alpha)\mathbf{e}_1 - l_2 \sin(\alpha)\mathbf{e}_2.\end{aligned}$$

- c: Is there an angle α for which the total torque vanishes. d: Express l_2 in l_1 and the masses m_1 and m_2 if it is given that the see saw is in balance at $\alpha = 0$.

Problem 74: Two particles on a rigid rod revisited

Consider two particles, A and B, at positions $\mathbf{x}_A$ and $\mathbf{x}_B$, of equal mass m and connected by a rigid rod of fixed length d , like in problem 33. We define the position, $\mathbf{X}$, of the *centre of mass* of these two particles by

$$\mathbf{X}(t) = \frac{1}{2}\{\mathbf{x}_A(t) + \mathbf{x}_B(t)\} . \quad (3.2.4)$$

The vector $\mathbf{d}(t)$ denoting the rod connecting particle A with B can be written as

$$\mathbf{d}(t) = \mathbf{x}_B(t) - \mathbf{x}_A(t) . \quad (3.2.5)$$

- a: Check the expression for $\mathbf{d}(t)$ with how it was given in problem 33 and express $\mathbf{x}_A$ and $\mathbf{x}_B$ in terms of $\mathbf{X}$ and $\mathbf{d}$.
b: Show that the total potential energy of the system is given by $U(\mathbf{x}_A, \mathbf{x}_B) = k\{\mathbf{x}_A^2 + \mathbf{x}_B^2\}$.
c: Express the total potential energy in terms of $\mathbf{X}$ and $\mathbf{d}$.
d: Let θ is the angle between $\mathbf{X}$ and $\mathbf{d}$. Express the total potential energy in terms of X , θ and d .
e: Sketch the system and the forces acting on the particles. Now compute $-\frac{\partial}{\partial X}U(X, \theta, d)$ and explain why this is the total force drawing the system towards the origin.
f: Compute $-\frac{\partial}{\partial X}U(X, \theta, d)$ and explain which part of this expression is the torque on the two particles.

Problem 75: Effective Potential

Recall problem 34 and look up the $\mathbf{x}(t)$ for an arbitrarily moving particle.

a: Show that angular momentum conservation would mean $\frac{d}{dt}\theta(t) = \frac{l}{mr^2}$, where l is the magnitude of the angular momentum.

b: Show that the kinetic energy can be rewritten as

$$\begin{aligned}\frac{1}{2}m\mathbf{v}(t)^2 &= \frac{1}{2}m\left(\frac{d}{dt}r(t)\right)^2 + \frac{1}{2}mr(t)^2\left(\frac{d}{dt}\theta(t)\right)^2, \\ &= \frac{1}{2}m\left(\frac{d}{dt}r(t)\right)^2 + \frac{l^2}{2mr(t)^2}.\end{aligned}\quad (3.2.6)$$

c: Suppose this particle moves in some potential energy field $U(\mathbf{x})$. Why is angular momentum conserved only when the potential depends on r *only*, i.e. $U(r)$?

d: Show that for this particle we *effectively* are dealing with a 1-dimensional problem (only the r -direction) of motion in an *effective* potential energy

$$U(r)_{eff} = U(r) + \frac{l^2}{2mr^2}.\quad (3.2.7)$$

Problem 76: The Bohr Atom

Early in the twentieth century it became clear that the classical dynamics of point particles was not sufficient to explain the oddities of the behaviour of atomic systems. By 1910 physicists had realised that atoms consists of a positively charged nucleus surrounded by orbiting negatively charged electrons. Yet, judging from the light emitted by atoms when they *fall* from an excited state back into their normal state, not all orbits were allowed for the electrons. Around 1915 Niels Bohr proposed a new model for the hydrogen atom in order to explain the structure of this emission spectrum. This problem deals with that model.

Hydrogen has a nucleus of charge $+e$ and a single electron of charge $-e$. The potential energy of that electron only depends on the distance r between the electron and the nucleus. It is called the *Coulomb potential* and given by

$$U(r) = -\frac{ke^2}{r}.\quad (3.2.8)$$

a: Explain why the angular momentum of the electron is conserved?

b: What is the *effective potential* the electron moves in?(see also problem 37 if you are unfamiliar with the term.)

c: Draw the potential and indicate in the drawing where it has a minimum. Call the radius of this minimum r_0 .

d: If it is given that the electron is in a *circular* orbit, then explain why that means that it is at the minimum of effective potential energy. Calculate r_0 in terms of the magnitude of angular momentum $|L|$.

e: Bohr proposed a *quantization* rule for the magnitude of angular momentum $|L|$

$$|L| = n\hbar,\quad (3.2.9)$$

where $\hbar$ is Planck's constant and n is some positive integer called the *principal quantum number*. Assume that the electron moves on a perfect circle. What are the allowed values for r_0 in terms of n ?

f: What are the allowed energies of the electron in terms of n ?

g: Explicitly calculate the ground-state energy of hydrogen corresponding to $n = 1$ in Joules.

Problem 77: The Quantum Harmonic Oscillator

We

can apply the method above also to a much simpler system; the 2d harmonic oscillator.

- a: Check out the potential of the 2d harmonic oscillator (see also problem **29**), and give the effective potential for this system (see also problem **38**).
- b: Determine the r_0 for which the effective potential takes on a minimum value.
- c: Use the general relationship $|L| = mr^2\omega$ for $r = r_0$ to express ω in terms of k and m .
- d: Use Bohr's quantization condition to show that the allowed energy values correspond to $E_n = n\hbar\omega$, for integer values of n .
- e: Give an argument why this suggests that for the 1d harmonic oscillator the allowed energies should be $E_n = \frac{n}{2}\hbar\omega$ (which is infact correct).

3.2.1. Orbits in the Solar System

This section is different from the ones you have seen so far. Yo are expected to work your way through this section by reading the text and doing the Problems in between.

Consider the Solar System with the Sun, with mass M , in the origin of the coordinate sustem. If a planet in the Solar System has a mass m and is at a position $\mathbf{x}(t)$ at time t , then the gravitational force acting on that planet exerted by the Sun is given by Newton's Law of Gravity

$$\mathbf{f}(\mathbf{x}) = -\frac{GmM}{\mathbf{x}(t)^2}\hat{\mathbf{x}}(t) . \quad (3.2.10)$$

The orbit of the planet is determined by this force through Newton's second law yielding

$$\begin{aligned} m\frac{d}{dt}\mathbf{v}(t) &= -\frac{GmM}{\mathbf{x}(t)^2}\hat{\mathbf{x}}(t) , \\ \mathbf{v}(t) &= \frac{d}{dt}\mathbf{x}(t) . \end{aligned} \quad (3.2.11)$$

This gravitational force is clearly a *central* force and thus the angular momentum of the planet

$$L = m\mathbf{x}(t) \wedge \mathbf{v}(t) . \quad (3.2.12)$$

is a conserved quantity.

Problem 78: Kepler's second Law

Kepler's second law states that the line connecting the Sun with the planet sweeps out equal surface areas in equal time-intervals.

- a: Argue that Kepler's second law states that area swept out by the position-vector per second is constant.
- b: Show that this is equivalent to the conservation of angular momentum.

Now note that we can write $\mathbf{x}(t) = x(t)\hat{\mathbf{x}}(t)$. This leads to

$$\mathbf{v}(t) = \left[\frac{d}{dt}x(t)\right]\hat{\mathbf{x}}(t) + x(t)\left[\frac{d}{dt}\hat{\mathbf{x}}(t)\right] ,$$

and if we substitute this back into our definition of angular momentum we find a simpler expression,

$$L = mx(t)^2\hat{\mathbf{x}}(t)\left[\frac{d}{dt}\hat{\mathbf{x}}(t)\right] ,$$

and note the use of the geometric product instead of the wedge!

Problem 79: Prove

Prove this last equation.

Now we can use the fact that for this planet we have

$$\frac{d}{dt}\mathbf{v}(t) = -\frac{GM}{\mathbf{x}(t)^2}\hat{\mathbf{x}}(t) ,$$

to obtain the expression

$$L\frac{d}{dt}\mathbf{v}(t) = GmM\left[\frac{d}{dt}\hat{\mathbf{x}}(t)\right] . \quad (3.2.13)$$

Using that L is time-independent this means that we can define a *new* vector which is also conserved,

$$\vec{\epsilon} = \frac{L\mathbf{v}(t)}{GmM} - \hat{\mathbf{x}}(t) . \quad (3.2.14)$$

This vector is called the *eccentricity vector*[‡] and it is very instrumental in determining the exact orientation of the orbit of the planet. It is also known as the *Laplace-Runge-Lenz* vector in some treatments.

Problem 80: Prove

Show that this eccentricity vector $\vec{\epsilon}$ is indeed a conserved quantity.

We can use L to find the orbit of the planet algebraically. Note that we have

$$\begin{aligned} L\mathbf{v}(t)\mathbf{x}(t) &= GmM(\hat{\mathbf{x}}(t) + \vec{\epsilon})\mathbf{x}(t) , \\ L\mathbf{v}(t) \cdot \mathbf{x}(t) + L\mathbf{v}(t) \wedge \mathbf{x}(t) &= GmMx(t)(1 + \vec{\epsilon} \cdot \hat{\mathbf{x}}(t) + \vec{\epsilon} \wedge \hat{\mathbf{x}}(t)) . \end{aligned}$$

In the last line of this equation we can split the scalar and bivector terms as they have to coincide. Furthermore we have, $\mathbf{v}(t) \wedge \mathbf{x}(t) = \frac{\tilde{L}}{m}$. These two results give for the scalar part

$$\begin{aligned} x(t) &= \frac{l^2}{m^2MG(1 + \vec{\epsilon} \cdot \hat{\mathbf{x}}(t))} , \\ l^2 &= L\tilde{L} . \end{aligned}$$

[‡]I indicate this vector with an "arrow" on top because the greek letters do not show that they are printed in bold format.

This equation allows us to write down an expressions for the orbit in terms of the angle θ between the time-dependent unit-vector $\hat{\mathbf{x}}(t)$ and the fixed eccentricity vector. We get

$$x(\theta) = \frac{l^2}{m^2 MG(1 + \epsilon \cos(\theta))} . \quad (3.2.15)$$

This expression gives us the distance $x(\theta)$ between the Sun and the planet as a function of the angle θ for an orbit in the plane represented by L .

Problem 81: Understanding the orbital equation

- a: Show that the expression for $x(\theta)$ indeed follows from the previous equations.
 b: Check that if $\epsilon = 1$ the denominator of the expression can become 0, and thus $x(\theta)$ can become infinite for one particular value of θ .
 c: If it is given that $\epsilon > 1$ find an equation for the *two* values of θ for which $x(\theta)$ becomes infinite.

It allows us to draw the orbit of the planet once its angular momentum and eccentricity are known! As these two are constants, they can be calculated from the initial position and initial velocity.

Using the magnitude ϵ of the eccentricity vector the possible orbits in the solar system can be classified into three main groups;

- $\epsilon < 1$: In this case the orbits are elliptic, becoming perfectly circular if $\epsilon = 0$.
- $\epsilon = 1$: In this case the orbits are parabolic. Objects which originate from the edges of the solar system (and were almost at rest there) typically follow parabolic orbits. You can consider it as an elliptic orbit with one point at infinity. Comets with extremely long periods are examples of these type of objects.
- $\epsilon > 1$: In this case the orbits are hyperbolic. These orbits have two different points at infinity. Here we are dealing with objects that come from the deep regions of interstellar space. Attracted by the Sun's gravity they fall towards it only to sweep past and and move away again to another part of the infinite reaches beyond the solar system.

Problem 82: Kepler's First Law

Kepler's first law says that *Planets move on elliptic orbits*. Let us verify that for $\epsilon < 1$ we are indeed dealing with an ellipse. Consider some point $\mathbf{b}$ away from the origin. Let $s(\mathbf{x})$ be the *sum* of the distances between $\mathbf{x}$ and the origin, and between $\mathbf{x}$ and $\mathbf{d}$.

- a: Give an expression for $s(\mathbf{x})$.
 b: Let ϕ be the angle between $\mathbf{x}$ and $\mathbf{d}$. Express $s(\mathbf{x})$ in terms of x , d and ϕ .
 c: An ellipse is the set of points for which $s(x, d, \phi)$ is constant. Consider s given and solve for x as a function of d , s and ϕ .
 d: Compare your expression for $x(s, d, \phi)$ with the orbital equation and comment on the validity of Kepler's first law.

We have now uncovered the orbit as a function of θ but ofcourse this angle depends on time, and as long as we do not know *how* we haven't completely solved the problem. Recall however that angular momentum was conserved. We had found

the expression

$$L = mx(t)^2 \hat{x}(t) \left[\frac{d}{dt} \hat{x}(t) \right] ,$$

Now use that the unit-vector is merely being rotated in the plane of the orbit, and denote the bivector representing this plane by J . Then we have

$$\begin{aligned} \hat{\mathbf{x}}(t) &= \exp\left(\frac{-\theta(t)}{2}J\right) \hat{\mathbf{e}} \exp\left(\frac{\theta(t)}{2}J\right) , \\ \Omega(t) &= \frac{d}{dt}\theta(t)J , \\ L &= mx(t)^2 \frac{d}{dt}\theta(t)J . \end{aligned}$$

From this last result we can read off that

$$\frac{d}{dt}\theta(t) = \frac{|L|}{2mx(\theta(t))^2} , \quad (3.2.16)$$

This is a nasty equation to solve. But it can be done exactly. However we will not attempt to completely integrate this here. The procedure works roughly as follows, the last equation is rewritten as

$$\int_0^t dt' = \int_{\theta(0)}^{\theta(t)} \frac{mx(\theta')^2}{|L|} d\theta' .$$

The integral on the left is easy, but the integral on the right requires substantially more effort to be evaluated.

Problem 83: Orders of magnitude

It is important to have some idea of the orders of magnitude of the forces in the Solar system.

- a: Compute the magnitude of the gravitational force that the Sun exerts on the Earth.
- b: Compute the magnitude of the gravitational force that the Earth exerts on the Moon.
- c: Compute the gravitational force that the Earth exerts on a person of 70 kg standing on the surface of the Earth.

Problem 84: Orbital data

Consider the orbital equation for $x(\theta)$. a: If we assume that $\epsilon < 1$, for which value of θ is the planet closest to the Sun? Call this distance p (This point is called the *perihelion*).

b: For which θ is the planet farthest from the Sun? Call this distance a (This point is called the *aphelium*).

c: Express the magnitude of the eccentricity ϵ in terms of a and p .

Problem 85: The Lunar Orbit

Let us use our knowledge to take a closer look at the Moon's orbit around the Earth. We will ignore the effect of the Sun on the orbit of the Moon.

- a: Look up the values for the maximal and minimal distances between the Earth and the Moon.
- b: Express the maximal and minimal distances $x(\theta)$ of an elliptic orbit in terms of eccentricity, and determine the eccentricity of the Moon's orbit around the Earth. (Hint: Consider the ratio of the two distances!)
- c: Use the eccentricity computed in b: and your value for the maximum distance between the Earth and the Moon to compute the magnitude of the Moon's angular momentum.
- d: If the Moon is closest to the Earth, why is the Moon's velocity perpendicular to the vector connecting Earth and Moon at that point?
- e: Compute the Moon's speed when closest to the Earth and also when farthest from the Earth.

Problem 86: Solar Scattering

Consider the orbital equation for $x(\theta)$ for a particle on a hyperbolic orbit. Such a particle enters and leaves the Solar system, we say it is *gravitationally scattered*. It enters from a direction θ_{in} and leaves in a direction θ_{out} . We choose the basis vector $\mathbf{e}_1$ to be parallel to $\vec{\epsilon}$ and $\mathbf{e}_2$ to be perpendicular to that.

- a: Show that the position vector of the particle is given by

$$\mathbf{x}(\theta) = x(\theta)\{\cos(\theta)\mathbf{e}_1 + \sin(\theta)\mathbf{e}_2\},$$

- and show that $\theta_{in} = -\theta_{out}$. b: Show that the velocity of the particle can be given as

$$\mathbf{v}(\theta) = \frac{|L|}{mx(\theta)^2} \frac{d}{d\theta} \mathbf{x}(\theta).$$

Use this to show that if the particle is very far away from the Sun that $\mathbf{v}$ and $\mathbf{x}$ becomes approximately parallel.

- c: If $\mathbf{x}$ and $\mathbf{v}$ would be really parallel then $L = 0$, which cannot happen due to angular momentum conservation. Assume however that the particle is not exactly on the line connecting the Sun and $\mathbf{x}(\theta_{in})$, but has a speed v_{in} (anti-)parallel to $\mathbf{x}(\theta_{in})$ and is a distance b away from that line. Why do the laws of physics require that

$$\begin{aligned} v_{in}^2 &= \mathbf{v}(\theta)^2, \\ |L| &= v_{in}b. \end{aligned}$$

- d: Explain why these two equations will allow us to find the magnitude of the eccentricity ϵ , and the angle θ_{out} in terms of b and v_{in} . (You do not have to compute these expressions, there is quite some algebra involved.)

Problem 87: Quantum Solar System

The force on a planet was given early on in this section. Now we know that the force derives from a potential as it is conservative.

a: Show that the gravitational potential energy of a planet at position $\mathbf{x}$, with the Sun in the origin, is given by

$$U_{grav}(\mathbf{x}) = -\frac{GmM}{\sqrt{\mathbf{x}^2}}. \quad (3.2.17)$$

b: Show that this is analogous to the Coulomb potential introduced in problem 38.

c: Calculate, by analogy, the energy-levels for planets around the Sun that would follow from the Bohr quantization rule.

d: What is the order of magnitude of n for the Earth, when you assume it is in a circular orbit?

e: What is the energy difference between this n and the next lower lying orbit?

Problem 88: Kepler's third Law

In this problem we take a closer look at the radial motion of planets.

a: Write down the effective potential $U_{eff}(r)$ for a planet in its orbit around the Sun at a radial distance r from the Sun.

b: If you assume that the orbit is circular with a radius $r = R$ then show that R corresponds to a minimum of this effective potential and find an expression for R in terms of the other variables.

c: Now we approximate the true orbit by writing $r = R + \delta$ assuming δ to be very small. Use Mathematica's "Series" command to calculate the effective potential including quadratic terms in δ . (So use "Series[$U_{eff}(R + \delta)$, { δ , 0, 2}]")

d: Compare this to the potential of the harmonic oscillator and, by analogy, derive an expression for the frequency of the *radial oscillations* around $r(t) = R$.

e: Compute the time needed for one oscillation from the frequency and compare the result with Kepler's 3rd law.

Problem 89: Additional Quantum numbers for the hydrogen atom

In problem a previous problem we discussed an approximation to the problem of radial motion in almost circular orbits. We found that the for small radial amplitudes we can approximate the radial motion as a 1d harmonic oscillator in the radial direction around a minimum of the effective potential.

a: Review problem 45 and problem 39 and give an estimate for the allowed energy values of the radial motion when taken to be quantized.

b: Give an expression for the total energy of a hydrogen atom involving the quantum number n for the orbital motion and a quantum number k for the oscillatory motion in the radial direction.

c: Compare the additional contribution of the radial motion to the main contribution of the orbital motion. Is the approximation any good?

Problem 90: Bohr was right for all the wrong reasons

At this point we need to say something about the "old quantum theory" of the atom by Bohr. His postulates to quantize angular momentum were in a sense very ad hoc. Moderate justification came only a few years later by an argument of De Broglie that associated the quantum number n to the number of times that the "quantum wavelength" of the electron would fit into the circumference of the classical orbit. Only after Schrödinger's breakthrough did it become clear that Bohr's quantization postulate is in fact very far from the real physical picture. In Bohr's model only states with angular momentum $L \neq 0$ can be identified with energy levels. In the correct treatment by Schrödinger it is seen that also states with $L = 0$ belong to the energy-levels derived by Bohr. So he was right, but for the wrong reasons. It is a remarkable fact of quantum mechanics that the energy-levels of hydrogen are in fact independent of the angular momentum values! This is a special symmetry property of the Coulomb potential. Without it the development of quantum mechanics would probably have taken much longer.

Write a 500 word essay on the Bohr model for the Atom.

3.3. Bodies with a finite number of constituents

In this section we will give a description of bodies that are made up of a large, possibly infinite, number of particles. The particles themselves are sometimes more complicated than point particles and so I will sometimes call them *constituents* to indicate that it is not always necessary to think of them in terms of point particles. Whatever picture you have, the important point is that in this chapter we will give tools to turn a *microscopic* description of a body into a macroscopic one. The result will be that we need to make additional assumptions about the interactions between the constituents before we can get a solvable set of equations. These additional assumptions take the form of *rigidity* in chapter 5, *elasticity* in chapter 6 and *fluidity* in chapter 7. Here we will start with introducing the main ideas. In Problems, dispersed through out the text, I will illustrate the ideas on a simple extended 2-body system: The Sun and Jupiter.

The simplest property of a body is its total mass M . In terms of the constituent masses m_j

$$M = \sum_j m_j \quad (3.3.1)$$

is just the sum of these masses.

Problem 91: The Sun and Jupiter

Look up the mass m_s of the Sun and the mass m_J of Jupiter and determine the total mass of this system.

Next let us choose a vector that properly represents the position of the body. We call this vector the *position of the center of Mass* and can express it in terms of the positions $\mathbf{r}_j(t)$ of the individual constituents as

$$\mathbf{X}(t) = \frac{1}{M} \sum_j m_j \mathbf{r}_j(t). \quad (3.3.2)$$

Problem 92: Sun-Jupiter centre-of-Mass

Let $\mathbf{r}_s(t)$ be the position of the Sun and $\mathbf{r}_J(t)$ the position of Jupiter. Define the centre of mass position $\mathbf{X}(t)$ for the system Sun + Jupiter.

Now the total momentum when expressed in terms of the momenta of the constituents can be rewritten as

$$\mathbf{P}(t) = \sum_j \mathbf{p}_j(t) , \quad (3.3.3)$$

$$\begin{aligned} &= \sum_j m_j \frac{d}{dt} \mathbf{r}_j(t) , \\ &= \frac{d}{dt} \sum_j m_j \mathbf{r}_j(t) , \\ &= M \frac{d}{dt} \mathbf{X}(t). \end{aligned} \quad (3.3.4)$$

Hence the total mass times the center-of-mass velocity represents the total momentum. Total mass M , center of mass position $\mathbf{X}(t)$ and total momentum $\mathbf{P}(t)$ thus describe a substantial part of the kinematics of an extended body.

Problem 93: Total momentum conservation for Sun-Jupiter

Show that the centre-of-mass momentum of the Sun-Jupiter system is conserved despite the fact that these objects exert gravitational forces on one another (hint: Use the gravitational force law given in chapter 3, and Newton's 3rd!).

We will now try to capture the *internal* kinematics of the extended body in terms of the positions of the constituents *relative to the centre-of-mass*. We define the relative positions through,

$$\mathbf{x}_j(t) = \mathbf{r}_j(t) - \mathbf{X}(t) . \quad (3.3.5)$$

Note that this yields that the relative momenta vanish,

$$\sum_j m_j \frac{d}{dt} \mathbf{x}_j(t) = 0. \quad (3.3.6)$$

Sometimes the centre-of-mass is also called the center-of-momentum.

Problem 94: Relative positions for Sun-Jupiter

If $\mathbf{x}(t)$ is the separation vector pointing from the Sun to Jupiter, and that the positions of the Sun, $\mathbf{x}_s(t)$, and Jupiter, $\mathbf{x}_J(t)$, are given relative to the centre-of-mass.

a: Show that these can be expressed in terms of $\mathbf{x}(t)$ as

$$\begin{aligned} \mathbf{x}_s(t) &= -\frac{m_J}{M} \mathbf{x}(t) , \\ \mathbf{x}_J(t) &= \frac{m_s}{M} \mathbf{x}(t) . \end{aligned}$$

In chapter 3 we tacitly assumed that $m_s \rightarrow \infty$, which is not correct, but not a bad approximation.

b: Argue that if we find the time-dependence of $\mathbf{X}(t)$ and of $\mathbf{x}(t)$ we have solved the problem of finding the orbits of Jupiter and the Sun.

We now compute the total kinetic energy of the system. It is defined as

$$E_k(t) = \sum_j \frac{1}{2} m_j \left(\frac{d}{dt} \mathbf{r}_j(t) \right)^2, \quad (3.3.7)$$

and if we substitute the relative and center of mass coordinates for the $\mathbf{r}_j(t)$ we obtain

$$E_k(t) = \frac{1}{2} M \left(\frac{d}{dt} \mathbf{X}(t) \right)^2 + \sum_j \frac{1}{2} m_j \left(\frac{d}{dt} \mathbf{x}_j(t) \right)^2. \quad (3.3.8)$$

The total kinetic energy splits up into a contribution from the centre-of-mass motion, and an internal energy given by the sum over all constituents.

Problem 95: Prove

Prove the separation of the kinetic energies.

Similarly we compute the total angular momentum. It is defined as

$$L = \sum_j \mathbf{r}_j(t) \wedge \mathbf{p}_j(t), \quad (3.3.9)$$

and if you do the algebra again you will find that

$$L = \mathbf{X}(t) \wedge \mathbf{P}(t) + \sum_j m_j \mathbf{x}_j(t) \wedge \frac{d}{dt} \mathbf{x}_j(t). \quad (3.3.10)$$

The first term is usually called the *orbital angular momentum* of the extended body. The second term represents the internal angular momentum, as the *spin* of the rigid body.

Problem 96: Sun-Jupiter: interparticle forces

The Sun and Jupiter exert forces on one another.

- a: Show that for the centre-of-mass coordinate Newton's 2nd law gives $M \frac{d^2}{dt^2} \mathbf{X}(t) = 0$.
b: Show that for the separation vector $\mathbf{x}(t)$ Newton's second law gives

$$\frac{m_s m_J}{M} \frac{d^2}{dt^2} \mathbf{x}(t) = - \frac{G m_s m_J}{\sqrt{\mathbf{x}^2(t)}} \hat{\mathbf{x}}(t).$$

- c: Explain why both Jupiter as well as the Sun will be in an elliptic orbit around the centre-of-mass?

We see that we have almost completed the Problem of completely separating the center-of-mass motion and the internal motion of the extended body. If you have completed all the Problems you will have also seen how this can be applied practically[§]. Can the separation be made complete? The answer is negative, in

[§]The motion of the Sun caused by Jupiter's gravitational pull is in principle observable. It is by this method that in the last 10 years well over 100 *extra-solar* planets have been discovered, i.e. planets orbiting other stars. Current observational techniques are not yet sensitive enough to detect planets of one Earth-mass.

general. External forces acting on the extended body will mix internal and center-of-mass degrees of freedom.

Torque again

If we consider the torque on each individual particle we would have to compute the total torque as,

$$T = \sum_j \mathbf{r}_j \wedge \mathbf{f}_j(\mathbf{r}_j) .$$

Splitting up the centre-of-mass and relative coordinates now gives

$$T = \mathbf{X} \wedge \sum_j \mathbf{f}_j(\mathbf{X} + \mathbf{x}_j) + \sum_j \mathbf{x}_j \wedge \mathbf{f}_j(\mathbf{X} + \mathbf{x}_j) .$$

Here we see the problem arise. ¶

Problem 97: The Torque of Gravity

Close to the Earth's surface the force of Gravity is constant and the force on the particle with mass m_j is given by $\mathbf{f}_j = -m_j g \mathbf{e}_1$, where $\mathbf{e}_1$ represents the upward direction.

a: Do falling bodies start to rotate by themselves? (Answer without calculating anything!)

b: Show that the second term in the expression for T indeed vanishes.

c: Show that if the centre-of-mass of the body remains fixed, then there must be an additional force at work.

The problem is in the force terms. Even if we assume that the force on particle j does not depend on the position of the other particles, we still need to make an approximation to get anywhere

$$\mathbf{f}_j(\mathbf{X} + \mathbf{x}_j) \approx \mathbf{f}_j(\mathbf{X}) + (\mathbf{x}_j \cdot \nabla) \mathbf{f}_j(\mathbf{X}) + \dots \quad (3.3.11)$$

where we ignore everything which depends on quadratic or higher powers of the $\mathbf{x}_j$. This approximation is related to the Taylor expansion you may have encountered in your maths classes.

Problem 98: Two particles on a massless rod

Recall that we studied the two particles on a massless rod in chapter 3. Both particles also experienced a force, the force on particle at $\mathbf{x}_j$ given by $\mathbf{f}_j = -k\mathbf{x}_j$. The rod separating the two particle was described by the vector $\mathbf{d}(t)$ pointing from particle A to particle B.

a: Show that the relative positions of particles A and B are $\mathbf{x}_A = -\frac{1}{2}\mathbf{d}(t)$ and $\mathbf{x}_B = \frac{1}{2}\mathbf{d}(t)$.

b: Use this to compute $\mathbf{f}_A(\mathbf{X} + \mathbf{x}_A)$ and $\mathbf{f}_B(\mathbf{X} + \mathbf{x}_B)$.

The main assumption behind this approximation is that the forces $\mathbf{f}_j$ vary only slowly across the body. If we use the above approximation we find, ignoring everything which is quadratic in the relative coordinates

$$T \approx \mathbf{X} \wedge \sum_j \mathbf{f}_j(\mathbf{X}) + \mathbf{X} \wedge \sum_j (\mathbf{x}_j \cdot \nabla) \mathbf{f}_j(\mathbf{X}) + \sum_j \mathbf{x}_j \wedge \mathbf{f}_j(\mathbf{X}) .$$

¶Now we don't worry about forces between the particles because I assume Newton's third law keeps things in balance. If that does not happen, internal dynamics can cause a body to start to rotate all by itself.

The first term is merely the torque produced by the total force on the body. The second term corresponds to a correction to the first term due to the inhomogeneity of the force field and the finite size of the extended body. In a gravitational context these forces are known as *tidal forces*. They also depends on the relative coordinates and thus allow for the exchange of angular momentum between the spin and the orbital parts. For the Earth-Moon system this is known to occur. The third term corresponds to torques relative to the centre-of-mass and thus drives the change of spin. Note that if the forces on the particles are all one and the same, i.e. $\mathbf{f}_j = \mathbf{f}$ then all but the first term vanish due to the fact that the relative positions sum to zero. If there is no net-force on the body, then the first term also vanishes.

Problem 99: Particles on a rod

Consider again the system from the last Problem. Now compute the approximate T .

Let us give a few examples of this:

- Consider a non-spinning ball rotating on a circular orbit around some force center. If no other forces act on this ball then nothing will make it start spinning, but now consider there is a gas present. The gas will cause friction on the ball's surface and this friction will be higher where the velocity is higher. The outer side of the ball travels at a higher velocity than the inner side and as a result the ball will loose orbital speed and start to rotate. There is a force transferring orbital angular momentum into spin.
- Consider the topspin of a tennis ball. The balls spin causes pressure differences on different sides of the ball. The effect is that a net force will be acting on the ball as long as it is spinning. Spinning tennis balls are reflected from their normal free-fall trajectories.

However the conceptual split between center-of-mass motion and internal motion is very useful. In the next two chapters we will devote all our attention to the internal motion for rigid bodies. These are called *rigid rotations*.

Problem 100: Differentiation with respect to Bivectors?

In the text we have seen that we can differentiate not only with respect to position vectors like $\mathbf{x}$, but also with respect to arbitrary vectors. The same can be done for bivectors. Let Ω be the bivector

$$\Omega = \omega_1 J_1 + \omega_2 J_2 + \omega_3 J_3 .$$

a: Give arguments why the following definition can represent the *differentiation with respect to a bivector* Ω ;

$$\partial_\Omega = \sum_{j=1}^3 -J_j \frac{\partial}{\partial \omega_j} . \quad (3.3.12)$$

b: Show that $\partial_\Omega \Omega = 3$.

c: Show that $\partial_\Omega \Omega^2 = 2\Omega$.

Problem 101: The Earth-Moon system revisited

We have seen in the Problems in this chapter that two bodies that exert a mutual gravitational force orbit in elliptical orbits around the centre-of-mass.

- a: What is the distance between the center of the Earth, and the centre-of-mass of the Earth-Moon system?
- b: Is the centre-of-mass of the Earth-Moon system inside or outside the Earth?
- c: Sketch the Earth-Moon system at some point in their orbit. Draw the gravitational force of the Moon on four points on the Earth's surface, 1) the point closest to it, 2) the point farthest from it, and 3) and 4) the two opposite points in the middle.

Problem 102: The Gravitational Field

The gravitational field $\mathbf{g}(\mathbf{x})$ around a point-particle of mass M is given by Newton's law of gravitation

$$\mathbf{g}(\mathbf{x}) = -\frac{GM}{x^2} \hat{\mathbf{x}}. \quad (3.3.13)$$

- a: Show that this gravitational field satisfies the property $\nabla \cdot \mathbf{g}(\mathbf{x}) = 0$ everywhere, except at $\mathbf{x} = 0$.
- b: Show that this gravitational field satisfies the property $\nabla \wedge \mathbf{g}(\mathbf{x}) = 0$ everywhere.
- c: Consider a sphere of radius R around the point-particle. Show that the product of $|\mathbf{g}(\mathbf{x})|$ on the sphere, with the area of the sphere equals $4\pi GM$ and thus is independent of the radius of the sphere.

3.4. Continuous Systems**3.5. Rigid Systems**

An extended system consists of many point particles, or effectively pointlike constituents. Each of these has a position $\mathbf{r}_i(t)$ that will depend on time. We call such a system a *rigid body* when the distance between any two point particles of the body is constant, i.e.

$$d_{ij} = \sqrt{(\mathbf{r}_i(t) - \mathbf{r}_j(t))^2} \quad (3.5.1)$$

does not depend on time. This reduces the motion of the body essentially to two types:

- Translations: all point particles move with the same velocity.
- Rotations: the motion can be described by applying one time-dependent Rotor to all initial-position vectors.

In the previous chapter we have seen that we can separate these two modes of motion into a centre-of-mass motion and internal motions relative to the centre-of-mass. For a rigid body the internal motions are confined to rotations. In this chapter we largely ignore the centre-of-mass motion and just study the rotational motion.

3.5.1. The Inertia Tensor

A rotation can be described by a rotor $R(t)$. The relative positions of the point particles in a rigid body will then change according to

$$\mathbf{x}_j(t) = R(t)\mathbf{x}_j(0)\tilde{R}(t) . \quad (3.5.2)$$

Note how the position vectors $\mathbf{x}_j(0)$, which are the positions of the point particles in the frame of the body itself are rotated into the position of the point particles as seen from the observers frame, who sees the body rotating. From this expression we can compute the relative velocities by straightforward differentiation, just like we did in chapter 2. By defining

$$\left\{ \frac{d}{dt} R(t) \right\} \tilde{R}(t) = -\frac{1}{2} \Omega_S(t) , \quad (3.5.3)$$

we can write

$$\frac{d}{dt} \mathbf{x}_j(t) = \frac{1}{2} \{ \mathbf{x}_j(t) \Omega_S(t) - \Omega_S(t) \mathbf{x}_j(t) \} . \quad (3.5.4)$$

This bivector $\Omega_S(t)$ represents the plane in which the rotation takes place *and* the angular velocity of the rotation as measured by the outside observer!

From the point of view of the rigid body at $t = 0$ we would have an angular velocity bivector $\Omega_B(t)$ given by

$$\Omega_B(t) = \tilde{R}(t) \Omega_S(t) R(t) , \quad (3.5.5)$$

this is just $\Omega_S(t)$ rotated back to the frame of the rigid body at $t = 0$. If we write down the equation for $R(t)$ in terms of $\Omega_B(t)$ we obtain

$$\frac{d}{dt} R(t) = -\frac{1}{2} R(t) \Omega_B(t) . \quad (3.5.6)$$

Our discussion of rigid body mechanics has one goal: finding a law of nature that allows us to calculate $\Omega_B(t)$, which then allows us to compute $R(t)$. The first place to look for such a law is in the box of *conservation laws*. So we consider angular momentum.

Problem 103: Rotating coordinates

In the derivation above we have introduced two bivectors $\Omega_S(t)$ describing the instantaneous angular velocity according to an *outside* observer, and $\Omega_B(t)$.

- a: Explain why $\Omega_S(t)$ represents the angular velocity as seen from a initial frame of the body at $t = 0$.
- b: Can we consider this the rest-frame of the rotating body?
- c: Suppose $\Omega_S(t) = \omega J_3$ for some constant angular speed ω . Give $R(t)$ and compute $\Omega_B(t)$.
- d: Can you explain the result under c:?

If we compute the internal angular momentum, i.e. the spin, of the rigid body we get

$$\begin{aligned}
 L(t) &= \sum_j m_j \mathbf{x}_j(t) \wedge \frac{d}{dt} \mathbf{x}_j(t) , \\
 &= \sum_j m_j \{R(t) \mathbf{x}_j(0) \tilde{R}(t)\} \wedge \frac{1}{2} \{\mathbf{x}_j(t) \Omega_S(t) - \Omega_S(t) \mathbf{x}_j(t)\} , \\
 &= R(t) \left\{ \sum_j m_j \mathbf{x}_j(0) \wedge \frac{1}{2} [\mathbf{x}_j(0) \Omega_B(t) - \Omega_B(t) \mathbf{x}_j(0)] \right\} \tilde{R}(t) .
 \end{aligned}$$

Notice the appearance of an interesting quantity inside the brackets. The expression

$$I(\Omega_B(t)) = \sum_j m_j \mathbf{x}_j(0) \wedge \frac{1}{2} \{\mathbf{x}_j(0) \Omega_B(t) - \Omega_B(t) \mathbf{x}_j(0)\} , \quad (3.5.7)$$

is called the *inertia tensor*. It relates the angular momentum $L(t)$ to a given bi-vector $\Omega_B(t)$ according to

$$L(t) = R(t) I(\Omega_B(t)) \tilde{R}(t) , \quad (3.5.8)$$

from which we see that $I(\Omega_B(t))$ is the angular momentum of the body rotated back to the $t = 0$ frame of the body. Working out the wedgeproduct we can rewrite the inertia tensor as

$$I(\Omega_B(t)) = \frac{1}{2} \sum_j m_j \{ \mathbf{x}_j(0)^2 \Omega_B(t) - \mathbf{x}_j(0) \Omega_B(t) \mathbf{x}_j(0) \} . \quad (3.5.9)$$

This quantity is straightforward to compute in many cases of practical interest. It is a linear map, not on vectors but rather on bivectors. The crucial property of the inertia tensor is that it only depends on the *time-independent properties* of the rigid body. All the time-dependence in I is a result of the time-dependence of $\Omega_B(t)$.

Problem 104: Two particles on a massless rigid rod

Suppose we have two particles, A and B, of mass m_A and m_B and suppose they are connected by a massless rod of length $2d$.

- a: Compute their centre-of-mass $\mathbf{X}(0)$ if their positions at $t = 0$ are $\mathbf{r}_A(0)$ and $\mathbf{r}_B(0)$.
b: Let $\mathbf{x}_A(0)$ and $\mathbf{x}_B(0)$ be their positions relative to the centre-of-mass, and assume that we have chosen a basis $\mathbf{e}_j$ such that both positions are along $\mathbf{e}_1$. Determine the numbers d_A and d_B in

$$\begin{aligned}
 \mathbf{x}_A &= d_A \mathbf{e}_1 , \\
 \mathbf{x}_B &= -d_B \mathbf{e}_1 .
 \end{aligned}$$

- c: Give the expression for the inertia tensor $I(\Omega)$ in this frame.

- d: There is an Ω for which $I(\Omega) = 0$. Find it and explain why it is like that.

Rotational Energy

Using the inertia tensor we can also give an expression for the rotational energy

of the body. This energy is the sum of the kinetic energies of all the separate constituents of the body. So we get when grinding through the algebra

$$\begin{aligned}
 E_r(t) &= \sum_j \frac{m_j}{2} \mathbf{v}_j(t) \cdot \mathbf{v}_j(t) , \\
 &= -\frac{1}{4} \{ \Omega_B(t) I(\Omega_B(t)) + I(\Omega_B(t)) \Omega_B(t) \} , \\
 &\equiv -\frac{1}{2} \Omega_B(t) \cdot I(\Omega_B(t)) .
 \end{aligned} \tag{3.5.10}$$

We had already derived this earlier in chapters 2 and 3. The minus sign is again caused by the fact that bivectors square to negative numbers, i.e. the rotational energy is a positive quantity.

Problem 105: Inertia and Kinetic Energy

Complete the above derivation by filling in the missing steps.

The inertia tensor satisfies an important property, it is *symmetric*. By that we means that for two arbitrary bivectors Ω_a and Ω_b we can show that

$$\Omega_a \cdot I(\Omega_b) = \Omega_b \cdot I(\Omega_a) . \tag{3.5.11}$$

Problem 106: Symmetry of the Inertia tensor

Use the definition of the inertia tensor, with the wedge-product written out, to prove this symmetry property.

This

property is very useful in computations, but also in establishing, for example that

$$\begin{aligned}
 \frac{d}{dt} E_r &= \left\{ \frac{d}{dt} \Omega_B(t) \right\} \cdot \partial_{\Omega_B(t)} E_r , \\
 &= \left\{ \frac{d}{dt} \Omega_B(t) \right\} \cdot I(\Omega_B(t)) , \\
 &= \Omega_B(t) \cdot I\left(\frac{d}{dt} \Omega_B(t)\right) .
 \end{aligned} \tag{3.5.12}$$

We will see in a later section that the term $I(\frac{d}{dt} \Omega_B(t))$ corresponds to the action of external forces, of *torques*, applied to the rotating body.

Problem 107: Rotational energy and angular velocity

In the last equation we have used the notion of *differentiation with respect to a bivector* that we briefly addressed in (a problem in) chapter 4.

a: Try to express the steps of the above differentiation in words.

b: Although it may be (a little to) challenging you may want to attempt to complete the derivation.

c: What could a *potential energy* look like in this case.

The physical meaning of the inertia tensor is the following: In general the angular momentum plane $L(t)$ and the angular velocity planes $\Omega_B(t)$ or $\Omega_S(t)$ are

not parallel! In the following section we will investigate under which conditions they can be taken to be parallel.

3.5.2. Principal Axes and Moments of Inertia

What are the conditions under which L and Ω_S are parallel? Well first note that if we have a bivector Ω_k for which

$$I(\Omega_k) = I_k \Omega_k , \quad (3.5.13)$$

for some number I_k , then for a rotation in the plane specified by Ω_k we indeed have

$$L = I_k \Omega_k . \quad (3.5.14)$$

The bivectors Ω_k that satisfy the former equation are called *principal planes of rotation*. More common is the term *principle axis* which refers to the vector $\mathbf{n}_k$ which are perpendicular to the principal planes, i.e.

$$\mathbf{n}_k = -i\Omega_k .$$

The numbers I_k are called the *moments of inertia*. Although I will not prove it here, it is possible to show that for the inertia tensor there always exist three perpendicular principal planes. So by a clever choice of basis we have

$$\begin{aligned} \Omega_B(t) &= \sum_{k=1}^3 \omega_k J_k , \\ I(\Omega_B(t)) &= \sum_{k=1}^3 \omega_k I_k J_k . \end{aligned}$$

The coursebook restricts its discussions of rigid body mechanics to rotations around principal axis only!

Moment of inertia for a principle axis

Earlier we derived an expression for the inertia tensor. Can we use it to find a simplified expression for the moment of inertia I_k for a given principle axis? Consider it's definition, but now for Ω_k

$$I(\Omega_k) = \frac{1}{2} \sum_j m_j \{ \mathbf{x}_j(0)^2 \Omega_k - \mathbf{x}_j(0) \Omega_k \mathbf{x}_j(0) \} .$$

Note that any constituent will only contribute to $I(\Omega_k)$ by the part $\mathbf{x}_{j//}(0)$ of its position vector $\mathbf{x}_j(0)$ which is in the plane Ω_k . So,

$$I(\Omega_k) = \sum_j m_j \mathbf{x}_{j//}(0)^2 \Omega_k .$$

So we get for the moment of inertia around this axis

$$I_k = \sum_j m_j \mathbf{x}_{j//}(0)^2 . \quad (3.5.15)$$

This is usefull only when Ω_k is a principle plane of rotation. If it is not, then we need additional forces, i.e. torques, to ensure that the rotation stays confined to this axis. We will see that in the next section.

Problem 108: Moment of inertia for a cylinder

Consider a massive cylinder with height h , radius d and a constant mass-density ρ .

a: Show that the mass of a thin shell of radius dr at a distance r from the central axis of the cylinder is given by $\rho 2\pi r h dr$.

b: Show that when the cylinder would rotate around its central axis, this shell would contribute a moment of inertia equal to $\rho 2\pi r^3 h dr$.

c: Show that the total moment of inertia of the cylinder is given by $I = \frac{\pi}{2} \rho d^4 h$.

d: Show that if M is the total mass of the cylinder we get $I = \frac{1}{2} M d^2$.

Parallel Axis

Now suppose that a body rotates in the plane J_k , i.e. $\Omega_B = \omega(t) J_k$, but not around the centre-of-mass, but about some other point that is at $\mathbf{a}$ relative to the origin. How does that affect L ? The answer to that question comes from calculating the inertia tensor for that case,

$$I_{\mathbf{a}}[\Omega_B(t)] = \frac{1}{2} \sum_j m_j ((\mathbf{x}_j(0) - \mathbf{a})^2 \Omega_B(t) - (\mathbf{x}_j(0) - \mathbf{a}) \Omega_B(t) (\mathbf{x}_j(0) - \mathbf{a})) \quad (3.5.16)$$

As $\mathbf{a}$ itself is in the plane J_k we can use that $\mathbf{a} J_k = -J_k \mathbf{a}$. The result then becomes

$$I_{\mathbf{a}}(\Omega_B) = I_0(\Omega_B) + \mathbf{a}^2 M \Omega_B . \quad (3.5.17)$$

This equation is called the *parallel axis theorem*, or *Steiner's theorem*.

Problem 109: Cylinders revisited

Consider the moment of inertia we computed for the rotation of the cylinder around its central axis. Now suppose we rotate the same cylinder around an axis parallel to its central axis, but at a distance s . What is the moment of inertia around that axis?

3.5.3. The Euler equations

What does the inertia tensor tell us about the way $\Omega_B(t)$ depends on time? The key here is the Torque and its effect on angular momentum. Let T be the torque produced by some outside force. If we compute the time-derivative of $L(t)$

$$\frac{d}{dt} L(t) = \frac{d}{dt} (R(t) I[\Omega(t)] \tilde{R}(t)) , \quad (3.5.18)$$

and go through the same routine as above we find

$$I\left[\frac{d}{dt} \Omega(t)\right] + \frac{1}{2} \{I[\Omega(t)]\Omega(t) - \Omega(t)I[\Omega(t)]\} = \tilde{R}(t) T(t) R(t) . \quad (3.5.19)$$

These differential equations for $\Omega(t)$ are called the *Euler equations*. It is evident that even torque-free rotation can become quite a complicated phenomenon to

describe when the inertia tensor of the body does not have a simple form. Below we will study one such example of torque-free motion.

Rotational Energy revisited

The Euler equations tell us that the total set of forces acting on the rotating body consists of two parts. There is the torque T , but also this term $\frac{1}{2}\{I[\Omega(t)]\Omega(t) - \Omega(t)I[\Omega(t)]\}$; what does it represent? Does it contribute to the change in total energy?

This last question is easily answered. As we considered the rate of change of rotational energy, we obtained

$$\frac{d}{dt}E_r(t) = \Omega_B(t) \cdot I\left(\frac{d}{dt}\Omega_B(t)\right) .$$

It is tedious but not extremely difficult to check that

$$\Omega_B(t) \cdot \frac{1}{2}\{I[\Omega(t)]\Omega(t) - \Omega(t)I[\Omega(t)]\} = 0 .$$

So the term $\frac{1}{2}\{I[\Omega(t)]\Omega(t) - \Omega(t)I[\Omega(t)]\}$ in the Euler equations does not contribute to a change in rotational energy. If it would have turned out different that would have meant we have a big problem because a free body cannot change its energy by itself!

Problem 110: A Helicopter

When the rotorblades of a helicopter are brought into motion they obviously gain angular momentum. But angular momentum is conserved for a helicopter.

- a: What is responsible for the torque on the rotor-blades that makes them pick up speed?
- b: What does angular momentum conservation require from the torques acting on the helicopter-cabin?
- c: Which part of the helicopter ensures that the cabin doesn't go into rotation either?

The term we are studying is a result of the fact that we are describing a rotating body while using a coordinate system (the body-axis rotated back to $t = 0$) that itself is rotating. Whenever you do physics in rotating systems these sort of fake forces crop up. They are like the *centrifugal force* experienced by people in a car going through a curve at high speed, or the *Coriolis force* due to the Earth's rotation that affects the motion of air and water, as well as Foucault's pendulum, on the Earth surface. So what we get is

$$\begin{aligned} \frac{d}{dt}E_r(t) &= \Omega_B(t) \cdot \tilde{R}(t)T(t)R(t) , \\ &= \Omega_S(t) \cdot T(t) . \end{aligned} \tag{3.5.20}$$

and this is the equation which would allow us to discuss *work done by Torque*. A discussion that we will not go into.

Problem 111: Everyday angular momentum physics

Angular momentum is an everyday phenomenon we have grown accustomed to.

a: When you drive a car at high speed and suddenly hit the brakes, the car's nose will dip down. Explain this as a result of angular momentum conservation.

b: When you drive your bike angular momentum conservation assists you in keeping your balance. Explain how.

c: When an olympic ice-dancer performs her pirouette, angular momentum conservation comes to her aide. Explain how.

3.5.4. The Symmetric Top

Consider a system for which two of the three moments of inertia are equal, i.e. $I_1 = I_2 \neq I_3$. This implies for an angular velocity $\Omega_B(t)$ given by

$$\Omega_B(t) = \omega_1(t)J_1 + \omega_2(t)J_2 + \omega_3(t)J_3 ,$$

that

$$I(\Omega_B(t)) = I_1\Omega_B(t) + (I_3 - I_1)\omega_3(t)J_3 .$$

When substituted in the Euler equations with vanishing Torque we get

$$I_1 \frac{d}{dt}\Omega_B + (I_3 - I_1)J_3 \frac{d}{dt}\omega_3(t) - \frac{1}{2}(I_3 - I_1)\omega_3(t)\{\Omega_B(t)J_3 - J_3\Omega_B(t)\} = 0 .$$

Note that any J_3 contribution to $\Omega_B(t)$ will not affect the last term of the Euler equation, so that we must have

$$\frac{d}{dt}\omega_3(t) = 0 .$$

So we can take the angular velocity in the J_3 -plane to be constant. As a result the equation simplifies to

$$\frac{d}{dt}\Omega_B - \frac{(I_3 - I_1)}{2I_1}\omega_3\{\Omega_B(t)J_3 - J_3\Omega_B(t)\} = 0 .$$

This equation can be easily solved. If we define the angular velocity

$$\omega = \frac{(I_3 - I_1)}{I_1}\omega_3 ,$$

then the solution is

$$\Omega_B(t) = \exp\left(-\frac{1}{2}\omega t J_3\right)\Omega_B(0) \exp\left(\frac{1}{2}\omega t J_3\right) . \quad (3.5.21)$$

So the angular velocity in the body-frame rotates in the J_3 -plane with a frequency determined by the relative difference between the moments of inertia.

To fully solve the problem we need to compute the rotor $R(t)$. The rotor equation in terms of $\Omega_B(t)$ is

$$\begin{aligned} \frac{d}{dt}R(t) &= -\frac{1}{2}R(t)\Omega_B(t) , \\ &= \frac{1}{2}R(t) \exp\left(-\frac{1}{2}\omega t J_3\right)\Omega_B(0) \exp\left(\frac{1}{2}\omega t J_3\right) . \end{aligned}$$

We can simplify this calculation by defining the new rotor $U(t)$ through

$$U(t) = R(t) \exp\left(\frac{1}{2}\omega t J_3\right) .$$

If we substitute this into the rotor equation, then after some algebra we find an equation for $U(t)$, reading

$$\frac{d}{dt}U(t) = -\frac{1}{2}U(t)\{\Omega_B(0) + \omega J_3\} .$$

This we can again solve to find $U(t)$ as

$$U(t) = \exp\left(-\frac{1}{2}\{\Omega_B(0) + \omega J_3\}\right) .$$

From this, and the definition of $U(t)$ we can retrieve $R(t)$. The result is, putting $R(0) = 1$,

$$R(t) = \exp\left(-\frac{1}{2}t\{\Omega_B(0) + \omega J_3\}\right) \exp\left(\frac{1}{2}\omega t J_3\right) . \quad (3.5.22)$$

This rotor generates two rotations. First of all there is the rotation in the initial plane of rotation. But secondly there is the additional rotation in the J_3 -plane, this is called *precession*. If we use the fact that

$$L(t) = L(0) = I_1\{\Omega_B(0) + \omega J_3\} ,$$

we can write the very appealing form

$$R(t) = \exp\left(-\frac{Lt}{2I_1}\right) \exp\left(\frac{1}{2}\omega t J_3\right) .$$

The rotation occurs in the plane of angular momentum, but there is an additional precession.

Earth's precession

The forces of the Moon and the Sun produce relatively weak torques on the Earth. To a good approximation we can there for consider the Earth a torque-free system. However, the Earth is not perfectly spherical but slightly flattened at the poles. As a result not all the moments of inertia are equal. Two of them are equal, but one is slightly different.

Problem 112: Flattened Earth

Find the degree to which the Earth is flattened by comparing the equatorial radius of the Earth to the polar radius. Can you give an estimate as to how this difference could affect the corresponding ratio of the moments of inertia?

We therefore consider the Earth to be a symmetric top. The numerical ratio of the moments of inertia of the Earth is

$$\frac{I_3 - I_1}{I_1} \approx 0.00327 . \quad (3.5.23)$$

As a result the angular speed of precession should be

$$\omega = 0.00327\omega_3 . \quad (3.5.24)$$

Now from experience we know that the rotation axis is not tilted strongly, and hence ω_3 should be equal to $\frac{2\pi}{24hrs}$ up to very high accuracy. We therefore find for the time for one complete precession

$$T_{prec} = \frac{2\pi}{\omega} \approx 306days . \quad (3.5.25)$$

So what's the situation?

Careful geographical and astronomical measurements over the past century reveal a rather chaotic precession with an amplitude of a few hundred meters. Careful analysis (Fourier analysis to be treated in the SCI 211 course) reveals that this precession has a component with a 365-day period. This one is believed to be caused mainly by the annual climatic cycle that transports large quantities of air and water across the globe. But next to that there is the so-called *Chandler wobble* with a period of 420 days. The current view holds it that this is the free-body precession we just derived. The main reasons for the mismatch in the period lie in the non-rigid nature of large parts of the Earth. An analysis of these periods also reveals strong damping effects due to friction between the Earth's mantle and its core. This would suggest the precession ceases after approximately 20 years. However this does not happen, indicating that there are still other forces at work that keep exciting the precessional motion of the Earth. It has been suggested that deep Earth quakes could generate changes in the Earth's inertia tensor that are significant enough to drive precessional motion. The *Tsunami* quake of December the 24th, 2004, was reportedly strong enough to actually have such an effect.

The Earth however is a very complicated system. Below its hard mantle it also consists of elastic and fluid components. The understanding of extended systems that we have achieved up to this point in this course is not good enough to describe, let alone understand these phenomena. In the next chapter we will turn our attention towards the description of non-rigid systems.

3.6. Assignments

Chapter 4

Deformable Systems

We have discussed rigid systems in the previous chapter. Rigidity is a good approximation when the differences between forces acting on different constituents are too small to cause a deformation of the body, i.e. a relative displacement of the constituents. However, many materials exist in which deformations do occur, either because the internal forces are relatively weak, such as in liquids or gelly-like substances, or because the external forces are very strong, such as for example the forces of plate-tectonics acting on the plates making up the Earth's surface. Finally there is a deeply fundamental reason to allow for non-rigidity: according to the theory of special relativity absolutely rigid bodies do not even exist. So there is a law of nature (the speed-of-light-limit) that puts a constraint on how rigid a body can be. In this chapter I want to give an account of how we describe such deformable systems in physics. We will study two types of systems; elastic media and fluids.

Elasticity

First of all we will turn to those systems where the internal forces resisting deformation depend on the amount of deformation. This leads us into the theory of elasticity, which we will only apply to linear media. The restriction to materials whose elastic response is linear to the forces applied is very useful for most practical applications. Furthermore, even materials whose elasticity properties are non-linear will behave in a linear fashion as long as the deformations remain sufficiently small. This restriction excludes a whole range of topics, such as the breaking and tearing of extended bodies. But it does allow us to study the, for example, the propagation of sound through an extended body. Furthermore it will allow us to uncover some central features that we will need in our discussion of fluids.

Fluids

Secondly we will look at those systems where the internal forces do not rely on the amount of deformation but rather *on the rate of deformation*. This is typical for what you consider a *fluid*. Take liquid water for example, basically you may deform it any way you want but the forces between water molecules do affect the rate at which these deformations take place. We will see that the stresses that build up in solids due to a deformation are transcribed in essentially unaltered form to the description of fluids, where they are the result of the rate of deformation. The role of the *elasticity* of a solid is, in a sense, taken over by the *viscosity* of a fluid.

However, the viscous forces are *dissipative*, i.e. turn energy into heat.

Phases

The distinction between elastic and fluid behaviour is not as clear as you may think. This is related to the fact that the distinction between the solid and liquid phase of materials is not always very sharp. There are materials which clearly are liquid or solid, but there are also those which, under certain conditions, display a smooth transition between the two phases. We defer that issue to the next chapters when we look again at the microscopic description of matter.

Goals 113: Deformations and Stress

The goal of the theory presented in this section is to obtain a law of nature that allows either the calculation of deformations when external forces are given, or the computation of those forces from observed deformations.

To attain this goal the first step is to find a proper description of deformation. The first two sections will be devoted to that. In the last section we will look at the forces related to deformations, called *stresses*.

4.1. Deformation

The main ingredient of the deformation of an elastic body is the change in the position of its constituents. Let us first consider an extended object of which the constituents have a position $\mathbf{x}_j$ when there are no forces and strains acting on the object. The index j merely labels the constituents. Now suppose that due to action of some force field the constituents shift their equilibrium position to $\mathbf{x}'_j$, this means that a displacement $\mathbf{d}_j$ has occurred given by

$$\mathbf{d}_j = \mathbf{x}'_j - \mathbf{x}_j .$$

It seems logical to assume that the new position $\mathbf{x}'$ of the constituent is in fact a function of the old position $\mathbf{x}$. Instead of messing around with vectors with all sorts of indices we could just as well focus on the *map from the old configuration to the new configuration*, i.e. $\mathbf{x}'(\mathbf{x})$. If we do that, then we can define what we could call a *displacement field* $\mathbf{d}(\mathbf{x})$ which assigns to $\mathbf{x}$ the displacement $\mathbf{d}$ of the constituent at $\mathbf{x}$, so

$$\mathbf{d}(\mathbf{x}) = \mathbf{x}'(\mathbf{x}) - \mathbf{x} . \quad (4.1.1)$$

Obviously a spatially constant displacement does *not* induce a deformation, and neither does an overall constant rotation. Hence, whatever we are going to use as a variable representing the actual deformation of the body it must revolve around the *variation* of this displacement throughout the body.

Consider two neighbouring constituents whose position was separated by an infinitesimal small vector $\mathbf{a}$. Their new positions are separated by

$$\begin{aligned} \mathbf{a}' &= \mathbf{x}'(\mathbf{x} + \mathbf{a}) - \mathbf{x}'(\mathbf{x}) , \\ &= \{\mathbf{d}(\mathbf{x} + \mathbf{a}) + \mathbf{x} + \mathbf{a}\} - \{\mathbf{d}(\mathbf{x}) + \mathbf{x}\} , \\ &\approx \mathbf{a} \cdot \nabla \mathbf{d}(\mathbf{x}) + \mathbf{a} . \end{aligned}$$

So we see that the change in the displacement is, approximately, given by

$$\Delta \mathbf{a} \equiv \mathbf{a}' - \mathbf{a} = \mathbf{a} \cdot \nabla \mathbf{d}(\mathbf{x}) .$$

This derivation displays that we are assuming that we are studying deformations that are small on the scale of the typical separation between the constituents. Obviously this excludes tearing and breaking, as under those conditions constituents become separated by large distances.

Problem 114: Example Deformations

Consider constituents separated by the vector $\mathbf{a} = \epsilon \mathbf{e}_1$ and study the effect of the following deformations by computing $\Delta \mathbf{a}$;

a: $\mathbf{x}'(\mathbf{x}) = c\mathbf{x} + \mathbf{d}$, where $\mathbf{d}$ and c are constants.

b: $\mathbf{x}'(\mathbf{x}) = c\mathbf{x}^2\mathbf{x}$, where c is a constant.

c: $\mathbf{x}'(\mathbf{x}) = \exp(-\alpha x_1 J_3)\mathbf{x} \exp(\alpha x_1 J_3)$, where α is a constant.

In each case try to picture what happens to the medium being deformed by making a sketch.

Deformation tensors

Deformation is thus described by means of a function that is *linear* in $\mathbf{a}$, and associates to it a vector. Such linear maps are often referred to as *tensors*, like the *inertia tensor* that we encountered in the previous chapter. This particular one is called the *deformation tensor* $\mathbf{D}$,

$$\mathbf{D}[\mathbf{a}] = \mathbf{a} \cdot \nabla \mathbf{d}(\mathbf{x}) . \quad (4.1.2)$$

Because the deformation-tensor spits out a vector, when its argument is a vector, I write it bold-faced.

I want to dissect this tensor into three pieces that describe three distinct types of deformations.

Local Rotations

First of all we consider the deformation be caused by an infinitesimal local *rotation*. In that case we expect the following form

$$\mathbf{D}_R[\mathbf{a}] = \frac{1}{2} \{ \mathbf{a} \omega(\mathbf{x}) - \omega(\mathbf{x}) \mathbf{a} \} ,$$

where $\omega(\mathbf{x})$ is the bivector representing the infinitesimal rotation. Note that due to the anti-symmetry of this expression we would have for any vector $\mathbf{b}$

$$\mathbf{b} \mathbf{D}_R[\mathbf{a}] - \mathbf{D}_R[\mathbf{a}] \mathbf{b} = (\mathbf{a} \wedge \mathbf{b}) \omega(\mathbf{x}) + \omega(\mathbf{x}) (\mathbf{a} \wedge \mathbf{b}) .$$

The last equality follows from testing what happens when the vectors $\mathbf{a}$ and $\mathbf{b}$ are *not* in the plane defined by the bivector $\omega(\mathbf{x})$, in which case the whole expression vanishes. Hence, what we are doing here is establishing the *component* of $\omega(\mathbf{x})$ that is in the plane $\mathbf{a} \wedge \mathbf{b}$. This bivector-part of the deformation tensor is fairly easy to compute from the expressions we have derived,

$$\mathbf{D}_R[\mathbf{a}] = \mathbf{a} \cdot \{ \nabla \wedge \mathbf{d}(\mathbf{x}) \} . \quad (4.1.3)$$

We call this the *rotational part* of the deformation tensor. In most elementary treatments on elasticity it is simply ignored. However, it does play a role in, for example, applications in geophysics.

Problem 115: Rotational Deformation

Reconsider the deformation c : from the previous Problem and determine the rotational part of the deformation tensor. Try to be clever and not to do a lot of computations.

4.1.1. The symmetric deformation tensor

Rotational deformations are not very common. But we now know how to isolate it from the other deformations, so we subtract it out of the deformation tensor, and define a new tensor

$$\mathbf{D}_S[\mathbf{a}] \equiv \mathbf{D}[\mathbf{a}] - \mathbf{D}_R[\mathbf{a}] . \quad (4.1.4)$$

Standard terminology is to simply call it *the deformation tensor*. The rotational part has been taken out, by hand, and it now contains only two more types of deformation.

Compression and Dilation

Now consider the deformation that is *isotropic*, i.e. that part which is the same in all directions. This type of deformation leads only to a change in volume at $\mathbf{x}$. We can get this part $D_V(\mathbf{x})$ by the following operation

$$D_V(\mathbf{x}) = \text{Tr}[\mathbf{D}_S] \equiv \sum_{j=1}^3 \mathbf{e}_j \cdot \mathbf{D}_S[\mathbf{e}_j] = \partial_{\mathbf{a}} \cdot D_S[\mathbf{a}] . \quad (4.1.5)$$

This part of the deformation tensor is a scalar. This is typically the sort of deformation that occurs when an isotropic outside pressure is exerted on the body. If we work out the definition we get

$$D_V(\mathbf{x}) = \nabla \cdot \mathbf{d}(\mathbf{x}) . \quad (4.1.6)$$

Now let us consider its exact relationship with volume changes.

An example of compression

Consider a small cube of which each side has the length a . So the sides are given by vectors of the type $a\mathbf{e}_j$, and we would describe this cube by the pseudoscalar $V_a = a^3 i$. Now the deformation yields new sides $\mathbf{e}'_j$ for the deformed cube given by

$$\mathbf{e}'_j - a\mathbf{e}_j = \mathbf{D}_S[a\mathbf{e}_j] . \quad (4.1.7)$$

Let us for simplicity assume that there is nothing but compression or dilation, then the change in length of each side of the cube is, with $a'_j = \mathbf{e}'_j \cdot \mathbf{e}_j$,

$$a' = a + a\{\mathbf{e}_j \cdot \mathbf{D}_S[\mathbf{e}_j]\} .$$

As a result we find for the new volume V'

$$\begin{aligned} V' &= a'_1 a'_2 a'_3 = a^3 (1 + \mathbf{e}_1 \cdot D_S[\mathbf{e}_1]) (1 + \mathbf{e}_2 \cdot D_S[\mathbf{e}_2]) (1 + \mathbf{e}_3 \cdot D_S[\mathbf{e}_3]) , \\ &\approx a^3 (1 + \sum_{j=1}^3 \mathbf{e}_j \cdot D_S[\mathbf{e}_j]) = a^3 (1 + D_V(\mathbf{x})) . \end{aligned} \quad (4.1.8)$$

The approximation that we make is that the relative changes in the vectors are small, then we can ignore square or cubic powers of such small numbers. As a result we find that the *relative change in volume* is given by

$$\frac{\Delta V}{V} = \frac{V' - V}{V} = D_V(\mathbf{x}) . \quad (4.1.9)$$

So the quantity $D_V(\mathbf{x})$ really represents local, infinitesimal, volume changes. Note that for compressions $D_V < 0$ and for dilations $D_V > 0$.

Problem 116: Bulk modulus

Consider a fluid that is under some pressure p , i.e. on every volume element V there is a force acting perpendicular to its surface such that its strength per m^2 is the pressure p . A small pressure increase Δp will lead to a small compression ΔV , if the material is *linear* then these two are related by

$$D_V = \frac{\Delta V}{V} = -B \Delta p , \quad (4.1.10)$$

where B is a material property called the *bulk modulus*.

- a: Whys is there a minus sign?
- b: Find the bulk modulus for water (book or BINAS) and compute the change ΔV for one liter of water if the pressure of the Earth's atmosphere doubles.
- c: Why must B be large for a material that is easy to compress?
- d: Find the most and the least compressible liquids in your data tables.

Only one part now remains in the deformation tensor. These deformations are called *shear* deformation,

$$\sigma[\mathbf{a}] = D_S[\mathbf{a}] - \frac{1}{3} D_V[\mathbf{x}] \mathbf{a} . \quad (4.1.11)$$

Shearing deformations are non-rotational and they are not compressions or dilations.

Problem 117: Shear Deformation is traceless

- a: Use the definition of the trace, and the definition of the tensor $\sigma[\mathbf{a}]$ to show that $Tr(\sigma[\mathbf{a}]) = 0$.
- b: Suppose the shearing deformation is only in the J_3 -plane. Why do we then have $\sigma_{11} = -\sigma_{22}$?

A typical example of a shearing deformation is what you do to a paper when you cut it with a pair of scissors. The paper tears when the shearing deformation becomes too large for the internal forces to maintain. Shear deformations are very

important in technical applications of physics in particular due to their role in breaking and tearing processes. A shearing deformation typically alters the surface area of a surface. Let us investigate that connection in detail.

An example of shear

Consider a small square with sides of length a in the $\mathbf{e}_1\mathbf{e}_2$ -plane. Its surface is described by the bivector $A = a^2 J_3$. Now after a deformation has occurred, the plane is described by a new bivector A' given by

$$A' = \mathbf{e}'_1 \wedge \mathbf{e}'_2 . \quad (4.1.12)$$

Those vectors can be written out in terms of the components

$$\mathbf{e}'_j = \sigma[\mathbf{e}_j] = \sigma_{1j}\mathbf{e}_1 + \sigma_{2j}\mathbf{e}_2 \quad (4.1.13)$$

The deformation of the square is signified by the angles between the old edges $\mathbf{e}'_j$ and the new ones $\mathbf{e}_j$. We write

$$\begin{aligned} \tan(\theta_1) &= \frac{\mathbf{e}'_1 \cdot \mathbf{e}_2}{\mathbf{e}'_1 \cdot \mathbf{e}_1} = \frac{\sigma_{12}}{1 + \sigma_{11}} , \\ \tan(\theta_2) &= \frac{\mathbf{e}'_2 \cdot \mathbf{e}_1}{\mathbf{e}'_2 \cdot \mathbf{e}_2} = \frac{\sigma_{21}}{1 + \sigma_{22}} . \end{aligned}$$

Now we assume these deformations are all very small, i.e. $\sigma_{11} \ll 1$ and $\sigma_{22} \ll 1$. If we also use that for small angles θ we may use the approximation

$$\tan(\theta) \approx \theta ,$$

we finally find, using the symmetry of the tensor, i.e. $\sigma_{12} = \sigma_{21}$,

$$\theta_1 = \theta_2 = \sigma_{12} . \quad (4.1.14)$$

Problem 118: Shear strain

Compare the above discussion to the discussion of shear deformations in the book. Describe in your own words how the two approaches are related.

4.2. Fluid Flow

Now how do we describe a system where the deformations themselves do not generate any counter forces, but where it is the *rate* of deformation that is the cause of stress. Here the assumption is that the positions of all the particles depends on time $\mathbf{x}_j(t)$. The deformation then takes place as time progresses, so with $\mathbf{x}'_j(t) = \mathbf{x}_j(t + dt)$, we expect that

$$\begin{aligned} \mathbf{d}_j(t) &= \mathbf{x}'_j(t) - \mathbf{x}_j(t) , \\ &\approx \mathbf{u}_j(t)dt , \end{aligned} \quad (4.2.1)$$

where $\mathbf{u}_j(t)$ is just the velocity of the j 'th particle. Now we do the same thing as we did earlier; instead of describing all particles separately we introduce a field. This time this is not the displacement field $\mathbf{d}(\mathbf{x})$ but a *flow* $\mathbf{u}(\mathbf{x}, t)$. As a result the deformation tensor becomes

$$D[\mathbf{a}] = \mathbf{a} \cdot \nabla \mathbf{u}(\mathbf{x}, t) dt ,$$

and the *rate of deformation* is described by the tensor

$$\tau[\mathbf{a}] \equiv \frac{D[\mathbf{a}]}{dt} = \mathbf{a} \cdot \nabla \mathbf{u}(\mathbf{x}, t) . \quad (4.2.2)$$

We can dissect this tensor in exactly the same way as we did earlier for deformations. For example,

$$\tau_R[\mathbf{a}] = \mathbf{a} \cdot \{ \nabla \wedge \mathbf{u}(\mathbf{x}, t) \} .$$

represents the *vorticity* of the flow $\mathbf{u}(\mathbf{x}, t)$, i.e. it is a measure for the amount of local, rotational motion in the flow. In fluid dynamics the quantity that is usually introduced is the bivector-field

$$\omega(\mathbf{x}, t) = \nabla \wedge \mathbf{u}(\mathbf{x}, t) , \quad (4.2.3)$$

called the *vorticity* of the flow. If the vorticity of a flow vanishes, the flow is called *irrotational*. Irrotational flows are generally easier to describe mathematically than rotational flows. In practice however, irrotational flows are not very common in physical systems. Still you can find whole textbooks on irrotational flows.

Just like elastic bodies can suffer compression and dilation, flows can also lead to a compression of the fluid. This is, once again described by

$$D_V(\mathbf{x}) = \nabla \cdot \mathbf{u}(\mathbf{x}, t) dt , \quad (4.2.4)$$

represents the local compression of the fluid. By dividing out the infinitesimal time-interval dt we obtain the rate of compression $\nabla \cdot \mathbf{u}(\mathbf{x}, t)$. If a fluid is *non-compressible* we have $\nabla \cdot \mathbf{u}(\mathbf{x}, t) = 0$. We say that *the fluid is divergence free*. Many liquids can be considered incompressible in practical applications, gasses usually aren't.

Problem 119: Potential Flows

If a fluid flow is irrotational, then it can be written as the gradient of some scalar function $\phi(\mathbf{x})$.

a: Show that if $\mathbf{u}(\mathbf{x}, t) = \nabla \phi(\mathbf{x}, t)$, then the flow is indeed irrotational.

b: Show that if an irrotational flow is *incompressible* then the function $\phi(\mathbf{x}, t)$ satisfies $\nabla^2 \phi(\mathbf{x}, t) = 0$. This equation is called the *Laplace equation*.

Conservation of Mass

If we let all this mass flow around due to deformations of the material how do we ensure that in our equations we don't lose anything? Mass ought to be conserved,

but how do we express that mathematically. The idea is the following: consider a mass-density $\rho(t, \mathbf{x})$ and a mass-current $\rho(t, \mathbf{x})\mathbf{u}(t, \mathbf{x})$. The rate at which the mass-density changes with time, at some point $\mathbf{x}$ must be completely due to the flow of the mass-current. No other sources or sinks are allowed. Mathematically this yields

$$\frac{\partial}{\partial t}\rho(t, \mathbf{x}) = -\nabla \cdot \{\rho(t, \mathbf{x})\mathbf{u}(t, \mathbf{x})\} \quad (4.2.5)$$

This equation is called the *continuity equation* or conservation of mass. Note that if we write it out completely we see that the rate-of-change of the density $\rho(t, \mathbf{x})$ is expressed by two terms; first of all there is the $\frac{\partial}{\partial t}\rho(t, \mathbf{x})$ which expresses the rate-of-change of the density of a fluid element while it is at $\mathbf{x}$. But there is a second term hidden in the r.h.s. of the equation. If we write that out using the product-rule we obtain two terms and one of them is $\mathbf{u}(t, \mathbf{x}) \cdot \nabla \rho(t, \mathbf{x})$. This expresses the rate-of-change of the density of the fluid element caused by it's motion.

Problem 120: Along with the Flow

Argue or show that if we move along with the flow, i.e. use a density $\rho(t, \mathbf{x}(t))$, that then the *total* time-derivative, $\frac{D}{Dt}$, of this density would be

$$\frac{D}{Dt}\rho(t, \mathbf{x}(t)) = \frac{\partial}{\partial t}\rho(t, \mathbf{x}) + \mathbf{u}(t, \mathbf{x}) \cdot \nabla \rho(t, \mathbf{x}(t)) .$$

As a result the continuity equation is written as

$$\frac{D}{Dt}\rho(t, \mathbf{x}(t)) = -\rho(t, \mathbf{x})\nabla \cdot \mathbf{u}(t, \mathbf{x}) , \quad (4.2.6)$$

in many fluid dynamics texts. The derivative can be interpreted as the time-derivative in a frame *moving along with the fluid*.

Problem 121: Continuity in a Pipe

Consider a flow $\mathbf{u}$ of water through a pipe. Let the pipe be cylindrical and straight along the $\mathbf{e}_1$ -direction / x -axis with a varying diameter $d(x)$.

a: Draw the system, and draw two crosssections A_1 and A_2 at the positions x_1 and x_2 .

If we ignore any *boundary layer*-effects it is reasonable to assume that $\mathbf{u}(x) = u_1(x)\mathbf{e}_1$ and that it only depends on x

b: We also assume the flow is *incompressible*. Argue that this means that $\frac{\partial}{\partial t}\rho(t, \mathbf{x}) = 0$ and that for a constant density ρ_0 the total mass M between A_1 and A_2 is given by the integral

$$M = \int \rho_0 \frac{\pi}{4} d(x)^2 dx ,$$

c: Show that the net amount of mass, ΔM , entering the tube between A_1 and A_2 per second is given by

$$\Delta M = \rho_0 \{u_1(x_1)A_1 - u_1(x_2)A_2\} ,$$

explain why $\Delta M = 0$ and use this to derive the continuity equation given in the textbook.

e: Show that if $x_2 = x_1 + h$ for an infinitesimal h , then

$$\frac{\Delta M}{A_1} = -\frac{\partial}{\partial x_1} \rho_0 u_1(x_1) .$$

Momentum Conservation

We have just seen how we can use all this fancy mathematical machinery to describe a *local* conservation law. The conservation of mass. Now what about the other conserved quantities? Energy? Momentum? Yes, they too satisfy local conservation laws. We will discuss momentum conservation here.

First we consider momentum. If we have a mass-density $\rho(\mathbf{x}, t)$ flowing with a velocity $\mathbf{u}(\mathbf{x}, t)$ this represents a *momentum-density* $\mathbf{p}(\mathbf{x}, t)$ given by

$$\mathbf{p}(\mathbf{x}, t) = \rho(\mathbf{x}, t)\mathbf{u}(\mathbf{x}, t) . \quad (4.2.7)$$

Now consider the density of momentum in the $\mathbf{e}_j$ -direction, $p_j(\mathbf{x}, t)$. It also flows with a velocity $\mathbf{u}(\mathbf{x}, t)$. So there exists a *momentum current* described by a set of vectors $\mathbf{P}_i$,

$$\mathbf{P}_i(\mathbf{x}, t) = \sum_{j=1}^3 \rho(\mathbf{x}, t) u_i(\mathbf{x}, t) \mathbf{u}(\mathbf{x}, t) . \quad (4.2.8)$$

The conservation law for this momentum then reads

$$\begin{aligned} \frac{\partial}{\partial t} \mathbf{p}(\mathbf{x}, t) &= - \sum_{i=1}^3 \nabla \cdot \mathbf{P}_i(\mathbf{x}, t) \mathbf{e}_i , \\ \frac{\partial}{\partial t} \rho(\mathbf{x}, t) \mathbf{u}(\mathbf{x}, t) &= - \sum_i \nabla \cdot [\rho(\mathbf{x}, t) u_i(\mathbf{x}, t) \mathbf{u}(\mathbf{x}, t)] \mathbf{e}_i . \end{aligned}$$

By working out the differentiations, and using the continuity equation, this expression can be simplified into the nicer form

$$\rho(\mathbf{x}, t) \frac{\partial}{\partial t} \mathbf{u}(\mathbf{x}, t) = -\rho(\mathbf{x}, t) \mathbf{u}(\mathbf{x}, t) \cdot \nabla \mathbf{u}(\mathbf{x}, t) . \quad (4.2.9)$$

If this momentum is conserved this implies that there are no forces acting on the fluid. So if there is a force-density $\mathbf{f}(\mathbf{x}, t)$ at work then we have

$$\rho(\mathbf{x}, t) \frac{\partial}{\partial t} \mathbf{u}(\mathbf{x}, t) + \rho(\mathbf{x}, t) \mathbf{u}(\mathbf{x}, t) \cdot \nabla \mathbf{u}(\mathbf{x}, t) = \mathbf{f}(\mathbf{x}, t) . \quad (4.2.10)$$

Note that, from the point of view of a frame moving along with the flow this gives

$$\rho(\mathbf{x}, t) \frac{D}{Dt} \mathbf{u}(\mathbf{x}, t) = \mathbf{f}(\mathbf{x}, t) ,$$

just Newton's second law. If we would have applied Newton's second law directly to the momentum-density we would have found the wrong result. Momentum in fluids not only changes due to forces, but it can also flow. When we flow along with the fluid it are only the forces that we see changing the local momentum-density.

Problem 122: Extended Object: Fluids

In chapter 4 we discussed extended objects at length introducing the centre-of-mass and other related variables. If we apply those ideas to fluids we can rederive all of the above equations.

a: Consider a lump of fluid with a volume δV to consist of N molecules with masses m_α , where $\alpha = 1, 2, \dots, N$. Write down the definition of centre-of-mass $\mathbf{X}$, the centre of mass velocity $\mathbf{V}$ and argue why $\rho(\mathbf{x}, t) = \frac{M}{\delta V}$.

b: What is the expression for the momentum-density of this lump of fluid?

c: Compute the expression for the i^{th} component of the momentum-flow in terms of the centre-of-mass velocity and the relative velocities $\mathbf{v}_\alpha$ of the particles? Show that it has an additional term given by

$$\tau_{ij} = \sum_{\alpha=1}^N \frac{m_\alpha}{\delta V} v_{\alpha i} v_{\alpha j} . \quad (4.2.11)$$

d: Show that this gives an additional term in the momentum-conservation equation equal to $\sum_{i,j} \frac{\partial}{\partial x_i} \tau_{ij} \mathbf{e}_j$.

The extra term you have found in the derivation in the previous Problem may disturb you. Apparently the derivation we did earlier for the momentum conservation only takes into account the centre-of-mass momentum. In the next chapter on thermodynamics we will actually be able to compute that this extra term reduces to $-\nabla p(\mathbf{x}, t)$, i.e. to the force caused by pressure gradients. The connection between the classical mechanics of extended systems and fluid dynamics is revealed as soon as we statistically average over the *relative* degrees of freedom of all the particles in the system. The important thing about this is that fluid dynamics is *not* a new branch of physics with distinct physical laws. Its laws follow from the underlying conservation laws of point-particle mechanics.

4.3. Stress

So far we have only discussed deformations, and in the case of liquids the rate of deformations, as physical concepts. The idea is that these notions are connected to the forces responsible for the deformations. As we have seen in previous chapter, forces can often be interpreted as the rate of change of some suitably defined energy in relation to a change in for example position. In the present context this suggests

looking at the rate of change of energy with the change in deformation. For elastic solids this is a good idea, for fluids we will need to adapt this approach a bit. However, that will turn out to be straightforward once we have seen how it works in the case of elasticity.

4.3.1. Stress tensor for elastic bodies

The stress tensor arises from a consideration of the change in energy of the system as a result of the deformation. As we saw this deformation is best described by the deformation tensor, and in fact there is every reason to believe that the separate parts, rotational, compressional and shear deformation, enter into this energy-function in different ways. Yet we will treat them collectively at first.

Let us focus on the entire deformation tensor D , that can be expressed in terms of its components $D_{ij} = \mathbf{e}_i \cdot D[\mathbf{e}_j]$. Note that we have from the definition

$$D_{ij}(t, \mathbf{x}) = \frac{\partial}{\partial x_i} d_j(t, \mathbf{x}) . \quad (4.3.1)$$

We expect the *energy-density* of an elastic body to depend on this tensor, so let $\epsilon(D(\mathbf{x}), \mathbf{x})$ describe the density of energy at $\mathbf{x}$. To obtain the total energy we need to *integrate* this density over the total volume V of the body

$$E_T[D] = \int_V \epsilon(D(\mathbf{x}), \mathbf{x}) dV . \quad (4.3.2)$$

Now the rate of change of this energy should correspond to the work done by the various forces acting on, and inside of, the body. The only thing that can change with time is the deformation, so we write

$$\begin{aligned} \frac{d}{dt} E_T[D] &= \int_V \frac{d}{dt} \epsilon(D(t, \mathbf{x}), \mathbf{x}) dV , \\ &= \int_V \sum_{i,j} \left[\frac{\partial}{\partial D_{ij}} \epsilon(D(t, \mathbf{x}), \mathbf{x}) \right] \frac{d}{dt} \frac{\partial}{\partial x_i} d_j(t, \mathbf{x}) dV , \end{aligned}$$

Now we introduce the shorthand

$$T_{ij}(\mathbf{x}) = \frac{\partial}{\partial D_{ij}} \epsilon(D(t, \mathbf{x}), \mathbf{x}) , \quad (4.3.3)$$

and we call this set of components the *stress tensor*. The meaning of these components is as follows: a small increase in the deformation D_{ij} will lead to a small change in the energy. The *stress* T_{ij} is positive when the energy increases under the deformation. Saying it in short-cut: stress "points in the direction of" energy-increase.

If we insert this in the previous expression, identify the time-derivative of the deformation as the velocity of the constituents during the deformation, and apply a *partial integration* then we get

$$\frac{d}{dt} E_T[D] = - \int_V \sum_{i,j} \left[\frac{\partial}{\partial x_i} T_{ij}(\mathbf{x}) \right] u_j(t, \mathbf{x}) dV .$$

The total amount of work done is the integral of a *work-density*, and this work-density can be recognised as the dot-product of the velocity of the constituents $\mathbf{u}(\mathbf{x})$ at $\mathbf{x}$ with something that must be a *force-density* $\mathbf{f}(\mathbf{x})$,

$$\mathbf{f}(\mathbf{x}) = \sum_{i,j}^3 \frac{\partial}{\partial x_i} T_{ij}(\mathbf{x}) \mathbf{e}_j . \quad (4.3.4)$$

Problem 123: Ordinary potential Energy

Consider a particle moving under the influence of some potential energy $U(\mathbf{x})$.

a: Show that the rate-of-change of the potential energy is given by

$$\frac{d}{dt} U(\mathbf{x}(t)) = -\mathbf{F}(\mathbf{x}(t)) \cdot \mathbf{v}(t) .$$

b: Justify the choice of sign connecting the components of the T with $\mathbf{f}(\mathbf{x})$ from the above example.

What we have found is that the upon the introduction of an energy as a function of the deformation the forces on, and inside of, the body are expressed in terms of the components of yet another tensor, the *stress-tensor*. This is a very powerfull concept of which I can only hope to show a few elementary aspects.

For non-rotational deformations T_{ij} is symmetric

If we assume that there are no rotational deformations, or if we simply completely ignore them, then we can take $D = D_S$, now because for the symmetric deformation tensor we have $D_{ij} = D_{ji}$ we get

$$T_{ij} = T_{ji} . \quad (4.3.5)$$

In that case we say that the stress tensor is symmetric. There are several physical reasons for assuming that this will generally be the case. Just like with the inertia tensor, that was also symmetric, the symmetry guarantees that there are three real *eigenvalues* T_{ii} such that with a suitable choice of coordinates the all $T_{ij} = 0$ if $i \neq j$. These eigen-values are usually called *principal stresses* and getting the tensor in this form ofcourse greatly simplifies any analysis. Note that in that case the force-density reduces to

$$\mathbf{f}(\mathbf{x}) = \sum_{i=1}^3 \frac{\partial}{\partial x_i} T_{ii}(\mathbf{x}) \mathbf{e}_i . \quad (4.3.6)$$

If we split out the trace of this diagonal stress-tensor we get a quantity that describes the change in energy of the system due to compression. Recalling that $D_V < 0$ for compressions, we see that the trace of the tensor T will thus be negative if a compression leads to an energy increase. Now the energy-increase due to volume change is related to the pressure; the pressure is positive if compression leads to energy increase. So we define

$$\begin{aligned} p(\mathbf{x}) &= - \sum_{j=1}^3 T_{jj}(\mathbf{x}) , \\ \sigma_{jj}(\mathbf{x}) &= T_{jj}(\mathbf{x}) + \frac{1}{3} p(\mathbf{x}) , \end{aligned}$$

then one part of the force-density is simply the gradient

$$\mathbf{f}_p(\mathbf{x}) = -\nabla p(\mathbf{x}) . \quad (4.3.7)$$

It is not unexpected that part of the force acting locally is equal to minus the gradient of the pressure! The remaining part of the deformation was the shear deformation. Correspondingly the associated stresses are called *shear stress*. The stress tensor that only contains shear stress is symmetric and traceless.

Forces on and forces inside

With the above expression we can wonder what the total force on the body is going to be. If we try to calculate it something remarkable happens. Let ∂V represent the boundary of the body, then it is possible to prove that

$$\mathbf{F} = \int_V \sum_{i,j} \frac{\partial}{\partial x_i} T_{ij}(\mathbf{x}) \mathbf{e}_j dV = \int_{\partial V} \sum_{i,j} n_i(\mathbf{x}) T_{ij}(\mathbf{x}) \mathbf{e}_j dS ,$$

where the components $n_j(\mathbf{x})$ are the components of the normal unit-vector at $\mathbf{x}$ on the surface ∂V , and dS represents a infinitesimal area-element at $\mathbf{x}$). In words; The total force on a body equals the integral of the stress-tensor across the surface of the body. So to establish the total force on a body we do not need to know exactly what happens inside, it is sufficient if we know what happens on its boundary! As it stands here this should be a bit of a surprise. However, when you realise that part of the force-density was due to the pressure-gradient, it may feel less suspicious that part of the total force on the body is found by integrating the pressure across the body's surface. The more general reason behind all this is the following; if a force is exerted somewhere in the body this causes a change in the momentum and energy of the constituents. However, as energy and momentum are conserved this energy must have come from somewhere. Any momentum and energy that transferred onto a constituent somewhere in the body must have first entered the body through its surface. That is the force that the surface-integral of the stress-tensor records.

Equilibrium and conservative elasticity

If the forces inside of the body are in equilibrium then we now see how we can specify that, namely as

$$\mathbf{Q}(\mathbf{x}, t) + \sum_i \frac{\partial}{\partial x_i} T_{ij}(\mathbf{x}) = 0 . \quad (4.3.8)$$

The forces $\mathbf{Q}(\mathbf{x}, t)$ are external forces acting on the body, the stress-tensor describes the forces due to elasticity. If the elastic forces are conservative this requires that

$$\nabla \wedge \mathbf{f}(\mathbf{x}) = 0 . \quad (4.3.9)$$

Conservative elastic forces are those for which a deformation that after some time returns the system to its original state does not do work in total. Bodies in which some form of deformation remains, for example due to tearing and breaking, have

non-conservative forces at work inside of them. Also bodies whose temperature changes during a deformation display this as a non-conservative mechanical force. If all forms of energy and momentum are properly included, energy is always conserved. This also means that if non-conservative external forces act on the body, the elastic forces maintaining equilibrium obviously need to respond in a non-conservative manner. The stress tensor must therefore be a very flexible concept that allows such behaviour. If it cannot, due to the structure and composition of the body, external non-conservative forces will result in a loss of equilibrium.

Linear materials

For many materials, up to a very good approximation, the relation ship between the Stresses T_{ij} and the deformations D_{ij} is linear, i.e. there exists a linear map $C[...]$ such that

$$T = C[D] , \quad (4.3.10)$$

written out in terms of components this gives the horrible expression

$$T_{ij} = \sum_{mn} C_{ijmn} D_{mn} .$$

In a linear and *isotropic* elastic solid, the stress and deformation tensors can be made diagonal by one and the same choice of basis. In that case we can write

$$\begin{aligned} T_{11} &= (\lambda + 2\mu)D_{11} + \lambda D_{22} + \lambda D_{33} , \\ T_{22} &= \lambda D_{11} + (\lambda + 2\mu)D_{22} + \lambda D_{33} , \\ T_{33} &= \lambda D_{11} + \lambda D_{22} + (\lambda + 2\mu)D_{33} , \end{aligned} \quad (4.3.11)$$

where the constants λ and μ are called the *Lamé constants* of the material. If we invert these relations we obtain the following set

$$\begin{aligned} D_{11} &= \frac{1}{E}T_{11} - \frac{\nu}{E}\lambda T_{22} - \frac{\nu}{E}T_{33} , \\ D_{22} &= -\frac{\nu}{E}T_{11} + \frac{1}{E}T_{22} - \frac{\nu}{E}T_{33} , \\ D_{33} &= -\frac{\nu}{E}T_{11} - \frac{\nu}{E}T_{22} + \frac{1}{E}T_{33} , \end{aligned} \quad (4.3.12)$$

where the constant E is called the *Young modulus* and ν is the *Poisson ratio*. Ofcourse the E and ν can be expressed in terms of the λ and the μ .

Problem 124: Inverting the Equations

Plug the first set of relations between the T_{ii} and the D_{jj} in Mathematica and solve for the D_{jj} to find the expressions of E and ν in terms of λ and μ .

An example with Hooke's Law

Consider a cylindrical body with its central axis in the $\mathbf{e}_1$ -direction, made from a linear, isotropic, elastic material. Now we put it under uni-axial stress, i.e. stress for which only one component is not 0; say $T_{11}(\mathbf{x}) \neq 0$ and $T_{22} = T_{33} = 0$. We do so by pulling it in (opposite) $\mathbf{e}_1$ -directions at both ends of the cylinder. Let one end of the cylinder be, and remain, at $x_1 = 0$.

We can deduce from the first set of equations that

$$D_{22} = D_{33} = -\frac{\lambda + 2\mu}{2\lambda} D_{11} .$$

From the second set of equations we immediately get that $T_{11} = ED_{11}$ which is called Hooke's Law. So we have computed all the components of the deformation in terms of the only non-vanishing stress. So we now know that

$$\begin{aligned} \frac{\partial}{\partial x_i} d_j(\mathbf{x}) &= 0, i \neq j, \\ &= \frac{\lambda + 2\mu}{2\lambda} ET_{11}(\mathbf{x}), i = j = 2, 3, \\ &= ET_{11}(\mathbf{x}), i = j = 1. \end{aligned}$$

If we assume that the body is in equilibrium then we must have

$$0 = \sum_i \frac{\partial}{\partial x_i} T_{ij}(\mathbf{x}) = \frac{\partial}{\partial x_1} T_{11}(\mathbf{x}),$$

and so the stress T_{11} can only depend on x_2 and x_3 , i.e. $T_{11}(x_2, x_3)$. As the equations for the components of $\mathbf{d}(\mathbf{x})$ tell us that the d_1 component only depends on x_1 , the d_2 component only on x_2 we are left with no other choice than to put T_{11} equal to a constant. As a result the deformations become

$$\begin{aligned} d_1 &= ET_{11}x_1, \\ d_2 &= -\frac{\lambda + 2\mu}{2\lambda} ET_{11}x_2, \\ d_3 &= -\frac{\lambda + 2\mu}{2\lambda} ET_{11}x_3. \end{aligned}$$

The deformation in the $\mathbf{e}_1$ direction leads to an dilation of the cylinder. The deformations in the other directions show that it squeezes together as a result of the strain.

4.4. Dynamics in Elastics and Fluids

Now that we have found ways of describing forces, or better *force-densities* we should start to wonder about how we can connect this to Newton's second law. We will do that in this last section of this chapter. But beware; our final result will probably not make you very happy. Because although we will reach our goal in the end we face an equation which by all means could be termed one of the most complicated equations in classical physics. It's properties have still not all been understood, and to a great extent hardly any satisfactory realistic solutions are known. Yet it is the equation that ultimately describes all the physics of extended bodies, including the Earth with its Oceans, Ice and Atmosphere. For some general inspiration check out www.navier-stokes.net.

Let us return to the conservation of momentum equation, but this time we include the appropriate forces. We find in general

$$\rho(\mathbf{x}, t) \frac{\partial}{\partial t} \mathbf{u}(\mathbf{x}, t) + \rho(\mathbf{x}, t) \mathbf{u}(\mathbf{x}, t) \cdot \nabla \mathbf{u}(\mathbf{x}, t) = \sum_{i,j} \frac{\partial}{\partial x_i} T_{ij}(\mathbf{x}) \mathbf{e}_j. \quad (4.4.1)$$

The choice of stress tensor essentially determines the type of system we're describing. In this section we will look at two straightforward choices.

A linear elastic medium

For a linear homogeneous elastic material we would have

$$T_e(\mathbf{x}, t) = C[D_S(\mathbf{x}, t)] ,$$

ignoring any rotational deformations. Assuming that the material is isotropic and that we have chosen a basis for which both tensors are diagonal yields the relations derived earlier between the T_{jj} and the D_{jj} . Using that

$$D_{jj} = \frac{\partial}{\partial x_j} d_j(\mathbf{x}, t) ,$$

we can write for, for example, T_{11}

$$\begin{aligned} T_{11} &= (\lambda + 2\mu) \frac{\partial}{\partial x_1} d_1(\mathbf{x}, t) + \lambda \frac{\partial}{\partial x_2} d_2(\mathbf{x}, t) + \frac{\partial}{\partial x_3} d_3(\mathbf{x}, t) , \\ &= \lambda \nabla \cdot \mathbf{d}(\mathbf{x}, t) + 2\mu \frac{\partial}{\partial x_1} d_1(\mathbf{x}, t) . \end{aligned}$$

As a result we find that

$$\sum_{i,j} \frac{\partial}{\partial x_i} T_{ij} \mathbf{e}_j = \lambda \nabla [\nabla \cdot \mathbf{d}(\mathbf{x}, t)] + 2\mu \sum_i \mathbf{e}_i \frac{\partial^2}{\partial x_i^2} d_i(\mathbf{x}, t) .$$

Hence the equation of motion can be written as

$$\begin{aligned} \rho(\mathbf{x}, t) \frac{\partial}{\partial t} \mathbf{u}(\mathbf{x}, t) + \rho(\mathbf{x}, t) \mathbf{u}(\mathbf{x}, t) \cdot \nabla \mathbf{u}(\mathbf{x}, t) \\ = \lambda \nabla [\nabla \cdot \mathbf{d}(\mathbf{x}, t)] + 2\mu \sum_i \mathbf{e}_i \frac{\partial^2}{\partial x_i^2} d_i(\mathbf{x}, t) . \end{aligned}$$

Let me give an example of the applications of this complicated equation.

Sound and Seismic waves

Consider the situation that the deformations are very small such that any contribution quadratic in the deformation $\mathbf{d}$ can be safely neglected. Furthermore we also assume that any density fluctuations are relatively small compared so that $\rho(\mathbf{x}, t) \approx \rho_0$. The equation of motion for the displacement field can then be simplified into,

$$\rho_0 \frac{\partial^2}{\partial t^2} \mathbf{d}(\mathbf{x}, t) = \lambda \nabla [\nabla \cdot \mathbf{d}(\mathbf{x}, t)] + 2\mu \sum_i \mathbf{e}_i \frac{\partial^2}{\partial x_i^2} d_i(\mathbf{x}, t) .$$

Now note that we have

$$\nabla \nabla \mathbf{d} = \nabla [\nabla \cdot \mathbf{d}] + \nabla [\nabla \wedge \mathbf{d}] ,$$

and when we assume that there are no rotational deformations we can write

$$\rho_0 \frac{\partial^2}{\partial t^2} \mathbf{d}(\mathbf{x}, t) = \lambda \nabla^2 \mathbf{d}(\mathbf{x}, t) + 2\mu \sum_i \mathbf{e}_i \frac{\partial^2}{\partial x_i^2} d_i(\mathbf{x}, t) .$$

This equation predicts the presence of waves in the elastic medium. There are two kinds of waves and they turn out to propagate with different speeds. First of all there are the *transversal waves* for which the deformation $\mathbf{d}$ is perpendicular to the direction $\mathbf{k}$ in which they propagate. If we insert the following trial solution into the equation of motion

$$\begin{aligned}\mathbf{d}(\mathbf{x}, t) &= \mathbf{d}_0 \cos(\omega t - k\mathbf{e}_1 \cdot \mathbf{x}) , \\ \mathbf{d}_0 &= d_2\mathbf{e}_2 + d_3\mathbf{e}_3 ,\end{aligned}$$

then we find

$$\rho_0\omega^2 = \lambda k^2 .$$

This represents a wave of frequency ω that propagates at a speed $v = \sqrt{\frac{\lambda}{\rho_0}}$. If we would consider a *longitudinal wave* where a trial solution may look like

$$\mathbf{d}(\mathbf{x}, t) = d_1\mathbf{e}_1 \cos(\omega t - k\mathbf{e}_1 \cdot \mathbf{x}) ,$$

we obtain, after insertion of the trial solution into the wave-equation,

$$\rho_0\omega^2 = (\lambda + 2\mu)k^2 .$$

These waves propagate with a speed given by $v = \sqrt{\frac{\lambda+2\mu}{\rho_0}}$. So the Lamé constants λ and μ can thus be retrieved from measuring the speed of these waves in the elastic solid. These waves are usually called *p-waves* in seismology. Another interesting wave phenomenon occurs when the deformations becomes divergence-free, i.e. when $[\nabla \cdot \mathbf{d}] = 0$, but when we do allow for rotational deformations, i.e. $[\nabla \wedge \mathbf{d}] \neq 0$. Then we can rederive a wave-equation that has so-called *S-waves* as solutions. They turn out to propagate with a speed $v = \sqrt{\frac{\mu}{\rho_0}}$.

4.4.1. Navier-Stokes and Bernoulli's equation

After having dealt with waves in elastic solids we should now want to do the same for fluids. Maybe it is good to stress here that the terms *fluids* is used for practically everything that is not solid. So a fluid is not necessarily a liquid, a gas may serve as a fluid as well. The following choice of stress tensor

$$T_f(\mathbf{x}, t) = \eta\tau_{ij} - p(\mathbf{x}, t)\delta_{ij} , \quad (4.4.2)$$

where $\delta_{ij} = 1$ if $i = j$ and 0 otherwise, describes a fluid with pressure $p(\mathbf{x}, t)$ and viscosity η . If we assume that the flow is incompressible, and we allow for conservative external forces we obtain the equation

$$\rho(\mathbf{x}, t)\frac{\partial}{\partial t}\mathbf{u}(\mathbf{x}, t) + \rho(\mathbf{x}, t)\mathbf{u}(\mathbf{x}, t) \cdot \nabla\mathbf{u}(\mathbf{x}, t) = -\nabla p(\mathbf{x}, t) + \eta\nabla^2\mathbf{u}(\mathbf{x}, t) - \nabla U(\mathbf{x}) . \quad (4.4.3)$$

This equation is usually referred to as the *Navier Stokes* equation. The functions $U(\mathbf{x})$ represents the potential energy due to external forces. Some authors use that term for the above equation and the continuity equation and the energy-conservation all together. What's in a word. The field of fluid dynamics has so

many applications that the proliferation of different names is almost unavoidable.

Energy conservation in stationary flows

For a stationary flow the velocity $\mathbf{u}$ is time-independent. This does not mean that nothing is moving; that we would call a *static* situation. The Navier-Stokes equation and the continuity equation then reduce to

$$\begin{aligned}\rho(\mathbf{x})\mathbf{u}(\mathbf{x}) \cdot \nabla \mathbf{u}(\mathbf{x}) &= -\nabla p(\mathbf{x}) + \eta \nabla^2 \mathbf{u}(\mathbf{x}) - \nabla U(\mathbf{x}) , \\ \nabla \cdot \rho(\mathbf{x})\mathbf{u}(\mathbf{x}) &= 0 .\end{aligned}$$

The continuity equation for an incompressible fluid then forces the density to be constant along the flow, i.e. $\mathbf{u}(\mathbf{x}) \cdot \nabla \rho(\mathbf{x}) = 0$. As viscosity is a force that dissipates energy, i.e. if we seek for a proper stationary flow viscosity should become negligible. Hence we can make the further simplification

$$\rho(\mathbf{x})\mathbf{u}(\mathbf{x}) \cdot \nabla \mathbf{u}(\mathbf{x}) + \nabla p(\mathbf{x}) + \nabla U(\mathbf{x}) = 0 .$$

Now if we take the dot-product of this expression with the velocity $\mathbf{u}(\mathbf{x})$ we can write the result as

$$\mathbf{u}(\mathbf{x}) \cdot \nabla \left[\frac{1}{2} \rho(\mathbf{x}) \mathbf{u}(\mathbf{x})^2 + p(\mathbf{x}) + U(\mathbf{x}) \right] = 0 .$$

In other words, along the flow the quantity inside of the brackets remains constant. So, if we have two points $\mathbf{x}_1$ and $\mathbf{x}_2$ along a *flow-line*, i.e. along the trajectory of a fluid-element as it flows, we should have

$$\frac{1}{2} \rho(\mathbf{x}_1) \mathbf{u}(\mathbf{x}_1)^2 + p(\mathbf{x}_1) + U(\mathbf{x}_1) = \frac{1}{2} \rho(\mathbf{x}_2) \mathbf{u}(\mathbf{x}_2)^2 + p(\mathbf{x}_2) + U(\mathbf{x}_2) . \quad (4.4.4)$$

This equation expresses the conservation of energy for stationary flows and is known as Bernoulli's equation. You'll find applications of it in the textbook.

4.5. Assignments

Chapter 5

Thermodynamical Systems

Thermodynamics, or the theory of heat, originated in the eighteenth and nineteenth century as a very successful physical theory without a clear idea about what the fundamental constituent *heat* really was. One early theory was the hat was infact a liquid called the *caloric fluid*. However, experiments due to Joule revealed the fact that heat was some form of energy, and that systems exist that may dispose of their internal heat by doing work. The replacement of the unit *calories* by that of *Joules* can be considered a late tribute to this fundamental discovery. We call systems that convert heat to some kind of usefull work *engines*. A historical example is the *steam engine* which converts the heat of steam into motion driving machines. It is no coincidence that the invention of engines and the development of thermodynamics went hand in hand. In view of the major social and cultural impact of industrial revolution it is fair to say that the consequences of the development of thermodynamics for society were huge indeed. Never before had physics and engineering played such a leading role in societal change. This example of the impact of science on society has probably only been surpassed in intensity by the still ongoing technological revolution caused by the joint venture of electrodynamics and quantum mechanics.

Thermodynamical systems and Statistics

Thermodynamics attempts to describe the real world, and realistic phenomena that we encounter in our daily lives. As a result we have to face the fact that these systems consist of very many particles. A typical amount of, say, air to breathe consists of around 10^{23} molecules. A microscopic approach simply must fail due to the large amount of particles involved, even when we consider all these molecules to be rigid (which they are not, ofcourse).

In the previous chapter on deformable bodies we have seen an approach in which the individual consituents were replaced by a description in terms of *field* quantities such as the flow $\mathbf{u}(\mathbf{x}, t)$ or the deformation $\mathbf{d}(\mathbf{x})$. Although we have only scratched the surface of that discipline you may have noticed that some problems remains almost unphraseable in that language. For example, the energy balance in a fluid is determined in part by quantities such as pressure- and the temperature gradients. The problem is that although pressure and temperature are amendable to a field description also, their dynamics are determined by the properties of the underlying microscopic particles. For example, fluid dynamics cannot tell us whether water is liquid, solid or vapour. The phase of water is determined by the external conditions, but also by the forces between the water molecules. In

meteorology it is of the utmost relevance to be able to compute what phase water is in as it flows along with currents in oceans or the atmosphere. So it is needed that the Navier-Stokes equations are amended with so-called *equations of state* that relate, for example, temperature, pressure and density of matter.

But there is more. For example, in the equations of the previous chapters we used quantity like *viscosity*, the bulk modulus or the Young modulus. These so-called *transport coefficients* describe the response of systems to small disturbances from equilibrium. Within the framework of fluid dynamics or elasticity they are *input* and not computable. If physics is to be a fundamental science it should offer a method of understanding where these coefficients come from. This again brings us back to the fundamental constituents and the forces between them.

So for all these various reasons we should now finally turn to the problem of how to understand macroscopic systems in terms of their microscopic constituents. In this chapter we will study various aspects of the *old* theory of thermodynamics from a modern, molecular, viewpoint. Subsequently we will devote a chapter to working out some of the consequences of the fundamental, statistical, microscopic foundations of the theory. The foundations were not uncovered until the very end of the nineteenth century when the concepts of *molecules* returned atomism to physics, soon followed by the oblivious quanta of the quantum revolution. Towards the end of the twentieth century computational power became available to actually perform the calculations necessary to understand *in detail* the macroscopic properties of systems with more than several thousand microscopic constituents.

5.1. Change and Coarsegraining

We have seen in the previous chapters that conservation laws play a pivotal role in understanding the workings of complicated physical systems. Infact, their role in chemistry, biology and Earth sciences cannot be overestimated either. Talking about conservation laws requires us to talk about *changes* in quantities. Now changes are always changes of one quantity with respect to another. Yet in thermodynamic systems with many, very many, particles there are so many variables that can change. Which are the useful and important ones? How can we safely and systematically ignore the unimportant ones?

Suppose we have some physical quantity B depending on a number of variables, say a_1 , a_2 and a_3 . If we want to study the change $dB(a_1, a_2, a_3)$ in the quantity $B(a_1, a_2, a_3)$ due to changes da_1 , da_2 and da_3 , in the variables we can write

$$dB(a_1, a_2, a_3) = da_1 \frac{\partial}{\partial a_1} B(a_1, a_2, a_3) + da_2 \frac{\partial}{\partial a_2} B(a_1, a_2, a_3) + da_3 \frac{\partial}{\partial a_3} B(a_1, a_2, a_3) .$$

In thermodynamical system the largest possible set of variables is that of the positions and velocities of *all* the constituents. So in principle we have many, very many, variables a_j where the index j runs from 1 to possibly a very large number $N_{approx} 10^{24}$. It is expedient to introduce a vectorspace with a very large number of dimensions N , to introduce a set of orthonormal basis vectors $\mathbf{e}_j$, and to introduce the vectors

$$\partial_{\mathbf{a}} = \sum_{j=1}^N \mathbf{e}_j \frac{\partial}{\partial a_j} ,$$

$$\mathbf{da} = \sum_{j=1}^N \mathbf{e}_j da_j .$$

This allows us to write

$$dB(\mathbf{a}) = \mathbf{da} \cdot \partial_{\mathbf{a}} B(\mathbf{a}) , \quad (5.1.1)$$

without us needing to worry about the exact number of variables involved. * It is fairly easy to see that if we have two *scalar* quantities $A(\mathbf{a})$ and $B(\mathbf{a})$ then we find for the change of the sum of both quantities

$$d\{A(\mathbf{a}) + B(\mathbf{a})\} = dA(\mathbf{a}) + dB(\mathbf{a}) , \quad (5.1.2)$$

and also

$$d\{A(\mathbf{a})B(\mathbf{a})\} = B(\mathbf{a})dA(\mathbf{a}) + A(\mathbf{a})dB(\mathbf{a}) . \quad (5.1.3)$$

Problem 125: Differential gymnastics

a: Prove the two equations above.

b: Show that if a scalar quantity $A(\mathbf{a})$ depends on the variables $\mathbf{a}$ through two quantities $T(\mathbf{a})$ and $V(\mathbf{a})$ then we can write

$$dA(T(\mathbf{a}), V(\mathbf{a})) = \frac{\partial}{\partial T} A(T, V) dT(\mathbf{a}) + \frac{\partial}{\partial V} A(T, V) dV(\mathbf{a}) . \quad (5.1.4)$$

We will assume that the microscopic variables $\mathbf{a}$ always completely describe the *state* of the system. We call such a description the *microstate* of the system. However, most of the microstates will resemble each other up to a very large extent. If, for example, we simply swap two molecules in the air in this classroom that does constitute a change in the microstate, but it has no important effect on the physical situation. Thus for most applications it is not necessary, not possible and not desirable to know the exact microstate of the system. Usually we try find a set of variables, such as for example the volume of a system, its temperature, the number of particles in the system, that are good enough despite the fact that they're not exact. The description of a system in terms of those variables is called a *macrostate*.

Thermodynamics attempts at describing the dynamics of macrostates of matter. Statistical mechanics provides the connection between the macrostates and the microstates. By itself thermodynamics suffers from the same disease that also affected fluid dynamics; the appearance of ad hoc constants or quantities that require further explanation. Statistical mechanics provides such explanation and makes the macroscopic properties of matter computable from the microscopic properties of the constituents.

*In case you were wondering in previous chapters why would introduce all this fancy mathematics: well, notice that you have just entered the world of 10^{24} -dimensional spaces.

Problem 126: Micro and Macro

A mole of gas locked in some container typically consists of approximately 10^{24} molecules.

a: Why does the full specification of the state of the gas require us to know about $6 \cdot 10^{24}$ physical quantities to infinite precision.

b: The specification any one of these quantities to some fairly modest accuracy easily requires 100 bytes of computer memory. Estimate the memory capacity necessary to specify the complete microstate of the gas.

c: Explain the role of momentum and collisions in the process by which a gas exerts pressure on the walls of its container.

d: Do we need to know the full state of the gas in order to be able to find the pressure, or can you name a quantity whose average value would in fact be enough?

The transition from microstates to macrostates is sometimes referred to as *coarsegraining*. Until we have a precise prescription how to do this, we can only use it as a concept. Before we go and derive the prescription we first study thermodynamics ignoring the underlying microscopic structures.

5.2. The Laws of Thermodynamics

How many laws does thermodynamics have? Well, a few and there is some disagreement among physicists about the exact number. All agree on the first and the second law, and historically they were arrived at first. But in later times the completeness of these two was questioned, and a *zeroth law* was introduced to make things complete. Some however will still argue that it is superfluous. Nevertheless I will give you the zeroth law as it is so tauntingly obvious to most human beings that they never considered making it a law of nature.

The Zeroth Law

Heat always flows from regions with a high temperature to those with a lower temperature.

Maybe it seems odd to you that something so obvious is made into a law of nature. But it isn't all that easy to establish this law from the other laws of thermodynamics, especially without much of a clue as to what heat really is. I will gladly accept the luxury of taking it in as another law, simply as a pedagogical reminder that sometimes the obvious is embarrassingly complicated.

Problem 127: A Simple System

Consider a set of N balls and two containers A and B . Now each ball can have one of two colours, *red* = -1 and *yellow* = $+1$. We assume all balls are distinguishable. The microstate of a single ball is given by specifying it's colour and the container it is in.

a: Show a single ball can be in any of 4 microstates.

b: Show that a set of N balls can be in any of 4^N microstates.

Now suppose we average the colour of the balls to assign a colour to the containers. If the average of the balls in a container is between -1 and -0.5 we say the container is Red, if the average is between 0.5 and 1 we say the container is Yellow, and otherwise we say the container is Orange. A macrostate is specified by giving the colour of both containers.

c: Show that we have 6 different macrostates.

d: If the chances for any single ball to be red are 50 percent, do you think all macrostates are equally likely?

The First Law

Consider a, possibly very complex, physical system. We assume that this system contains energy, probably in various different kinds. Furthermore we will assume that this system can change its energy only in two ways; either by absorbing or releasing a small amount of heat dQ , or by doing a small amount of work dW . Energy conservation then tells us that

$$dE(\mathbf{a}) = dQ(\mathbf{a}) + dW(\mathbf{a}) , \quad (5.2.1)$$

and this is the First Law of Thermodynamics. We will assume that the energy of a system is a function of the variables $\mathbf{a}$ which is such that to every *microstate* we can associate one energy $E(\mathbf{a})$. This implies that the change $dE(\mathbf{a})$ can be written as

$$dE(\mathbf{a}) = d\mathbf{a} \cdot \partial_{\mathbf{a}} E(\mathbf{a}) . \quad (5.2.2)$$

If the change in some variable da_j leads to a change in energy then the work the system does in this process is not necessarily described by a change in some other function. To understand what this means consider the situation in classical mechanics: not every change in the energy of an object is the result of a change in potential or kinetic energies. Sometimes there are *forces that do work* that are not derivable from some underlying potential energy, i.e. non-conservative forces. This can ofcourse also occur on the microscopic level, so we will write in general

$$dW(\mathbf{a}) = -\mathbf{F}(\mathbf{a}) \cdot d\mathbf{a} , \quad (5.2.3)$$

and we call the quantity $\mathbf{F}(\mathbf{a})$ a *generalised force*. A change in the volume of a system will induce many small changes in many of the variables in $\mathbf{a}$ and All these changes are part of the vector $d\mathbf{a}$ and as a result the system will respond with a complicated large number of forces all contained in $\mathbf{F}(\mathbf{a})$. Together these forces give rise to the *pressure*. Now the vector that corresponds to volume-changes would be $\partial_{\mathbf{a}} V(\mathbf{a})$, so the magnitude of the change in $d\mathbf{a}$ that results from the volume change is that part of $d\mathbf{a}$ that is in the direction of $\partial_{\mathbf{a}} V(\mathbf{a})$. So with the

coarse-graining concept in mind we could write

$$\begin{aligned} dV(\mathbf{a}) &= \mathbf{da} \cdot \partial_{\mathbf{a}} \widehat{V}(\mathbf{a}) , \\ p(\mathbf{a}) &= \mathbf{F}(\mathbf{a}) \cdot \partial_{\mathbf{a}} \widehat{V}(\mathbf{a}) . \end{aligned} \quad (5.2.4)$$

So, the result of the work done due to an infinitesimal volume change can be expressed as

$$dW(\mathbf{a}) = -p(\mathbf{a})dV(\mathbf{a}) . \quad (5.2.5)$$

So far this is still an *exact* and microscopic expression. It becomes coarsegrained as soon as we ignore the underlying dependence on the microscopic variables $\mathbf{a}$. Recall that when the volume change is a compression that $dV(\mathbf{a}) < 0$.

Not all changes in energy occur due to work. If we add or subtract particles from the system the energy will also change. Let $N_j(\mathbf{a})$ denote the number of particle of type j , and let $\mu(\mathbf{a})_j$ denote the energy increase of the system if we add a single particle of type j . Then a change dN_k will lead to a change in energy given by

$$dW(\mathbf{a}) = \sum_j \mu_j(\mathbf{a})dN_j(\mathbf{a}) , \quad (5.2.6)$$

which is sometimes referred to as *chemical work*. The function $\mu(\mathbf{a})_j$ is called the *chemical potential* of the particles of type j . The first law including chemical work reads

$$dE(\mathbf{a}) = dQ(\mathbf{a}) - p(\mathbf{a})dV(\mathbf{a}) + \sum_j \mu_j(\mathbf{a})dN_j(\mathbf{a}) . \quad (5.2.7)$$

Problem 128: Work and Volume change

In many cases of interest the systems we are dealing with are gases. As you know gases exert a pressure on the walls of the containers that hold them. Consider a cube of which one side-panel can be displaced inwards and outwards.

a: If p is the pressure of the gas inside the cube and A is the surface area of the movable panel. Explain why the total force on the panel is given by $F = pA$.

b: If we displace the panel by an infinitesimal amount dx inwards, then show that the total change in volume is $dV = Adx$.

c: If we assume the displacement is so small that the pressure does not change. Then show that the infinitesimal amount of work dW that we must do is given by $dW = -pdV$.

The remaining source of energy change, the heat-exchange $dQ(\mathbf{a})$, remains elusive to us for now. Intuitively we ofcourse know what it means. But how to exactly put this in terms states or forces we do not know yet. For this we will need the *second law of thermodynamics* that will relate $dQ(\mathbf{a})$ to the change in a new quantity called *Entropy*.

5.3. Entropy and Heat Engines

Before we can uncover the second law of thermodynamics we must get a bit more familiar with the first law. Suppose we have some engine that steadily runs through a cycle of processes. We call it a *heat engine* because it takes in energy in

the form of heat. If you are thinking of, say, a jet engine taking in kerosine then the heat-engine within the jet-engine is the thing that takes in the heat from the burned kerosine. In physics we don't care about aircraft design, nor the design of aircraft engines. So don't take this word too literal.

A typical heat engine absorbs an amount of heat Q_1 from some reservoir with a high-temperature, then does an amount of work W , and dumps whatever heat Q_2 is left into a cooler reservoir. We call such an engine *reversible* if its mode of operation can be reversed.

Problem 129: Engine terminology

- a: Identify the three elements in the above process in the way a steam engine works. Or if you find this too outdated, try to do this for a normal car engine.
- b: What part of the steam engine is the actual heat-engine and can it work in reversed mode?
- c: Can a full jet-engine work in reversed mode?

After completing a full cycle the engine is back in its initial state, i.e. all state-variables have returned to their original values. As a result the energy of the engine after the cycle has been completed has not changed, i.e. $dE(\mathbf{a}) = 0$. So energy conservation tells us that

$$Q_1 - (W + Q_2) = 0 .$$

Now we define the *efficiency* η of the engine as

$$\eta = \frac{W}{Q_1} . \quad (5.3.1)$$

Suppose that we use the work W to drive another reversible heat engine with an efficiency η' working between the same two reservoirs, but now in the reversed mode: it takes an amount of heat Q'_2 from the cool reservoir, absorbs the work W to raise the amount of heat to Q'_1 which it then dumps into the hot reservoir. Let the efficiency of this engine be η' . Can $\eta' \neq \eta$? Note that we have

$$Q'_1 = \frac{W}{\eta'} = Q_1 \frac{\eta}{\eta'} .$$

If $\eta > \eta'$ then we would have constructed, by combining the two engines, a heat engine that dumps more heat into the first reservoir than it absorbs from it, i.e. it transfers a net amount of heat from the colder second reservoir to the hotter first reservoir. This violates the zeroth law and hence must be ruled out. The only other option is that the engine actually *creates* energy which is ruled out by the first law. So we must conclude that for the two engines $\eta = \eta'$. The efficiencies of reversible heat engines working between the same set of reservoirs is identical.

Problem 130: Efficiency

- a: Show that if in the text we would have taken $\eta' < \eta$ reversing both engines again leads to a conflict with the zeroth law.
- b: Argue that efficiency is never larger than 1.
- c: Argue that for normal, less than perfect, refrigerators this means that they dump more heat into the kitchen than they extract from the content of the fridge.
- d: Argue why the temperature in your kitchen will increase if you leave the fridge open for a long time.

Absolute Temperature

Let us again consider some heat-engine parable. This time the heat-engine working between two heat-reservoirs is seconded by a second engine which comprises a two-step engine. In the first step a quantity of heat Q'_1 is taken from the first reservoir, used to do work and a quantity of heat Q_0 is dumped into a second heat-engine. The second engine also does work and dumps whatever heat Q'_2 is left into the cool reservoir. Now suppose you have chosen some temperature scale, like Celcius or Fahrenheit, or whatever you like.

The first reservoir is at a temperature t_1 , the second reservoir is at a temperature t_2 and the heat that is exchanged inside of the second engine is set at a temperature t_0 . We can expect that the ratios of exchanged heat are some function of the two temperatures involved. For the first engine we have

$$\frac{Q_2}{Q_1} \equiv R(t_2, t_1) = 1 - \eta .$$

Due to the fact that engines working between equal heat reservoirs must have equal efficiencies we also must have

$$R(t_2, t_1) = \frac{Q'_2}{Q'_1} = \frac{Q'_2 Q_0}{Q_0 Q'_1} = R(t_2, t_0) R(t_0, t_1) .$$

For these ratios to satisfy this product rule in all generality there should exist some function $f(t)$ such that

$$R(t_2, t_1) = \frac{f(t_2)}{f(t_1)} .$$

This function of temperature we call *the absolute temperature* T . As we are never able to extract more heat from an object than that object possesses we automatically have that T is always larger than or equal to 0.

To get a better understanding of heat-exchange it is extremely usefull to define a quantity $dS(\mathbf{a})$ called the *entropy*-change under heat-exchanges by a small amount $dQ(\mathbf{a})$ given by

$$dS(\mathbf{a}) = \frac{dQ(\mathbf{a})}{T(\mathbf{a})} . \quad (5.3.2)$$

Now we should wonder whether this change $dS(\mathbf{a})$ can be related to the change of some specific quantity $S(\mathbf{a})$. An engine consideration will help us to answer that question.

For the engine discussed previously we see that at the beginning of its cycle it picks up a certain amount of Entropy, and it disposes of a second amount if

entropy just before the end of its cycle. We just saw that we have an absolute temperature T which is such that

$$\frac{Q_2}{Q_1} = \frac{T_2}{T_1} .$$

Using this property of the absolute temperature we can compute the entropy change of the engine. As the engine returned to its initial state after completion of the cycle the total entropy change is

$$dS = \frac{Q_1}{T_1} - \frac{Q_2}{T_2} = 0 ,$$

i.e. it has not changed at all. This strongly supports the idea that the Entropy of a system only depends on the state of the system. In other words, we postulate the existence of a quantity called the *entropy* of the system $S(\mathbf{a})$. for which

$$d\mathbf{a} \cdot \partial_{\mathbf{a}} S(\mathbf{a}) = \frac{dQ(\mathbf{a})}{T(\mathbf{a})} .$$

Its relevance follows from the following, the second law of thermodynamics.

Second law of Thermodynamics

The second law of thermodynamics states that for an isolated system that evolves only under the influence of internal processes the change in Entropy is greater than or equal to 0.

There has been, and still is much debate about this law. Many see in it the explanation for the *arrow of time*, i.e. for our perception that time always runs from the past to the future. Still others claim that this law offers no explanation, but rather requires more explanation. The fundamental issue here is that this law seems to imply that nature works *one-way* whereas Newton's second law works both ways. During the times that the microscopic nature of macroscopic systems was ill-understood this did not mean very much as there was no reason to believe that Newton's second law holds for this odd caloric matter called heat. However this changed drastically in the second half of the nineteenth century. Not only was Newtonian physics seen to be applicable to molecules and alike, thermodynamics even could be derived from it as a statistical theory.

Order and Disorder

An increase in Entropy is usually associated with a decrease in order. In the next chapter we will have the change to make these notions more precise. For now let's work with an example: if snowflakes, with their perfectly ordered arrangements of water molecules, are put into cold but very dry air they spontaneously turn to watervapor. This change is allowed by the second law as it is entropy increasing. But very visually it also increases disorder.

Breaking the Glass

The second law makes many predictions that seem totally obvious. For example, consider a table with a glass on one edge. The second law asserts that the

evolution in which the glass is disturbed by some random, local, fluctuation in air-pressure, falls and breaks is perfectly okay. The shattered glass represents a state of higher entropy. Because of that, the same second law also says that the reverse process will not happen. And I guess none of us have ever seen the remnants of a glass jump together and reform into a glass. So the second law just seems to be common sense. The statistical approach will reveal a slightly different view. The reverse process is allowed but it can be shown to be extremely unlikely. Hence, we would need to wait an incredibly long time, much longer than the current age of the universe, to see it occur. Effectively this seems to banish entropy decreasing phenomena from our visible universe. This approach to the second law makes it an *average* law, or a statistical law which is not fundamentally true but only in a probabilistic sense. Though this is a very pragmatic and useful view, to many it is not entirely satisfactory. It solves a problem of principles by turning it into a matter of probabilities. At the same time it raises the question *why* the universe started out in a state of low entropy, i.e. an extremely unlikely state?

Equilibrium

With the second law at our disposal we can formulate an important concept. We say that a system is *in equilibrium* when the entropy has attained its maximum value. If that has happened all evolution will come to a halt. This notion will be of great use in the subsequent developments, but also in making the connection with statistical arguments.

5.4. Thermodynamic potentials

We have, until now, discussed two functions that depend on the state of a system; the energy of a system $E(\mathbf{a})$ and its entropy $S(\mathbf{a})$. Thermodynamics discusses these things in terms of coarse-grained variables. So let us make that transition now. We first rewrite the first law in terms of the change in Entropy and the mechanical work done by pressure,

$$dE(\mathbf{a}) = T(\mathbf{a})dS(\mathbf{a}) - p(\mathbf{a})dV(\mathbf{a}) + \sum_j \mu_j(\mathbf{a})dN_j(\mathbf{a}) .$$

Then we stare at this equation for a few minutes and make a *coarse-graining assumption*:

$$\begin{aligned} E(\mathbf{a}) &= E(S(\mathbf{a}), V(\mathbf{a}), N(\mathbf{a})) , \\ T(\mathbf{a}) &= T(S(\mathbf{a}), V(\mathbf{a}), N(\mathbf{a})) , \\ p(\mathbf{a}) &= p(S(\mathbf{a}), V(\mathbf{a}), N(\mathbf{a})) . \end{aligned} \tag{5.4.1}$$

In other words we assume that all appreciable and important changes in the energy occur due to changes in temperature, the number of particles and the changes in volume. As soon as we have a precise and well-defined method of coarse-graining we can check whether this assumption is any good. As a result of this assumption we can rewrite the first law as

$$dE(S, V, N) = T(S, V, N)dS - p(S, V, N)dV + \mu(S, V, N)dN , \tag{5.4.2}$$

where for simplicity we have also assumed there is only one species of particles. So the assumption results in a notion of energy that is a function of volume V ,

Entropy S and the number of available particles N , i.e. $E(S, V, N)$. Quantities like temperature, pressure and chemical potential result from taking derivatives

$$\begin{aligned}\frac{\partial}{\partial S}E(S, V, N) &= T(S, V, N) , \\ \frac{\partial}{\partial V}E(S, V, N) &= -p(S, V, N) ,\end{aligned}\tag{5.4.3}$$

$$\frac{\partial}{\partial N}E(S, V, N) = \mu(S, V, N) .\tag{5.4.4}$$

As a result of this we can derive a number of useful relations that can be tested experimentally such as for example

$$\frac{\partial}{\partial V}T(S, V, N) = -\frac{\partial}{\partial S}p(S, V, N) .$$

These kind of relations are called *Maxwell relations*.

Problem 131: Maxwell Relations

Derive the Maxwell relation given above and find at least one more Maxwell relation.

Although you may have a good feeling about the physical nature of energy, volume and particle numbers I am quite sure you are less familiar with Entropy. So it may seem reasonable to suggest that we could also describe the system with Energy as a variable, rather than Entropy. If we go back to the first law we see that we could have also written it as

$$dS = \frac{1}{T}dE + \frac{p}{T}dV - \frac{\mu}{T}dN .\tag{5.4.5}$$

This suggests that we can interpret the entropy as a function of energy E , volume V and particle number N , i.e. $S(E, V, N)$. Note that in that case we should also use $T(E, V, N)$, $p(E, V, N)$ and $\mu(E, V, N)$. Can we just simply swap variables and functions like that? The answer is that we are indeed allowed to do that as long as the relationship between entropy and energy is invertible. A sufficient condition for this is that the derivative

$$\frac{\partial}{\partial S}E(S, V, N) = T(S, V, N) \neq 0 .$$

Under most circumstances this requirement is satisfied.

Now suppose that we are studying a large thermodynamical system, A, that contains a small sub-system, A', and suppose that the two can only exchange heat, i.e. the volume of both is fixed so no work can be done and no particles can be exchanged between the two. We assume that both systems are in equilibrium with one another, i.e. the total entropy has obtained a maximum value.

Problem 132: Equilibrium and equal Temperatures

We could wonder whether the temperatures in A and A' are equal. Can you find an answer?

- a: Assume that the two systems have different temperatures T and T' . Calculate the change in total entropy due to the exchange of a small amount of heat dQ between the two systems.
 b: Show that the second law requires the heat to flow from the hot to the cold system.
 c: Show that the requirement of equilibrium can only be satisfied if both systems have the same temperature.

Now suppose the smaller system absorbs an amount of heat dQ from the large system. For the total change in entropy we have

$$0 = dS_{A+A'}(\mathbf{a}, \mathbf{a}') = dS_A(\mathbf{a}) + dS_{A'}(\mathbf{a}') = \frac{1}{T(\mathbf{a}')} \{-dE_{A'}(\mathbf{a}') + T(\mathbf{a}')dS_{A'}(\mathbf{a}')\} .$$

In other words we can define a new quantity, $F_{A'}(T, V, N)$, the so-called Helmholtz free-energy of the system A'

$$F_{A'}(T, V, N) \equiv E_{A'}(S, V, N) - TS_{A'}(E, V, N) .$$

For this free energy we find that it reaches a *minimum* value when a system with fixed volume and fixed number of particles is in equilibrium at some constant temperature. The isolated system $A + A'$ must satisfy $dS_{A+A'} \geq 0$ and as a result we get $dF_{A'} \leq 0$. So in general we define the new thermodynamical potential, the *Helmholtz free energy* $F(T, V, N)$ as,

$$F(T, V, N) = E(S, V, N) - TS(E, V, N) . \quad (5.4.6)$$

Note that if we formally differentiate $F(T, V, N)$ with respect to E we get

$$\frac{\partial}{\partial E} F(T, V, N) = 1 - T \frac{\partial}{\partial E} S(E, V, N) = 1 - 1 = 0 ,$$

where we've identified T with the appropriate temperature. This result is consistent with the fact that we treat F as depending only on T , V and N . You may check for yourself that differentiating with respect to S is also consistent. If we write out what the differential dF of the free energy is, and use the first law, we find

$$dF = -SdT - pdV + \mu dN . \quad (5.4.7)$$

This procedure of going from one function $E(S, V, N)$ to another $F(T, V, N)$ by trading the dependence on S for a dependence on the associated variable T is called a *Legendre transformation*. These Legendre transformations play an important role in many branches of physics.

Problem 133: Free energy and Enthalpy

- a: Derive the form of dF given in the text.
 b: Consider the following function $H = E + PV$ and show that its differential satisfies $dH = TdS + Vdp + \mu dN$. This H is called the *Enthalpy*.
 c: Argue that $H(S, p, N)$ reaches an extremal value for systems with a fixed number of particles and at a constant pressure.
 d: Study the so-called Gibbs free energy $G = H - TS$.

Specific Heat and Heat-Capacity

Obviously the amount of heat stored in a certain amount of matter does not only depend on its temperature but also on the actual amount of matter, and the type of matter involved. To describe this it is useful to define the so-called *specific heat* of a material. We call specific heat at fixed volume c_V : the amount of heat dQ that needs to be transferred to a material *per kilogram* to raise the temperature by one degree Kelvin dT while the volume remains fixed. We want to keep the volume fixed to ensure that the system does not do any work. Hence, if we want to increase the temperature of matter with a density ρ in a fixed volume V by ΔT then the amount of heat we must transfer is given by

$$\Delta Q = \rho V c_V \Delta T . \quad (5.4.8)$$

The number $C_V = \rho V c_V$ is often called the *heat-capacity* of the system. Often the specific heat is a function of the temperature, i.e. $c_v(T)$. If we now raise the temperature of a system, with a fixed volume, we also increase its entropy. From

$$dS = \frac{dQ}{T} = \frac{\rho V c_V(T)}{T} dT ,$$

we get

$$S(T_2) - S(T_1) = \int_{T_1}^{T_2} \frac{\rho V c_V(T)}{T} dT . \quad (5.4.9)$$

Once we are able to compute specific heats, we can also find the entropy changes.

Next to the specific heat at constant volume there is also a specific heat at constant pressure c_p . In this case, as the volume is allowed to change, work can be done. So normally $c_p \geq c_V$.

Problem 134: Specific heats

Give arguments why you would expect that $c_p \geq c_V$.

5.5. Statistical Physics

In the previous chapter you have seen how thermodynamics works on the basis of three laws and that the illusive quantity *Entropy* plays a very fundamental role. Obviously it is then disappointing that we seem to know so little about what Entropy actually is. Furthermore we have formulated thermodynamics essentially without making any referral to the microscopic structure of matter. Therefore it seems reasonable to suspect that we can make a lot of progress if we somehow manage to find a relation between macrostate and microstates. Said differently: how does coarsegraining work?

5.5.1. Collisions and Equilibrium

When thinking about the words *statistical physics* it is tempting to consider the origin of the *statistics* to be non-physical. Statistics is a branch of mathematics, you may think, so that is where we should look for the right rules to find out how to statistically describe nature. Partially that is indeed the case, but there is a

very strong physics component in the statistics we encounter in nature. In this section we will derive the statistical rules for physical systems from a consideration of conservation laws. Before we get there we will need a tiny bit of probability theory.

Consider some physical system that is known to be able to exist in a large number, N , of different microstates ψ_n . If we know the *probability* $P(\psi_n)$ of the microstate ψ_n this obviously tells us how likely it is, in an absolute sense, that we will find the physical system in that state. Because the physical system must always be in one of the states ψ_n we must require that

$$\sum_{n=1}^N P(\psi_n) = 1 . \quad (5.5.1)$$

Now suppose that when the physical system is in a state ψ_j it has a total energy $E(\psi_j)$. Once we know the probability for each microstate we can compute an average energy $\langle E \rangle$ given by

$$\langle E \rangle = \sum_{n=1}^N E(\psi_n) P(\psi_n) . \quad (5.5.2)$$

We can calculate in the same way the average $\langle \kappa \rangle$ for any other property $\kappa(\psi_j)$ of the system of which we know how it depends on the choice of the microstate ψ_j . The mean value is what we would expect to find when we perform a measurement of $\kappa(\psi_j)$ on many copies of the same physical system and subsequently average all the results. The average does not tell us anything about the mean *deviation* from the average. If we compute the average of $E - \langle E \rangle$ we find

$$\langle (E - \langle E \rangle) \rangle = 0 . \quad (5.5.3)$$

Problem 135: The average deviation vanishes

Show that the average deviation vanishes.

As a result this is not a good measure for the width of the probability density. Usually one uses the standard deviation σ , for example for the energy,

$$\sigma(E) = \sqrt{\langle (E - \langle E \rangle)^2 \rangle} . \quad (5.5.4)$$

Problem 136: Standard Deviation and RMS

a: Show that $\langle (E - \langle E \rangle)^2 \rangle = \langle E^2 \rangle - \langle E \rangle^2$.

b: Why is $\sigma(E)$ sometimes called the *root-mean-squared* deviation of the energy?

Finally there is one more concept which is often very useful, namely that of *correlation*. If two quantities, say κ_1 and κ_2 , are said to be correlated, then we mean that the values that κ_1 can take are not fully independent of the values taken

by κ_2 and vice versa. An average quantity which measures this type of correlation can be defined through the function $C(\kappa_1, \kappa_2)$ given by

$$C(\kappa_1, \kappa_2) = \langle (\kappa_1 \kappa_2 - \langle \kappa_1 \rangle \langle \kappa_2 \rangle) \rangle . \quad (5.5.5)$$

Problem 137: Vanishing Correlation and independence

a: Give an argument based on the definition of the function $C(\kappa_1, \kappa_2)$ that shows that when the quantities κ_1 and κ_2 are not correlated then $C(\kappa_1, \kappa_2) = 0$.

Consider a probability density $P(\kappa_1, \kappa_2)$ that only depends on the values the two quantities take in a given microstate.

b: Show that if this probability density is of the form $P(\kappa_1, \kappa_2) = f_1(\kappa_1)f_2(\kappa_2)$ for two probability densities f_j then $\langle \kappa_j \rangle = \sum_{\kappa_j} \kappa_j f_j(\kappa_j)$, where the sum is over all the values that κ_j can take.

c: Show that in this case we have $C(\kappa_1, \kappa_2) = 0$.

In the previous Problem you will have shown an important property of probability distributions; if the probability-densities of quantities factorize then the quantities are not correlated.

Now often we do not really know the probability but we know a statistical weight $w(\psi_n)$ of the microstates. Weights do not have to sum to 1 like probabilities must; but that is the only difference between them. So the probability is always proportional to the weight. In fact if we define the sum of all the weights Z as

$$Z = \sum_{n=1}^N w(\psi_n) , \quad (5.5.6)$$

then we can write the probability for the state ψ_j to occur as

$$p(\psi_j) = \frac{w(\psi_j)}{Z} . \quad (5.5.7)$$

The sum of all the weight, Z , is usually called the *partition function* of the physical system. It will turn out to contain all the thermodynamical information about the physical system.

Now let us determine the weights for the microstates of some physical system. As an explicit example we will consider a gas of N essentially free, physically identical but distinguishable, particles. Furthermore we assume the system is in some form of *equilibrium*, i.e. its macroscopic properties do not change with time. We will assume that these particles can engage in very brief contact-interactions called *collisions*. During collisions they are allowed to exchange energy, momentum and angular momentum while obeying the usual conservation laws. The weight of a particular microstate of the system can depend on the positions and momenta of all the particles in the system; $w(\mathbf{x}_1, \mathbf{p}_1, \dots, \mathbf{x}_N, \mathbf{p}_N)$. For the sake of simplicity we will assume that weight factorizes, i.e.

$$w(\mathbf{x}_1, \mathbf{p}_1, \dots, \mathbf{x}_N, \mathbf{p}_N) = f(\mathbf{x}_1, \mathbf{p}_1) \dots f(\mathbf{x}_N, \mathbf{p}_N) . \quad (5.5.8)$$

We have also assumed that all particles, as they are physically identical, have the same single-particle weightfunction $f(\mathbf{x}, \mathbf{p})$ depending on their position and momentum.

Equilibrium

Now collisions are constantly happening and as a result the system is continuously changing its microstate as the particles going into a collision will typically come out of it with different energies and momenta. Assuming it is still in equilibrium does not require us to forbid collisions, rather it requires that the total weight of the microstates *before* and *after* the collisions is equal. Consider two particles with momenta $\mathbf{p}_{a1}$ and $\mathbf{p}_{b1}$. As we will assume that they are point-particles the collision takes place at a single point $\mathbf{x}$. After the collision the two particles have momenta $\mathbf{p}_{a2}$ and $\mathbf{p}_{b2}$. If the system is in equilibrium, the contributions of these two particles to the total weight of the microstate before and after the collision must be equal, i.e.

$$f(\mathbf{x}, \mathbf{p}_{a1})f(\mathbf{x}, \mathbf{p}_{b1}) = f(\mathbf{x}, \mathbf{p}_{a2})f(\mathbf{x}, \mathbf{p}_{b2}) .$$

Note that when we take the natural logarithm on both sides this gives us

$$\log(f(\mathbf{x}, \mathbf{p}_{a1})) + \log(f(\mathbf{x}, \mathbf{p}_{b1})) = \log(f(\mathbf{x}, \mathbf{p}_{a2})) + \log(f(\mathbf{x}, \mathbf{p}_{b2})) .$$

The quantity $\log(f(\mathbf{x}, \mathbf{p}))$ is thus apparently a *additively* conserved quantity in collisions. If we make a pessimistic guess then we could consider only energy E to be conserved in collisions. In that case we would have

$$\log(f(\mathbf{x}, \mathbf{p})) = \alpha - \beta E , \quad (5.5.9)$$

where the numbers α and β are the same for all particles. If these numbers depend on $\mathbf{x}$ we say the system is in *local equilibrium*. If they are constants we talk about *global equilibrium*. So we find for the weight contributed by a single, free, particle with the energy E_j to the total system

$$f(\mathbf{x}, \mathbf{p}) = \exp(\alpha - \beta E_j) . \quad (5.5.10)$$

Problem 138: Chemical potential

Show that if we introduce a new parameter μ such that $\alpha = \beta\mu$ then μ must have the units of an energy.

This constant μ can be interpreted as the energy that it takes to add a particle to the system. It is usually called the *chemical potential*.

As a result we find for the total weight of the microstate of N particles

$$\begin{aligned} w(\psi) &= \exp(\alpha - \beta E_1) \dots \exp(\alpha - \beta E_N) , \\ &= \exp(\alpha N - \beta \sum_{j=1}^N E_j) = \exp(\alpha - \beta E(\psi)) . \end{aligned} \quad (5.5.11)$$

This weight is called the Boltzmann weight of the microstate. Although our derivation of it may seem very particular to one special system, free particles, this is not

the case. If you go through the concepts we used carefully you'll see that it allows for a lot of room for changes while still giving the same final result.

Problem 139: A more general Weight

In the derivation of the Boltzmann weight we assumed only energy was conserved in collisions. But ofcourse in most cases so is momentum.

a: Show that momentum conservation allows us to write $\log(f(\mathbf{x}, \mathbf{p})) = \beta\mu - \beta E + \beta \mathbf{u} \cdot \mathbf{p}$, where the vector $\mathbf{u}$ is the same for all particles.

b: Assume that $\mu = 0$ and show that if E is the standard kinetic energy $E = \frac{\mathbf{p}^2}{2m}$ then the weight is maximal for a momentum $\mathbf{p} = m\mathbf{u}$.

c: Give an argument based on your results under b: that suggests that $\mathbf{u}$ is simply the average velocity of the N particles.

5.5.2. The Partition function and averages

Now that we have the weight of a microstate at our disposal we can turn to finding the probability of each microstate. As we already discussed, the first step in this process is determining the partition function. So we define

$$Z[\beta, \mu, V] = \sum_{\psi} \exp(\beta\mu N(\psi) - \beta E(\psi)) , \quad (5.5.12)$$

where a sum over all allowed microstates is implied, and where $N(\psi)$ is the number of particles in the microstate ψ . Suppose we are able to actually compute the sum, then the probability of the microstate ψ would be given by

$$P(\psi) = \frac{\exp(\beta\mu N(\psi) - \beta E(\psi))}{Z[\beta, \mu, V]} . \quad (5.5.13)$$

We can use this to derive a very interesting and compact formula for the average energy $\langle E \rangle$. Note that

$$\begin{aligned} \langle E \rangle &= \sum_{\psi} \frac{E(\psi) \exp(\beta\mu N(\psi) - \beta E(\psi))}{Z[\beta, \mu, V]} , \\ &= \sum_{\psi} - \frac{\frac{\partial}{\partial \beta} \exp(\beta\mu N(\psi) - \beta E(\psi))}{Z[\beta, \mu, V]} , \\ &= - \frac{1}{Z[\beta, \mu, V]} \frac{\partial}{\partial \beta} Z[\beta, \mu, V] , \\ &= - \frac{\partial}{\partial \beta} \log(Z[\beta, \mu, V]) . \end{aligned} \quad (5.5.14)$$

The average total energy thus simply is a derivative with respect to $\log(Z)$. How is this with other quantities?

Generalised Forces

Recall that in the chapter on thermodynamics we defined generalised forces as the derivatives of the total energy with respect to some external variable. Let us see how we can formulate this in the statistical context.

Suppose that the energies of the allowed microstates depend on a number of external variables x_j . Then we can define the corresponding *generalised force* F_j through

$$F_j \equiv -\frac{\partial}{\partial x_j} \langle E \rangle, \quad (5.5.15)$$

$$\begin{aligned} &= -\frac{\partial}{\partial x_j} \sum_{\psi} \frac{E(\psi) \exp(\beta \mu N(\psi) - \beta E(\psi))}{Z[\beta, \mu, V]}, \\ &= \sum_{\psi} \frac{\frac{1}{\beta} \frac{\partial}{\partial x_j} \exp(\beta \mu N(\psi) - \beta E(\psi))}{Z[\beta, \mu, V]}, \\ &= \frac{\partial}{\partial x_j} \frac{1}{\beta} \log(Z[\beta, \mu, V]). \end{aligned} \quad (5.5.16)$$

Again we see that the logarithm of the partition function play an important role, but now in the combination $\frac{1}{\beta} \log(Z)$.

Problem 140: Generalised Forces derived

The derivation above contains a few steps that weren't written out explicitly. Makes this derivation complete.

Before we investigate what this function means let us first write down one special version of the above equations which will be useful later. Probably the most important generalised force is the *pressure*, p , of the gas. We have

$$p(\beta, \mu, V) \equiv \frac{\partial}{\partial V} \frac{1}{\beta} \log(Z[\beta, \mu, V]). \quad (5.5.17)$$

Problem 141: Particle numbers

Consider a system where the number of particles N is allowed to vary from one microstate to another. Use the partition function to show that

$$\langle N \rangle = \frac{\partial}{\partial \mu} \frac{1}{\beta} \log(Z[\beta, \mu, V]). \quad (5.5.18)$$

We see from these examples that the logarithm of the partition function is a very interesting quantity. Does it also have a direct physical meaning?

The partition function sum was written as a sum over all microstates ψ . But suppose that we would have in our possession a function $\Omega(E)$ that specifies the *number of microstates* corresponding to a certain total energy E . Then the partition function sum could be written as, taking $\mu = 0$ for simplicity,

$$Z = \int \Omega(E) \exp(-\beta E) dE. \quad (5.5.19)$$

Now typically the number of accesible states $\Omega(E)$ will be a strongly increasing function of E , whereas the Boltzman weight is a strongly decreasing function of E .

Problem 142: Number of states at fixed Energy

Consider a system of 1 free particle, i.e. the particle only has kinetic energy.

- a: Show that if E is fixed then the allowed momenta $\mathbf{p}$ lie on a 3d sphere with a radius $\sqrt{2mE}$.
- b: Give an argument why a sphere, with a radius R in n dimensions has a surface area proportional to R^{n-1} and check this for a 3d sphere.
- c: Give an argument why the total number of microstates, with a total energy E , available for a single, free, particle is proportional to E .
- d: Give an argument why for n , free, identical, particles with a total energy E the allowed momenta lie on a sphere in $3n$ -dimensional space.
- e: Give an argument why for n , free, particles the total number of available microstates with a fixed total energy E is proportional to $E^{\frac{3n-1}{2}}$.
- f: Why is it justified to assume that the total number of available microstates grows very rapidly with E ? Why is it justified to totally ignore the 1 in the exponent $(3n - 1)$ for physically relevant systems such as a mole of gas?

It seems natural to expect that for some value E' the product of the two functions will attain a maximum that gives the dominating contribution to the partition function, i.e.

$$Z \approx c_0 \Omega(E') \exp(-\beta E') ,$$

where c_0 is some constant that possibly depends on β .

Problem 143: The Maximum at first Guess and T

Consider a gas of n free particles and let the total number of microstates with an energy E be given by $\Omega(E) = c_1 E^{\frac{3n}{2}}$, where c_1 is some constant.

- a: Show that the function $\Omega(E') \exp(-\beta E')$ attains its maximum at $E' = \frac{3}{2} n \frac{1}{\beta}$.
- b: Compare this expression to the total energy for an ideal gas given in the course book and deduce that $\frac{1}{\beta} = kT$.

Now note that the above results imply that

$$kT \log(Z) \approx kT \log(\Omega(E')) - E' .$$

if we assume that $\log(c_0) \approx 0$. This results gives an interesting association. Comparing to the definition of free energy given in the thermodynamics chapter this suggests

$$F[\beta, V] = -\frac{1}{\beta} \log(Z[\beta, \mu, V]) , \quad (5.5.20)$$

$$S[E, V] = k \log(\Omega(E')) . \quad (5.5.21)$$

In the statistical approach Entropy is associated to the total number of microstates available at a given energy! These equations essentially bridge the gap between thermodynamics and classical mechanics.

Problem 144: Checking c_1

In the above derivation we have assumed that c_1 is very small compared to, say, $\Omega(E')$ and $\beta E'$. But can we show this to be reasonable?

a: Use the value for E which we found dominates the partition function and insert it into the approximate expression for Z .

b: Use Mathematica to compute the integral $\int_0^\infty E^{\frac{3n}{2}} \exp(-\beta E)$ and give another expression for Z based on the integral.

c: Use the results under a: and b: to estimate the value of c_1 and check its relative size to $\beta E'$ and $\Omega(E')$. Was our approximation reasonable?

Using these relations we can write down

$$p(\beta, \mu, V) = -\frac{\partial}{\partial V} F[\beta, \mu, V] , \quad (5.5.22)$$

$$\langle N \rangle = -\frac{\partial}{\partial \mu} F[\beta, \mu, V] . \quad (5.5.23)$$

In statistical mechanics the partition functions is directly connected to the free energy. Entropy is a measure of the available number of microstates for a physical system. Research in statistical mechanics therefor revolves around computing partition functions of the systems of interest. In this text we will only present one illustrative but simple example.

5.6. An Ideal Gas

Now you will want to see that the whole machinery actually works. Let us consider the ideal gas in a box of volume V ; i.e. a gas of free point particles with mass m that engage in mutual collisions. If we have a fixed number, N , of such particles in the gas then the total energy in any microstate reads

$$E = \sum_{j=1}^N \frac{\mathbf{p}_j^2}{2m} . \quad (5.6.1)$$

As it is independent of the position of the particles, the integration over all possible positions will simply yield a volume-factor V . Also, because the weight will factorise the integrations over all the allowed momenta will become $3N$ gaussian integrals. Thus the computation of the partition functions becomes

$$\begin{aligned} Z[\beta, V] &= \int d^3x_1 \dots d^3x_N d^3p_1 \dots d^3p_N \exp(-\beta E) , \\ &= \left[\int d^3x d^3p \exp\left(-\beta \frac{\mathbf{p}^2}{2m}\right) \right]^N , \\ &= V^N \left[\int_{-\infty}^{\infty} dp \exp\left(-\beta \frac{p^2}{2m}\right) \right]^{3N} , \\ &= V^N \sqrt{\frac{2\pi m}{\beta}}^{3N} . \end{aligned} \quad (5.6.2)$$

Armed with this result let us first compute the average energy of the ideal gas,

$$\langle E \rangle = -\frac{\partial}{\partial \beta} \log(Z[\beta, V]) = \frac{3}{2} N \frac{1}{\beta} . \quad (5.6.3)$$

This reproduces the results from the book. Let us also compute the pressure of this gas,

$$p(\beta, V) = -\frac{\partial}{\partial V} F[\beta, V] = \frac{N}{\beta V} . \quad (5.6.4)$$

It is not difficult to check that the above result is equivalent to the well-known ideal gas law

$$pV = NkT . \quad (5.6.5)$$

Note that we have now actually *derived* the ideal gas law from the principles of ordinary classical mechanics. Such equations are called *equations of state*. For systems that contain interacting particles they become a whole lot more complicated to compute. Yet some straightforward adaptations can be made. Note that the volume appearing in the equation above is the total volume available for a single particle. If we assume that the gas molecules actually do take up some volume themselves, say a volume v per molecule, then we can correct for this by replacing $V \rightarrow (V - b)$, where $b = Nv$. Such a volume correction occurs because of short-range repulsive forces between the gas molecules. There are however also long-range attractive forces between molecules. They tend to reduce the pressure by an amount proportional to the square of the density, $\frac{N^2}{V^2}$. As a result we can put things together to form

$$(p + \frac{aN}{V^2})(V - v) = kT , \quad (5.6.6)$$

which is the so-called van der Waals gas-law. It describes fairly realistic gasses and even allows for a qualitative description of the phasetransitions between liquid and gas phase.

The whole exercise from week 1 until now may have been tough for you. But the reward is that you now have in your toolbox a set of methods and techniques that will allow you to tackle almost any realistic physical problem. Ofcourse it still will take a lot of exercising and gaining intuition about how these tools work, before you will feel fully proficient with them. At the same time, if you proceed in physics you will also quickly encounter situations in which you will need the ability to turn back to the fundamental concepts behind these methods in order to find out how they must be adapted to more challenging physical systems.

5.7. Assignments

Bibliography

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